Introduction to OceanSDM

2026-07-21

Predicting the dynamic, three-dimensional distributions of marine species is challenging given most SDM tools currently can only predict static 2D habitats (but see an example of voluModel package,Owens & Rahbek, 2023). Here, we present a novel package ‘OceanSDM’ to fill this gap. This package includes 18 user-friendly functions helping researchers to complete tasks from initial data extraction to final publication-quality figure creation. It can be split into three major sections: (1) Prepare species and environmental data, (2) Build 2D SDMs for each temporal bin and generate related figures, and (3) Estimate suitable/preferred depth and map the 3D habitat for each temporal bin. Here I illustrate the utility of this package with a pelagic shark (whale shark). For the temporal bin, I used quarterly bins (four bins).

Here are the packages you need.

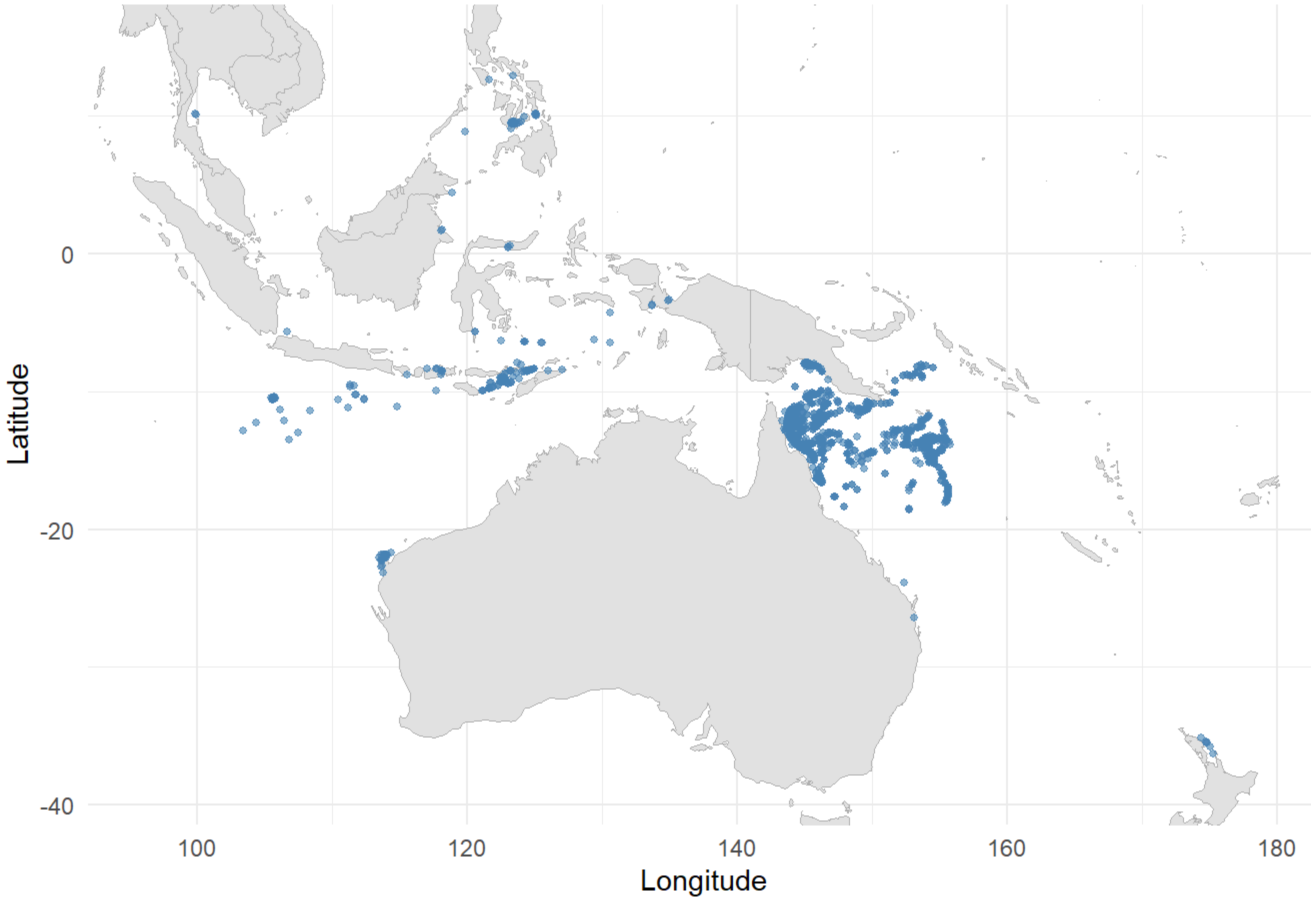
```
# Install the package if you haven't done this yet.
# if (!require("remotes")) install.packages("remotes")
# remotes::install_github("HarryZhang0130/OceanSDM",dependencies = TRUE) # When you are asked to update dependent packages, you can skip all but make sure you installed the 'sdm' package (>= 1.2=59).
library(OceanSDM) # this package
library(sdm) # for using horizontal species distribution modelling functions
library(lubridate) # for date format
```

1. Prepare species and environmental data

1.1 Download species data from GBIF and OBIS using ob_data()

The Global Biodiversity Information Faculty (GBIF) and Ocean Biodiversity Information System (OBIS) are the two major sources of species occurrences at the global scale. The function ob_data() helps users automatically download the geo-referenced occurrences of either a single or many species. Here I demonstrated with an example of smaller temporal and spatial ranges than the real case in the paper to save the running time for creating this vignette.

```
 ws_occ<-ob_data(
    key<-'Rhincodon typus' ,# It can be a vector of the Latin names of all focal species
    sp_geometry <- "POLYGON ((90 -41, 90 40, 180 40, 180 -41, 90 -41))", # Replace this to "POLYGO
N ((90 -41, 90 40, 180 40, 180 -41, 90 -41))" to simulate my real case study!!!
    start_date<-"2015-01-01", # Replace this to "2015-01-01" to simulate my real case study!!!
    end_date<-"2024-12-30",
    event_date<-'2015-01-01,2024-12-30') # Replace this to '2015-01-01,2024-12-30' to simulate my
real case study!!!
#> Retrieved 688 records of approximately 688 (100%)
#>
#> Info: Standard column 'species' already present and used.
#>
#> Info: Standard column 'decimalLatitude' already present and used.
#>
#> Info: Standard column 'decimalLongitude' already present and used.
#>
#> Info: Standard column 'eventDate' already present and used.
#>
#> Info: Standard column 'species' already present and used.
#>
#> Info: Standard column 'decimalLatitude' already present and used.
#>
#> Info: Standard column 'decimalLongitude' already present and used.
#>
#> Info: Standard column 'eventDate' already present and used.
#> Original occurrences:
#> 'data.frame':    5267 obs. of  7 variables:
#>  $ species         : chr  "Rhincodon typus" "Rhincodon typus" "Rhincodon typus" "Rhincodon typu
s" ...
#>  $ decimalLatitude : num  10.1 -15.5 10.1 -36.3 -35.5 ...
#>  $ decimalLongitude: num  125.1 145.6 99.9 175.3 174.8 ...
#>  $ date            : num  19727 19739 19741 19747 19728 ...
#>  $ year            : num  2024 2024 2024 2024 2024 ...
#>  $ month           : num  1 1 1 1 1 1 1 1 1 1 ...
#>  $ day             : num  5 17 19 25 6 31 4 4 4 4 ...
#> NULL
#> Plot occurrences on worldmap:
```




```
# # save the occurrences
write.csv(ws_occ$sp_occ,"F:/whaleshark_sdm/occ/R_typus.csv",row.names = F)
```

1.2 Download .nc files of marine environmental predictors using env_download()

The Copernicus Marine Service (CMS) offers dynamic, three-dimensional environmental data that can be extracted with the function env_download(). But please note you must first install the copernicusmarine toolbox (and configure its running environment with conda or other softwares, see details at https://help.marine.copernicus.eu/en/articles/7970514-copernicus-marine-toolbox-installation (https://help.marine.copernicus.eu/en/articles/7970514-copernicus-marine-toolbox-installation)) on which this function was built. Here I only showed a case of extracting sea surface temperature at smaller temporal and geographic ranges than the real case I used in the paper.

```
## Set up the path of the cms downloading tool and output directory
path_cms<-"C:/shark/miniconda/envs/copernicusmarine/Scripts/copernicusmarine.exe" # replace to you
r own directory
out_dir <- "F:/whaleshark_sdm/cmems" # replace to your own directory
```

1.2.1 Extract surface layer of marine environmental predictors

```
## Note here is a simplified example for fast creating the vignettes
productId <- "cmems_mod_glo_phy-mnstd_my_0.25deg_P1M-m"
variable <- " --variable thetao_mean"
date_min <- ymd(20240101) # start_date, it should be ymd(20150101) in the paper
date_max <- ymd(20241201) # end_date
lon<-c(90,180) # longitudinal range
lat<-c(-41,40) # latitudinal range
## sea surface temperature
depth <- list(0.51, 2) # depth_min, depth_max
out_name <- paste("IndoWPac_temp_1m_y2024.nc", sep="") # it should be paste("IndoWPac_temp_1m_y201
5_y2024.nc", sep="") in the paper
env_download(path_cms,out_dir,productId,variable,
              date_min,date_max,lon,lat,depth,out_name)
#> [1] "\"C:/shark/miniconda/envs/copernicusmarine/Scripts/copernicusmarine.exe\" subset -i cmems_
mod_glo_phy-mnstd_my_0.25deg_P1M-m -x 90 -X 180 -y -41 -Y 40 -t 2024-01-01 -T 2024-12-01 -z 0.51 -
Z 2 -v  --variable thetao_mean -o \"F:/whaleshark_sdm/cmems\" -f IndoWPac_temp_1m_y2024.nc"
#> [1] 127
## sea surface height
# variable <- " --variable zos_mean"
# out_name <- paste("IndoWPac_slh_1m_y2015_y2024.nc", sep="")
# env_download(path_cms,out_dir,productId,variable,
#              date_min,date_max,lon,lat,depth,out_name)
## mixed layer depth
# variable <- " --variable mlotst_mean"
# out_name <- paste("IndoWPac_mlt_1m_y2015_y2024.nc", sep="")
# env_download(path_cms,out_dir,productId,variable,
#              date_min,date_max,lon,lat,depth,out_name)
## salinity
# variable <- " --variable so_mean"
# out_name <- paste("IndoWPac_so_1m_y2015_y2024.nc", sep="")
# env_download(path_cms,out_dir,productId,variable,
#              date_min,date_max,lon,lat,depth,out_name)
```

1.2.2 Extract multiple depth layers of ocean temperature

This step is NOT necessary for the 2D SDM construction, but is required for estimating suitable/tolerable depths based on species realized climatic niche and the three-dimensional distributions of the focal climatic variable (here, temperature) over the chosen temporal bins (here, four quarters). Here given the size of the multi-depth layers of data, I split it into four .nc files (0-200m, 200-400m, 400-1050m, 1050-1946m). The maximum depth (1946m) was based on the known depth records of the whale shark in the study area. The codes were set to ‘not run’ with the hash tag while creating this vignette as it was time consuming. Users can simply remove these hash tags (Using ‘Control+Shift+C’ for Windows or ’Command+Shift+C” for Mac) and run the code.

```
# variable <- " --variable thetao_mean"
# out_dir = "F:/whaleshark_sdm/cmems"
# productId = "cmems_mod_glo_phy-mnstd_my_0.25deg_P1M-m"
# date_min = ymd(20150101) # start_date
# date_max = ymd(20241201) # end_date
# lon<-c(90,180)
# lat<-c(-41,40)
# depth <- list(0.50, 200) # depth_min, depth_max
# out_name <- paste("IndoWPac_temp_200m_y2015_y2024.nc", sep="")
# env_download(path_cms,out_dir,productId,variable,
#              date_min,date_max,lon,lat,depth,out_name)
# depth <- list(200, 400) # depth_min, depth_max
# out_name <- paste("IndoWPac_temp_200_400m_y2015_y2024.nc", sep="")
# env_download(path_cms,out_dir,productId,variable,
#              date_min,date_max,lon,lat,depth,out_name)
# depth <- list(400, 1050) # depth_min, depth_max
# out_name <- paste("IndoWPac_temp_400_1050m_y2015_y2024.nc", sep="")
#env_download(path_cms,out_dir,productId,variable,
#              date_min,date_max,lon,lat,depth,out_name)
# depth <- list(1050, 1946) # depth_min, depth_max
# out_name <- paste("IndoWPac_temp_1050_1946m_y2015_y2024.nc", sep="")
# env_download(path_cms,out_dir,productId,variable,
#              date_min,date_max,lon,lat,depth,out_name)
```

1.2.3 Extract sea surface data for net primary productivity and O2 from another CMS dataset

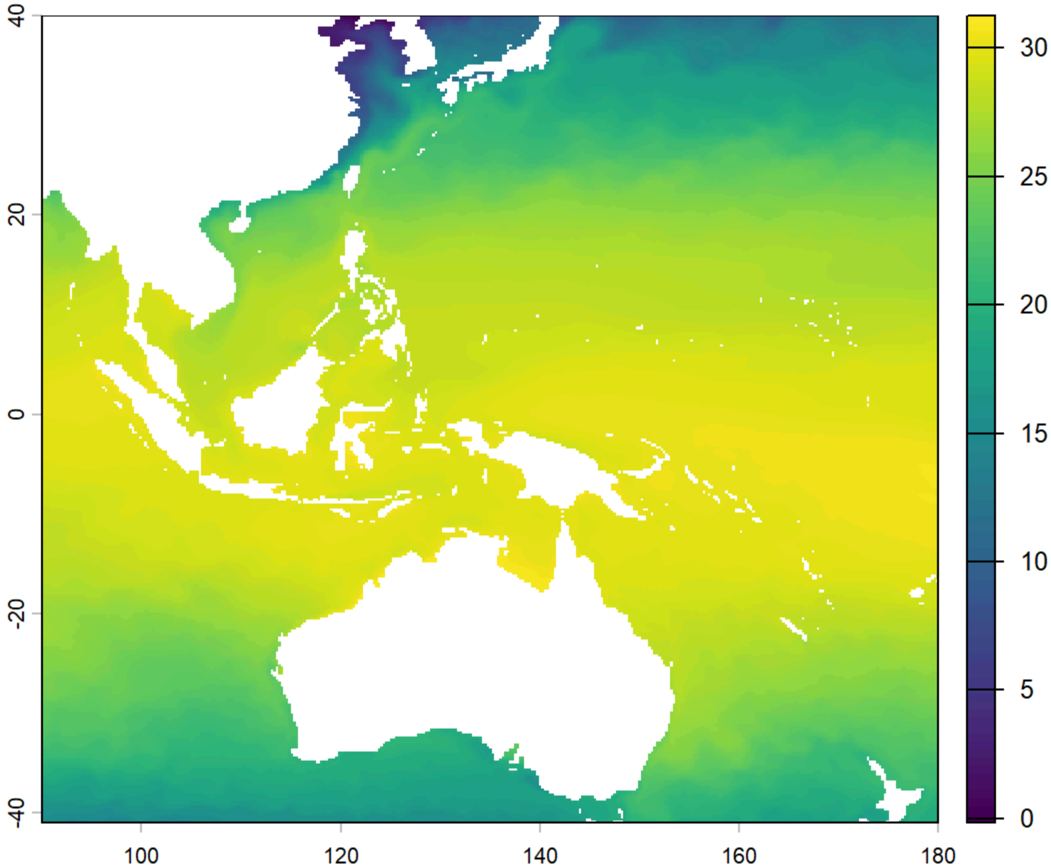
```
# out_dir <- "F:/cmems"
# productId <- "cmems_mod_glo_bgc_my_0.25deg_P1M-m"
# date_min <- ymd(20150101) # start_date
# date_max <- ymd(20241201) # end_date
# lon<-c(90,180)
# lat<-c(-41,40)
# depth <- list(0.51, 2) # depth_min, depth_max
# variable <- " --variable nppv"
# out_name <- paste("IndoWPac_npp_1m_y2015_y2024.nc", sep="")
# env_download(path_cms,out_dir,productId,variable,
#              date_min,date_max,lon,lat,depth,out_name)

# variable <- " --variable o2"
# out_name <- paste("IndoWPac_o2_1m_y2015_y2024.nc", sep="")
# env_download(path_cms,out_dir,productId,variable,
#              date_min,date_max,lon,lat,depth,out_name)
```

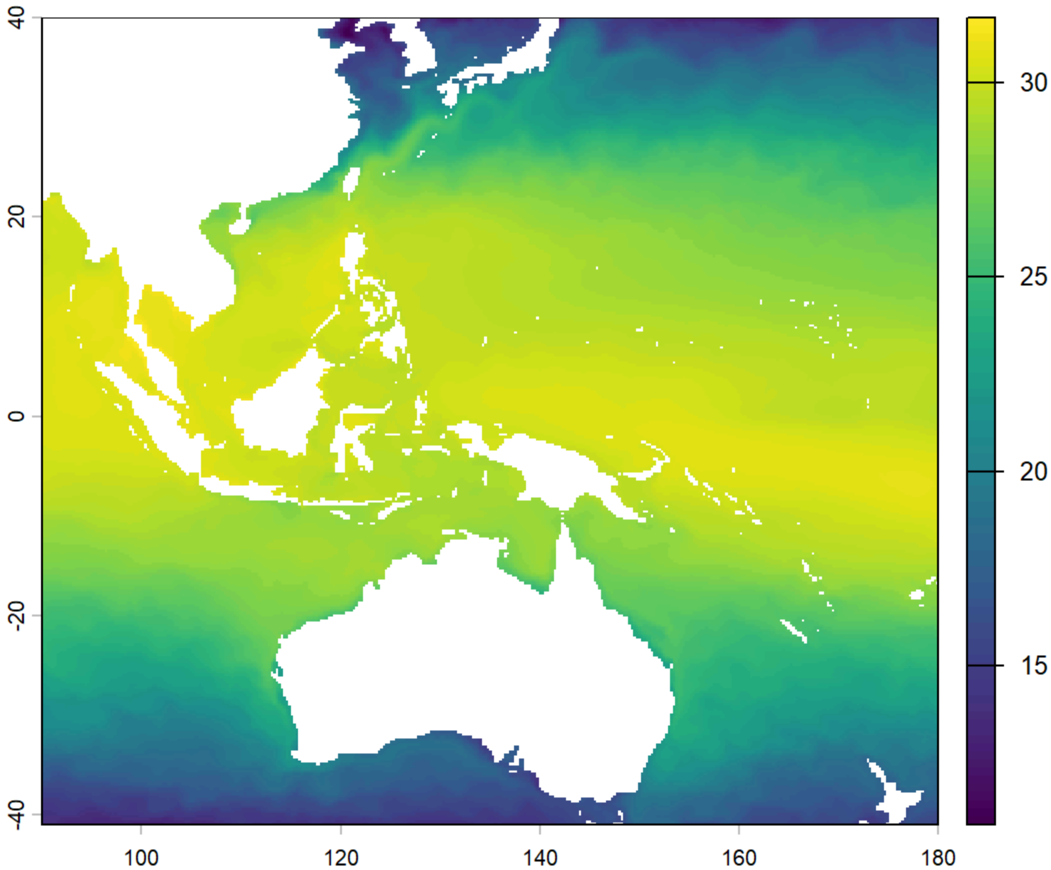
For extracting predictor datasets of future projections, please check https://esgf-node.llnl.gov/search/cmip6/ (https://esgf-node.llnl.gov/search/cmip6/) There is also an R package for downloading CMIP6 data: https://github.com/marine-ecologist/CMIP6/blob/main/CMIP6_projections.R (https://github.com/marine-ecologist/CMIP6/blob/main/CMIP6_projections.R)

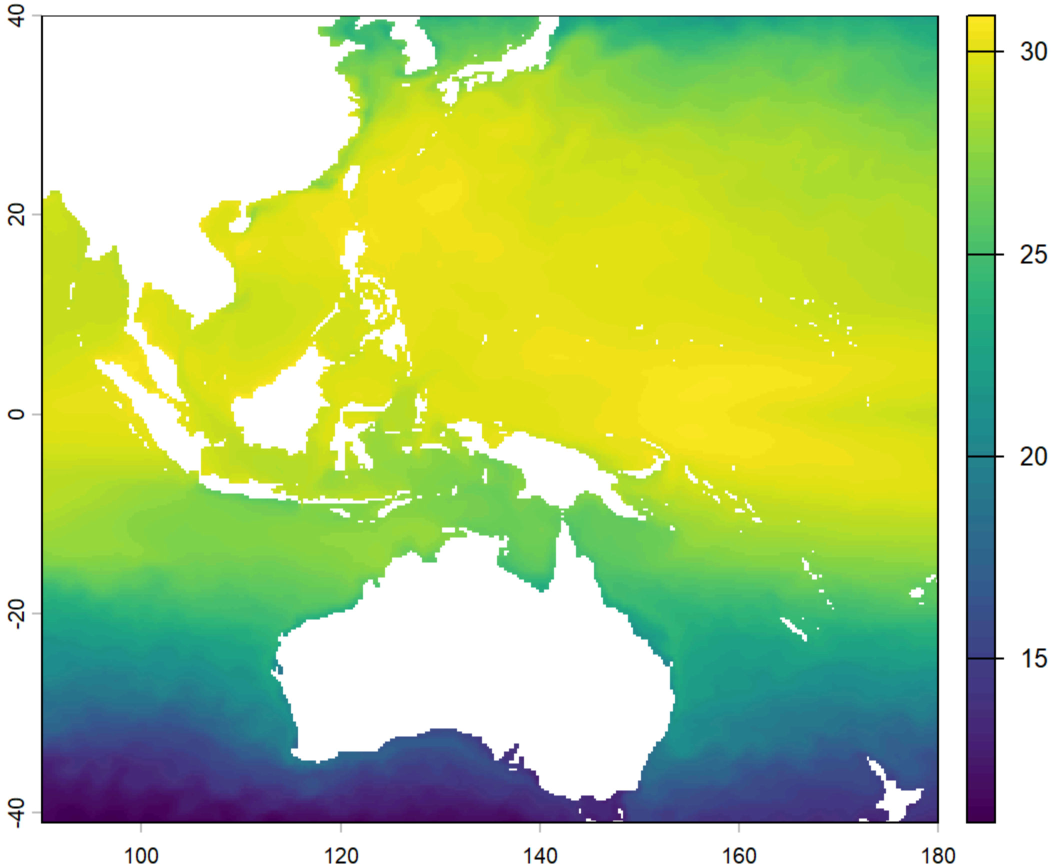
1.2.4 Prepare temporally-binned predictors using env_bin()

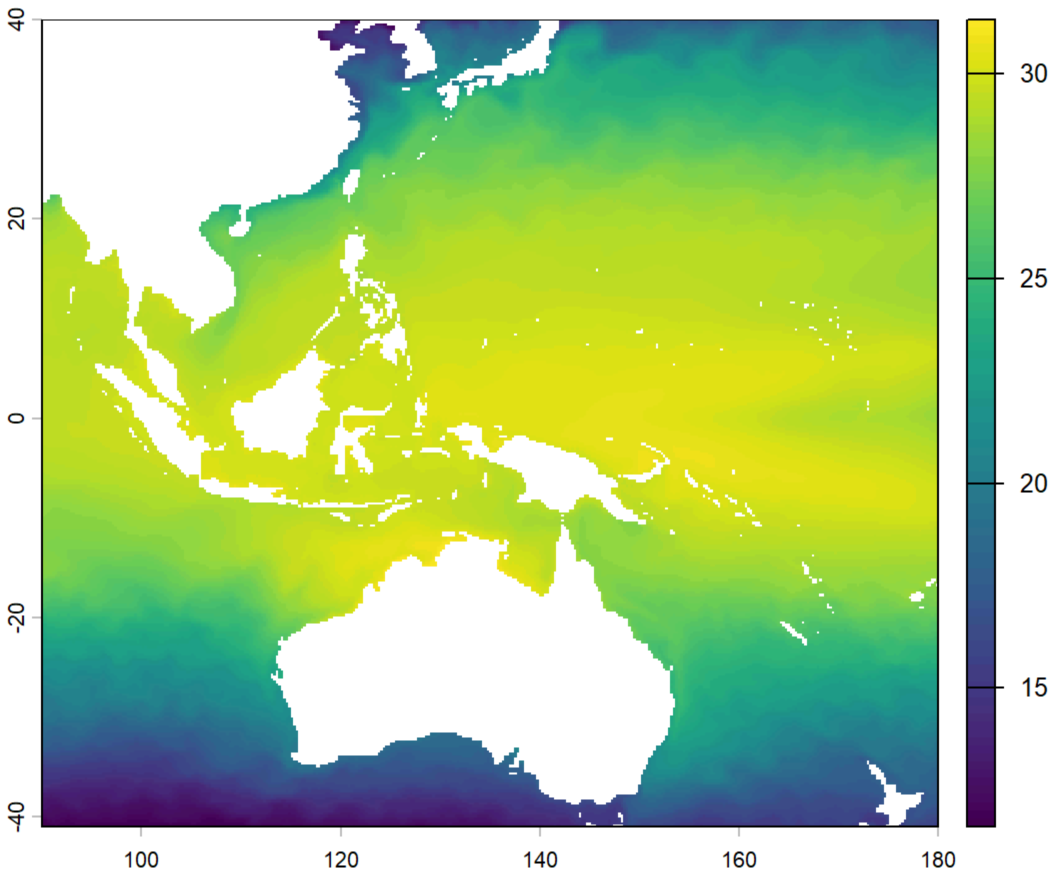
To predict the dynamic distributions of marine species across the 2D space, all predictors should be aggregated into mean values within the specific temporal bins.

```
gc() # clear the R environment
#>            used  (Mb) gc trigger  (Mb) max used  (Mb)
#> Ncells  8218968 439.0   12737551 680.3 12737551 680.3
#> Vcells 33310453 254.2   63733182 486.3 66193628 505.1
## sea surface temperature (Note this is a simplified example of using downloaded data in 1.2.1)
env_bin(sp_trait="pelagic",
        env_path="F:/whaleshark_sdm/cmems/IndoWPac_temp_1m_y2024.nc",
# in the paper, I used "F:/whaleshark_sdm/cmems/IndoWPac_temp_1m_y2015_y2024.nc"
       out_dir="F:/whaleshark_sdm/bin2",
# in the paper, I used "F:/whaleshark_sdm/bin". here I set the directory to a temporal folder to r
eceive the output from this example.
        var_name="thetao_mean",
        lon_min=90,
       lon_max=180,
        lat_min=-41,
        lat_max=40, # in the paper, I used 40
       date_min=ymd(20240101), # in the paper, I used ymd(20150101)
        date_max = ymd(20241201),
        temporal_bin="quarterly")
#> === env_bin: Starting environmental binning ===
#>  Species trait: pelagic
#>  Temporal binning: quarterly
#>  Variable: thetao_mean
#>  Date range: 2024-01-01 to 2024-12-01
#>  Output directory: F:/whaleshark_sdm/bin2
#> Opening NetCDF file...
#>  Time steps: 12
#>  Depth levels: 1 (using surface layer for pelagic)
#> Processing pelagic (using surface layer)
#> --- Quarterly binning: computing quarterly means ---
```


40
20
0
-20
-40
100
120
140
160
180
30
25
20
15
10
5
0


40
20
0
-20
-40
100
120
140
160
180
30
25
20
15


40
20
0
-20
-40
100
120
140
160
180
30
25
20
15


40
20
0
-20
-40
100
120
140
160
180
30
25
20
15

```
#> --- Quarterly binning completed. 4 files generated. ---
#> === env_bin: Processing completed successfully ===
## sea surface height
# env_bin(sp_trait="pelagic",
#        env_path="F:/cmems/IndoWPac_slh_1m_y2015_y2024.nc",
#        out_dir="F:/whaleshark_sdm/bin",
#        var_name="zos_mean",
#        lon_min=90,
#        lon_max=180,
#        lat_min=-41,
#        lat_max=40,
#        date_min=ymd(20150101),
#        date_max = ymd(20241201),

#        temporal_bin="quarterly")
## mixed layer depth
# env_bin(sp_trait="pelagic",
#        env_path="F:/cmems/IndoWPac_mlt_1m_y2015_y2024.nc",
#        out_dir="F:/whaleshark_sdm/bin",
#        var_name="mlotst_mean",
#        lon_min=90,
#        lon_max=180,
#        lat_min=-41,
#        lat_max=40,
#        date_min=ymd(20150101),
#        date_max = ymd(20241201),

#        temporal_bin="quarterly")
## sea surface salinity
# env_bin(sp_trait="pelagic",
#        env_path="F:/cmems/IndoWPac_so_1m_y2015_y2024.nc",
#        out_dir="F:/whaleshark_sdm/bin",
#        var_name="so_mean",
#        lon_min=90,
#        lon_max=180,
#        lat_min=-41,
#        lat_max=40,
#        date_min=ymd(20150101),
#        date_max = ymd(20241201),

#        temporal_bin="quarterly")
## sea surface NPP
# env_bin(sp_trait="pelagic",
#        env_path="F:/cmems/IndoWPac_npp_1m_y2015_y2024.nc",
#        out_dir="F:/whaleshark_sdm/bin",
#        var_name="nppv",
#        lon_min=90,
#        lon_max=180,
#        lat_min=-41,
#        lat_max=40,
#        date_min=ymd(20150101),
#        date_max = ymd(20241201),
```

```
#        temporal_bin="quarterly")
## sea surface dissolved oxygen
# env_bin(sp_trait="pelagic",
#        env_path="F:/cmems/IndoWPac_o2_1m_y2015_y2024.nc",
#        out_dir="F:/whaleshark_sdm/bin",
#        var_name="o2",
#        lon_min=90,
#        lon_max=180,
#        lat_min=-41,
#        lat_max=40,
#        date_min=ymd(20150101),
#        date_max = ymd(20241201),

#        temporal_bin="quarterly")
```

Please note that env_bin() function will create sub-folders (e.g., Q1, Q2, Q3, Q4) for each temporal bin, and save the predictors to the corresponding sub-folders. You may, however, need to rename them to the more meaningful type if necessary. For instance, ocean temperature is usually named as “thetao_mean” in the original cms dataset, and you may change it to “Temperature”.

1.2.5 Download the depth dataset

The Bio-ORACLE provides statistics of ocean depths (mean, maximum, etc.). Here I did not run the code while creating this vignette, but you can run it after removing hash tags.

```
# devtools::install_github("bio-oracle/biooracler")
# library(biooracler)
# library(terra)
# list<-list_layers("Ocean depth")
# dataset_id <- list$dataset_id[158]
# latitude <- c(-41, 40)
# longitude <- c(90, 179.975)
# constraints <- list(latitude, longitude)
# names(constraints) <- c("latitude", "longitude")
# variables <- "bathymetry_mean"
# depth <- download_layers(dataset_id, variables, constraints)
# depth<-abs(depth) # convert values to positive.
# temp<-terra::rast("F:/whaleshark_sdm/bin/Q1/Temperature.asc")
# depth_resampled <- terra::resample(depth, temp, method = "bilinear")
# writeRaster(depth_resampled,"F:/whaleshark_sdm/bin/Depth.asc",
#             NAflag=-9999,overwrite=TRUE,filetype = "AAIGrid")
```

Please note that you need to add the depth_mean.asc sub-folders (e.g., Q1, Q2, Q3, Q4) for each temporal bin, and save the predictors to the corresponding sub-folders. You may, however, need to rename them to the more meaningful type if necessary. For instance, ocean temperature is usually named as “thetao_mean” in the original cms dataset, and you may change it to “temperature”. I have uploaded the prepared quarterly predictor datasets to the package, so I will get access to these system files directly as follows.

1.2.6 Plot environmental predictors (optional)

```
# Load required packages
# library(terra)
# library(ggplot2)
# library(cowplot)
# library(dplyr)
# library(scales)   # for squish
#
# # Main directory and quarters
# main_dir <- system.file("extdata", package = "OceanSDM") # replace to your own directory where t
he predictor data were stored
# quarters <- c("Q1", "Q2", "Q3", "Q4") # replace to the sub-folder for each quarter under the mai
n_dir
#
# # Loop over each quarter
# for (q in quarters) {
#   quarter_path <- file.path(main_dir, q)
#
#   # Read all .asc files in the quarter folder
#   files <- list.files(quarter_path, pattern = "\\.asc$", full.names = TRUE,
#               ignore.case = TRUE)
#   r <- terra::rast(files)
#   names(r) <- tools::file_path_sans_ext(basename(files))
#
#   n_layers <- nlyr(r)
#
# ## Convert raster layers to a data frame for ggplot
#   df_list <- list()
#   for (i in 1:n_layers) {
#     df <- as.data.frame(r[[i]], xy = TRUE, na.rm = FALSE)
#     colnames(df) <- c("x", "y", "value")
#     df$variable <- names(r)[i]
#     df_list[[i]] <- df
#   }
#   df_all <- bind_rows(df_list)
#
# # Create a list of ggplot objects with external labels
#   gg_list <- list()
#   for (i in 1:n_layers) {
#     layer_name <- names(r)[i]
#     df_sub <- filter(df_all, variable == layer_name)
# ## Compute 2% and 98% percentiles for color stretching (percent clip)
#     values_vec <- df_sub$value[!is.na(df_sub$value)]
#     if (length(values_vec) > 0) {
#       p2 <- quantile(values_vec, 0.02, na.rm = TRUE)
#       p98 <- quantile(values_vec, 0.98, na.rm = TRUE)
#       # If p2 == p98, avoid identical limits; use full range
#       if (p2 == p98) {
#         p2 <- min(values_vec, na.rm = TRUE)
#         p98 <- max(values_vec, na.rm = TRUE)
#       }
```

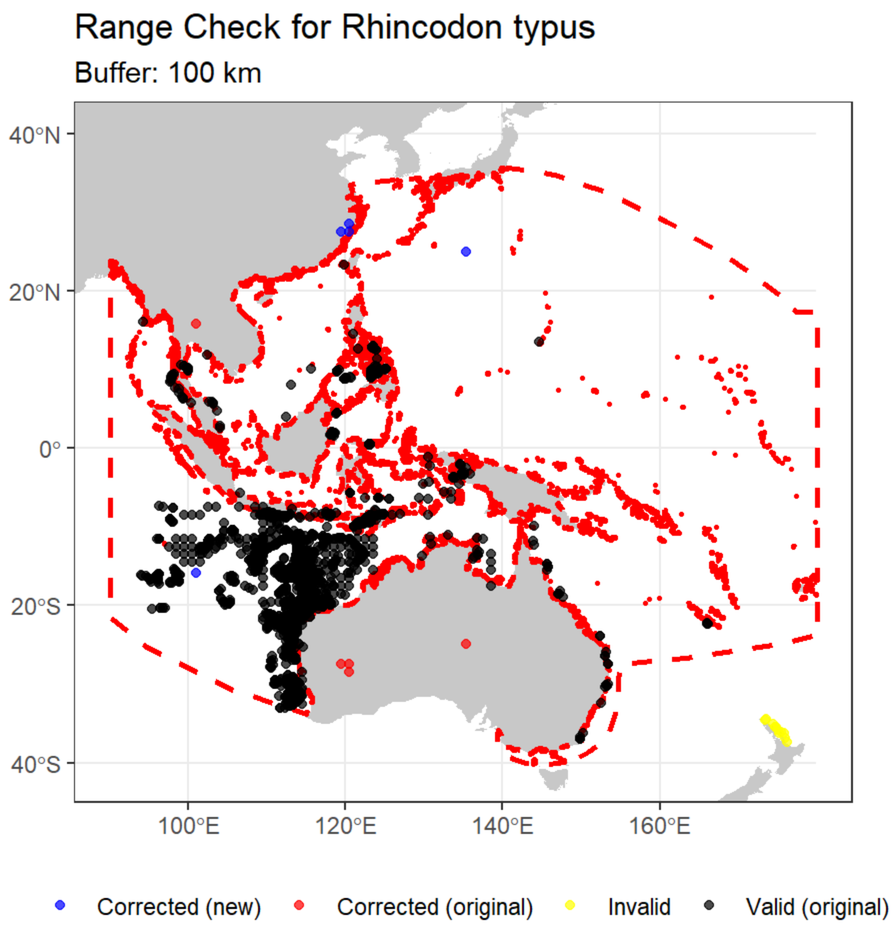
```
#     } else {
#       p2 <- 0; p98 <- 1  # fallback
#     }
#
#     p <- ggplot(df_sub, aes(x = x, y = y, fill = value)) +
#       geom_raster() +
#       scale_fill_viridis_c(
#         option = "inferno",
#         begin = 0.1,
#         end = 0.9,
#         limits = c(p2, p98),       # percent clip range
#         oob = squish,             # squish extremes to the nearest limit color
#         na.value = "transparent"
#       ) +
#       coord_fixed() +
#       labs(title = layer_name, fill = "Value") +
#       theme_minimal() +
#       theme(
#         plot.title = element_text(size =8, face = "bold", hjust = 0.5),
#         legend.position = "right",
#         legend.key.size = unit(0.25, "cm"),
#         legend.text = element_text(size = 4),
#         legend.title = element_text(size = 5),
#         legend.spacing = unit(0.1, "cm"),
#         axis.text = element_blank(),
#         axis.title = element_blank(),
#         axis.ticks = element_blank(),
#         panel.grid = element_blank(),
#         plot.margin = ggplot2::margin(t = 5, r = 5, b = 5, l = 5, unit = "pt")
#       )
#
#     ggdraw_obj <- ggdraw(p) +
#       draw_label(
#         label = paste0("(", letters[i], ")"),
#         x = 0, y = 1,
#         hjust = -0.05,
#         vjust = 1,
#         size = 8,
#         fontface = "bold"
#       )
#     gg_list[[i]] <- ggdraw_obj
#   }
#
# ## Arrange subplots
#   ncol <- ceiling(sqrt(n_layers))
#   nrow <- ceiling(n_layers / ncol)
#   combined <- plot_grid(plotlist = gg_list,
#                  ncol = ncol, nrow = nrow, align = "hv")
#
# ## Create title with "Environmental predictors of QX" and separate by ~2 lines
#   title_text <- paste0("Environmental predictors of ", q)
#   title_gg <- ggdraw() +
```

```
#     draw_label(title_text, x = 0.5, y = 0.5, size = 16, fontface = "bold") +
#     theme(plot.margin = ggplot2::margin(b = 20, t = 10, unit = "pt"))
#
# #   # Combine title and plot vertically
#   final_plot <- plot_grid(title_gg, combined, ncol = 1,
#                   rel_heights = c(0.12, 1), align = "v")
#
# ## Display the plot
#   print(final_plot)
# }
```

1.3 Check occurrence coordinates against species range map with range_check_single() (optional)

Sometimes, you may need to check if all your species occurrences fall within the species range map. You can use range_check_single() for a single species, or range_check() for multiple taxa.

```
# You can get access to the example occurrences of whale sharks in the system file folder extdata.
occ_path <-system.file("extdata", "R_typus.csv", package = "OceanSDM")

# You can get access to the example range map of whale sharks in the system file folder extdata.
range_path <-system.file("extdata", "R_typus_wip2.shp", package = "OceanSDM")
# But for creating this vignette, I loaded them from my PC.
#occ_path<-"F:/whaleshark_sdm/occ/R_typus.csv"
#range_path<-"F:/whaleshark_sdm/range_map/R_typus_wip2.shp"
result <- range_check_single(
  species_name="Rhincodon typus",
      occ_data = occ_path,
      range_path = range_path,
      buffer_km = 100,
  plot = T,
  xlim = c(90, 179.975),
  ylim = c(-41, 40)
  )
#> Processing species: Rhincodon typus - number of points: 5267
#>  Step 1: Checking points inside buffered range map
#>  Step 1 valid points: 5250 invalid points: 17
#>  Step 2: Trying sign combinations for coordinates
#>  Combination 3 corrected points: 5
#>  Final valid points: 5255 final invalid points: 12
```




```
occ_range<-rbind(result$valid_points,result$invalid_points) # Note here I kept the 'invalid_points' because they are actually valid seasonal sightings even though they fell beyond the IUCN range map.
# # save the occurrences
write.csv(occ_range,"F:/whaleshark_sdm/occ/R_typus_r.csv",row.names = F)
```

1.4 Adjust occurrence coordinates with correct_ob()

Previous SDM studies often find that some occurrences were not spatially matched with the environmental layer, but these points fall within a close distance to the environmental layer. This may result from the resolution uncertainty of the occurrences or the coarse resolution of environmental data. I might be worthy correcting them so they match each other, especially when your focal species are naturally rare and you want to keep as many occurrences as possible. Here I showed an example with temperature of the 1st quarter bin

```
## Define input paths
ev_layer<-system.file("extdata", "Q1/Depth.asc", package = "OceanSDM")
occ_range<-system.file("extdata", "R_typus_r.csv", package = "OceanSDM")
output_path<-"F:/whaleshark_sdm/occ/R_typus_adj.csv"
corrected_ob <- correct_ob(
  occ_path = occ_range,
  predictor_raster_path = ev_layer,
  max_distance_km = 56,
  out_dir = output_path,
  verbose = TRUE, plot=T,
  xlim = c(90, 179.975),
  ylim = c(-41, 40)
)
#> Reading predictor raster...
#> Found 833 points with NA predictor values
#> Point 4 corrected at distance 38.35 km
#> Point 9 corrected at distance 20.25 km
#> Point 10 corrected at distance 22.02 km
#> Point 11 corrected at distance 21.71 km
#> Point 12 corrected at distance 17.88 km
#> Point 14 corrected at distance 22.98 km
#> Point 15 corrected at distance 21.13 km
#> Point 16 corrected at distance 17.72 km
#> Point 17 corrected at distance 21.07 km
#> Point 18 corrected at distance 20.28 km
#> Point 19 corrected at distance 18.59 km
#> Point 20 corrected at distance 23.54 km
#> Point 21 corrected at distance 22.87 km
#> Point 22 corrected at distance 22.03 km
#> Point 23 corrected at distance 21.18 km
#> Point 24 corrected at distance 19.78 km
#> Point 25 corrected at distance 22.26 km
#> Point 26 corrected at distance 22.88 km
#> Point 27 corrected at distance 21.01 km
#> Point 28 corrected at distance 21.86 km
#> Point 29 corrected at distance 16.41 km
#> Point 30 corrected at distance 18.31 km
#> Point 31 corrected at distance 14.68 km
#> Point 32 corrected at distance 20.97 km
#> Point 33 corrected at distance 22.04 km
#> Point 34 corrected at distance 22.11 km
#> Point 35 corrected at distance 21.41 km
#> Point 36 corrected at distance 21.54 km
#> Point 37 corrected at distance 21.54 km
#> Point 38 corrected at distance 16.10 km
#> Point 39 corrected at distance 22.26 km
#> Point 40 corrected at distance 20.45 km
#> Point 41 corrected at distance 21.57 km
#> Point 42 corrected at distance 19.48 km
#> Point 43 corrected at distance 25.45 km
#> Point 44 corrected at distance 20.27 km
```

```
#> Point 45 corrected at distance 16.13 km
#> Point 46 corrected at distance 22.99 km
#> Point 48 corrected at distance 21.44 km
#> Point 50 corrected at distance 19.09 km
#> Point 51 corrected at distance 21.12 km
#> Point 52 corrected at distance 21.11 km
#> Point 53 corrected at distance 21.99 km
#> Point 54 corrected at distance 22.33 km
#> Point 55 corrected at distance 21.24 km
#> Point 56 corrected at distance 18.74 km
#> Point 57 corrected at distance 21.78 km
#> Point 58 corrected at distance 22.71 km
#> Point 59 corrected at distance 21.68 km
#> Point 60 corrected at distance 22.74 km
#> Point 61 corrected at distance 18.08 km
#> Point 63 corrected at distance 22.31 km
#> Point 64 corrected at distance 22.08 km
#> Point 65 corrected at distance 22.15 km
#> Point 66 corrected at distance 21.11 km
#> Point 67 corrected at distance 21.96 km
#> Point 68 corrected at distance 21.69 km
#> Point 69 corrected at distance 21.61 km
#> Point 70 corrected at distance 23.28 km
#> Point 71 corrected at distance 25.58 km
#> Point 72 corrected at distance 21.42 km
#> Point 73 corrected at distance 22.42 km
#> Point 74 corrected at distance 17.28 km
#> Point 75 corrected at distance 21.67 km
#> Point 76 corrected at distance 21.40 km
#> Point 77 corrected at distance 19.14 km
#> Point 78 corrected at distance 21.50 km
#> Point 80 corrected at distance 21.46 km
#> Point 81 corrected at distance 22.14 km
#> Point 82 corrected at distance 22.14 km
#> Point 83 corrected at distance 19.91 km
#> Point 84 corrected at distance 24.98 km
#> Point 85 corrected at distance 21.83 km
#> Point 86 corrected at distance 22.35 km
#> Point 88 corrected at distance 21.67 km
#> Point 89 corrected at distance 21.20 km
#> Point 90 corrected at distance 22.49 km
#> Point 91 corrected at distance 17.93 km
#> Point 92 corrected at distance 22.44 km
#> Point 94 corrected at distance 21.80 km
#> Point 95 corrected at distance 18.23 km
#> Point 96 corrected at distance 22.43 km
#> Point 97 corrected at distance 21.63 km
#> Point 98 corrected at distance 21.62 km
#> Point 99 corrected at distance 21.42 km
#> Point 101 corrected at distance 20.94 km
#> Point 102 corrected at distance 20.85 km
#> Point 103 corrected at distance 22.29 km
```

```
#> Point 104 corrected at distance 23.18 km
#> Point 105 corrected at distance 21.92 km
#> Point 106 corrected at distance 21.63 km
#> Point 107 corrected at distance 21.60 km
#> Point 108 corrected at distance 19.44 km
#> Point 109 corrected at distance 22.61 km
#> Point 110 corrected at distance 22.03 km
#> Point 111 corrected at distance 24.60 km
#> Point 112 corrected at distance 22.08 km
#> Point 114 corrected at distance 28.57 km
#> Point 116 corrected at distance 19.08 km
#> Point 118 corrected at distance 21.85 km
#> Point 119 corrected at distance 20.42 km
#> Point 120 corrected at distance 22.52 km
#> Point 121 corrected at distance 22.13 km
#> Point 122 corrected at distance 14.06 km
#> Point 123 corrected at distance 20.53 km
#> Point 124 corrected at distance 22.88 km
#> Point 125 corrected at distance 22.14 km
#> Point 126 corrected at distance 21.68 km
#> Point 127 corrected at distance 22.73 km
#> Point 128 corrected at distance 21.84 km
#> Point 129 corrected at distance 20.06 km
#> Point 130 corrected at distance 16.01 km
#> Point 132 corrected at distance 21.22 km
#> Point 133 corrected at distance 22.07 km
#> Point 134 corrected at distance 22.45 km
#> Point 135 corrected at distance 23.11 km
#> Point 136 corrected at distance 21.30 km
#> Point 137 corrected at distance 21.96 km
#> Point 138 corrected at distance 21.99 km
#> Point 139 corrected at distance 19.48 km
#> Point 140 corrected at distance 21.15 km
#> Point 142 corrected at distance 21.84 km
#> Point 143 corrected at distance 18.17 km
#> Point 144 corrected at distance 21.61 km
#> Point 145 corrected at distance 22.66 km
#> Point 146 corrected at distance 21.88 km
#> Point 147 corrected at distance 22.48 km
#> Point 148 corrected at distance 21.66 km
#> Point 149 corrected at distance 17.38 km
#> Point 150 corrected at distance 23.18 km
#> Point 151 corrected at distance 20.18 km
#> Point 152 corrected at distance 22.12 km
#> Point 153 corrected at distance 22.98 km
#> Point 154 corrected at distance 21.38 km
#> Point 155 corrected at distance 22.65 km
#> Point 156 corrected at distance 20.79 km
#> Point 157 corrected at distance 22.99 km
#> Point 158 corrected at distance 20.31 km
#> Point 159 corrected at distance 19.91 km
#> Point 160 corrected at distance 22.50 km
```

```
#> Point 161 corrected at distance 20.80 km
#> Point 162 corrected at distance 21.56 km
#> Point 163 corrected at distance 21.15 km
#> Point 164 corrected at distance 21.25 km
#> Point 165 corrected at distance 22.88 km
#> Point 166 corrected at distance 22.17 km
#> Point 167 corrected at distance 22.51 km
#> Point 168 corrected at distance 21.48 km
#> Point 169 corrected at distance 21.24 km
#> Point 170 corrected at distance 20.41 km
#> Point 171 corrected at distance 20.46 km
#> Point 172 corrected at distance 22.37 km
#> Point 173 corrected at distance 20.96 km
#> Point 174 corrected at distance 22.60 km
#> Point 176 corrected at distance 19.80 km
#> Point 177 corrected at distance 14.66 km
#> Point 178 corrected at distance 22.40 km
#> Point 179 corrected at distance 16.17 km
#> Point 180 corrected at distance 23.62 km
#> Point 181 corrected at distance 25.20 km
#> Point 182 corrected at distance 23.30 km
#> Point 183 corrected at distance 19.51 km
#> Point 184 corrected at distance 22.14 km
#> Point 185 corrected at distance 20.39 km
#> Point 186 corrected at distance 19.69 km
#> Point 187 corrected at distance 22.15 km
#> Point 188 corrected at distance 23.47 km
#> Point 189 corrected at distance 16.57 km
#> Point 190 corrected at distance 23.22 km
#> Point 191 corrected at distance 21.68 km
#> Point 198 corrected at distance 22.12 km
#> Point 199 corrected at distance 21.89 km
#> Point 200 corrected at distance 22.97 km
#> Point 201 corrected at distance 20.64 km
#> Point 202 corrected at distance 25.50 km
#> Point 203 corrected at distance 22.37 km
#> Point 204 corrected at distance 21.54 km
#> Point 205 corrected at distance 21.61 km
#> Point 207 corrected at distance 22.61 km
#> Point 208 corrected at distance 22.12 km
#> Point 209 corrected at distance 21.53 km
#> Point 210 corrected at distance 21.20 km
#> Point 211 corrected at distance 21.74 km
#> Point 213 corrected at distance 21.69 km
#> Point 214 corrected at distance 22.16 km
#> Point 215 corrected at distance 22.24 km
#> Point 216 corrected at distance 21.60 km
#> Point 217 corrected at distance 21.05 km
#> Point 218 corrected at distance 20.50 km
#> Point 219 corrected at distance 21.31 km
#> Point 220 corrected at distance 20.26 km
#> Point 221 corrected at distance 18.70 km
```

```
#> Point 222 corrected at distance 22.90 km
#> Point 223 corrected at distance 22.00 km
#> Point 224 corrected at distance 21.95 km
#> Point 225 corrected at distance 22.70 km
#> Point 226 corrected at distance 21.20 km
#> Point 227 corrected at distance 20.78 km
#> Point 228 corrected at distance 19.11 km
#> Point 229 corrected at distance 22.18 km
#> Point 235 corrected at distance 15.12 km
#> Point 238 corrected at distance 18.29 km
#> Point 243 corrected at distance 23.05 km
#> Point 254 corrected at distance 26.69 km
#> Point 265 corrected at distance 22.53 km
#> Point 268 corrected at distance 19.52 km
#> Point 274 corrected at distance 17.05 km
#> Point 333 corrected at distance 17.02 km
#> Point 337 corrected at distance 21.54 km
#> Point 340 corrected at distance 18.51 km
#> Point 344 corrected at distance 23.21 km
#> Point 349 corrected at distance 15.68 km
#> Point 379 corrected at distance 36.94 km
#> Point 380 corrected at distance 17.00 km
#> Point 382 corrected at distance 19.56 km
#> Point 383 corrected at distance 32.47 km
#> Point 384 corrected at distance 37.40 km
#> Point 385 corrected at distance 46.05 km
#> Point 386 corrected at distance 14.48 km
#> Point 388 corrected at distance 21.65 km
#> Point 389 corrected at distance 42.31 km
#> Point 391 corrected at distance 14.41 km
#> Point 394 corrected at distance 18.92 km
#> Point 396 corrected at distance 26.96 km
#> Point 397 corrected at distance 23.68 km
#> Point 398 corrected at distance 41.06 km
#> Point 399 corrected at distance 19.65 km
#> Point 404 corrected at distance 49.14 km
#> Point 405 corrected at distance 53.82 km
#> Point 412 corrected at distance 18.58 km
#> Point 413 corrected at distance 19.22 km
#> Point 417 corrected at distance 27.97 km
#> Point 420 corrected at distance 40.54 km
#> Point 429 corrected at distance 36.61 km
#> Point 431 corrected at distance 18.38 km
#> Point 432 corrected at distance 15.78 km
#> Point 433 corrected at distance 19.09 km
#> Point 438 corrected at distance 15.80 km
#> Point 441 corrected at distance 45.12 km
#> Point 443 corrected at distance 38.93 km
#> Point 454 corrected at distance 41.92 km
#> Point 459 corrected at distance 23.79 km
#> Point 464 corrected at distance 21.72 km
#> Point 465 corrected at distance 21.55 km
```

```
#> Point 466 corrected at distance 18.43 km
#> Point 467 corrected at distance 22.31 km
#> Point 469 corrected at distance 18.79 km
#> Point 470 corrected at distance 26.45 km
#> Point 471 corrected at distance 23.23 km
#> Point 472 corrected at distance 21.90 km
#> Point 473 corrected at distance 20.99 km
#> Point 474 corrected at distance 22.33 km
#> Point 476 corrected at distance 21.66 km
#> Point 477 corrected at distance 23.08 km
#> Point 478 corrected at distance 19.85 km
#> Point 479 corrected at distance 22.36 km
#> Point 481 corrected at distance 23.84 km
#> Point 482 corrected at distance 21.95 km
#> Point 483 corrected at distance 21.79 km
#> Point 484 corrected at distance 21.27 km
#> Point 485 corrected at distance 23.71 km
#> Point 486 corrected at distance 21.55 km
#> Point 487 corrected at distance 20.56 km
#> Point 488 corrected at distance 23.26 km
#> Point 489 corrected at distance 21.99 km
#> Point 490 corrected at distance 23.20 km
#> Point 491 corrected at distance 18.44 km
#> Point 492 corrected at distance 20.88 km
#> Point 493 corrected at distance 22.21 km
#> Point 494 corrected at distance 22.58 km
#> Point 495 corrected at distance 25.78 km
#> Point 496 corrected at distance 19.68 km
#> Point 497 corrected at distance 21.53 km
#> Point 498 corrected at distance 21.90 km
#> Point 499 corrected at distance 23.10 km
#> Point 500 corrected at distance 24.81 km
#> Point 501 corrected at distance 23.85 km
#> Point 502 corrected at distance 22.70 km
#> Point 503 corrected at distance 22.06 km
#> Point 504 corrected at distance 23.32 km
#> Point 505 corrected at distance 21.65 km
#> Point 506 corrected at distance 21.07 km
#> Point 507 corrected at distance 22.25 km
#> Point 508 corrected at distance 19.28 km
#> Point 509 corrected at distance 25.04 km
#> Point 511 corrected at distance 20.71 km
#> Point 512 corrected at distance 21.99 km
#> Point 514 corrected at distance 22.97 km
#> Point 515 corrected at distance 21.20 km
#> Point 517 corrected at distance 22.92 km
#> Point 518 corrected at distance 21.70 km
#> Point 521 corrected at distance 21.97 km
#> Point 522 corrected at distance 22.62 km
#> Point 523 corrected at distance 21.70 km
#> Point 524 corrected at distance 25.11 km
#> Point 525 corrected at distance 21.28 km
```

```
#> Point 526 corrected at distance 22.86 km
#> Point 527 corrected at distance 21.99 km
#> Point 529 corrected at distance 20.10 km
#> Point 532 corrected at distance 19.03 km
#> Point 534 corrected at distance 24.90 km
#> Point 536 corrected at distance 21.36 km
#> Point 538 corrected at distance 20.74 km
#> Point 540 corrected at distance 19.15 km
#> Point 541 corrected at distance 25.84 km
#> Point 542 corrected at distance 20.76 km
#> Point 543 corrected at distance 20.82 km
#> Point 546 corrected at distance 22.40 km
#> Point 549 corrected at distance 22.01 km
#> Point 550 corrected at distance 21.06 km
#> Point 551 corrected at distance 20.10 km
#> Point 552 corrected at distance 22.05 km
#> Point 553 corrected at distance 25.22 km
#> Point 555 corrected at distance 22.40 km
#> Point 556 corrected at distance 21.17 km
#> Point 557 corrected at distance 24.24 km
#> Point 558 corrected at distance 21.15 km
#> Point 560 corrected at distance 22.27 km
#> Point 561 corrected at distance 21.86 km
#> Point 563 corrected at distance 21.73 km
#> Point 564 corrected at distance 21.46 km
#> Point 566 corrected at distance 19.77 km
#> Point 567 corrected at distance 21.99 km
#> Point 568 corrected at distance 22.09 km
#> Point 569 corrected at distance 25.85 km
#> Point 572 corrected at distance 26.01 km
#> Point 573 corrected at distance 24.24 km
#> Point 574 corrected at distance 21.28 km
#> Point 577 corrected at distance 21.60 km
#> Point 579 corrected at distance 17.43 km
#> Point 580 corrected at distance 21.72 km
#> Point 581 corrected at distance 23.00 km
#> Point 582 corrected at distance 18.39 km
#> Point 583 corrected at distance 21.87 km
#> Point 584 corrected at distance 21.17 km
#> Point 585 corrected at distance 22.05 km
#> Point 587 corrected at distance 24.76 km
#> Point 590 corrected at distance 25.79 km
#> Point 591 corrected at distance 22.10 km
#> Point 592 corrected at distance 20.54 km
#> Point 593 corrected at distance 27.24 km
#> Point 594 corrected at distance 20.81 km
#> Point 595 corrected at distance 22.64 km
#> Point 596 corrected at distance 20.36 km
#> Point 597 corrected at distance 20.11 km
#> Point 599 corrected at distance 22.09 km
#> Point 600 corrected at distance 22.10 km
#> Point 601 corrected at distance 22.21 km
```

```
#> Point 602 corrected at distance 21.67 km
#> Point 603 corrected at distance 21.32 km
#> Point 604 corrected at distance 23.16 km
#> Point 605 corrected at distance 18.69 km
#> Point 606 corrected at distance 17.26 km
#> Point 607 corrected at distance 19.34 km
#> Point 608 corrected at distance 22.08 km
#> Point 611 corrected at distance 21.86 km
#> Point 615 corrected at distance 22.17 km
#> Point 616 corrected at distance 21.52 km
#> Point 619 corrected at distance 17.98 km
#> Point 620 corrected at distance 21.62 km
#> Point 621 corrected at distance 24.76 km
#> Point 623 corrected at distance 19.61 km
#> Point 626 corrected at distance 18.59 km
#> Point 627 corrected at distance 19.25 km
#> Point 628 corrected at distance 19.88 km
#> Point 629 corrected at distance 18.10 km
#> Point 633 corrected at distance 19.70 km
#> Point 634 corrected at distance 22.29 km
#> Point 635 corrected at distance 24.16 km
#> Point 636 corrected at distance 21.52 km
#> Point 637 corrected at distance 25.81 km
#> Point 639 corrected at distance 24.89 km
#> Point 640 corrected at distance 22.05 km
#> Point 645 corrected at distance 23.83 km
#> Point 646 corrected at distance 23.57 km
#> Point 648 corrected at distance 21.04 km
#> Point 649 corrected at distance 21.39 km
#> Point 651 corrected at distance 25.37 km
#> Point 652 corrected at distance 22.86 km
#> Point 654 corrected at distance 22.74 km
#> Point 655 corrected at distance 22.51 km
#> Point 659 removed: nearest cell at 68.02 km (exceeds max 56.0 km)
#> Point 669 corrected at distance 21.66 km
#> Point 678 corrected at distance 18.14 km
#> Point 684 corrected at distance 15.17 km
#> Point 688 corrected at distance 48.82 km
#> Point 695 corrected at distance 24.90 km
#> Point 696 corrected at distance 24.79 km
#> Point 697 corrected at distance 17.52 km
#> Point 698 corrected at distance 34.65 km
#> Point 699 corrected at distance 20.39 km
#> Point 700 corrected at distance 16.91 km
#> Point 706 corrected at distance 17.10 km
#> Point 717 corrected at distance 23.91 km
#> Point 721 corrected at distance 18.47 km
#> Point 726 corrected at distance 18.97 km
#> Point 728 corrected at distance 18.97 km
#> Point 748 corrected at distance 21.15 km
#> Point 749 corrected at distance 15.89 km
#> Point 751 corrected at distance 15.89 km
```

```
#> Point 754 corrected at distance 15.89 km
#> Point 758 corrected at distance 15.89 km
#> Point 759 corrected at distance 18.97 km
#> Point 792 corrected at distance 18.82 km
#> Point 793 corrected at distance 18.83 km
#> Point 795 corrected at distance 18.82 km
#> Point 798 corrected at distance 21.15 km
#> Point 801 corrected at distance 21.15 km
#> Point 802 corrected at distance 18.97 km
#> Point 803 corrected at distance 21.15 km
#> Point 806 corrected at distance 18.83 km
#> Point 821 corrected at distance 15.89 km
#> Point 827 corrected at distance 18.97 km
#> Point 834 corrected at distance 18.46 km
#> Point 840 corrected at distance 18.97 km
#> Point 851 corrected at distance 18.82 km
#> Point 854 corrected at distance 18.97 km
#> Point 857 corrected at distance 15.10 km
#> Point 864 corrected at distance 20.19 km
#> Point 874 corrected at distance 18.46 km
#> Point 876 corrected at distance 15.89 km
#> Point 894 corrected at distance 18.46 km
#> Point 900 corrected at distance 21.15 km
#> Point 917 corrected at distance 16.58 km
#> Point 930 corrected at distance 15.67 km
#> Point 933 corrected at distance 13.83 km
#> Point 942 corrected at distance 14.68 km
#> Point 943 corrected at distance 16.58 km
#> Point 944 corrected at distance 14.68 km
#> Point 947 corrected at distance 24.82 km
#> Point 948 corrected at distance 14.68 km
#> Point 952 corrected at distance 14.68 km
#> Point 955 corrected at distance 14.68 km
#> Point 957 corrected at distance 26.01 km
#> Point 967 corrected at distance 14.68 km
#> Point 968 corrected at distance 24.82 km
#> Point 971 corrected at distance 24.82 km
#> Point 973 corrected at distance 14.76 km
#> Point 974 corrected at distance 14.68 km
#> Point 977 corrected at distance 14.68 km
#> Point 978 corrected at distance 14.42 km
#> Point 985 corrected at distance 14.69 km
#> Point 998 corrected at distance 19.92 km
#> Point 999 corrected at distance 24.82 km
#> Point 1007 corrected at distance 24.82 km
#> Point 1008 corrected at distance 19.92 km
#> Point 1009 corrected at distance 19.92 km
#> Point 1010 corrected at distance 14.68 km
#> Point 1013 corrected at distance 16.01 km
#> Point 1016 corrected at distance 14.76 km
#> Point 1018 corrected at distance 16.71 km
#> Point 1019 corrected at distance 16.83 km
```

```
#> Point 1022 corrected at distance 19.92 km
#> Point 1025 corrected at distance 14.62 km
#> Point 1031 corrected at distance 15.52 km
#> Point 1039 corrected at distance 14.68 km
#> Point 1046 corrected at distance 14.68 km
#> Point 1047 corrected at distance 24.82 km
#> Point 1048 corrected at distance 17.67 km
#> Point 1049 corrected at distance 16.67 km
#> Point 1050 corrected at distance 15.78 km
#> Point 1051 corrected at distance 14.68 km
#> Point 1053 corrected at distance 14.68 km
#> Point 1055 corrected at distance 21.15 km
#> Point 1059 corrected at distance 21.73 km
#> Point 1062 corrected at distance 16.71 km
#> Point 1064 corrected at distance 14.68 km
#> Point 1068 corrected at distance 19.92 km
#> Point 1073 corrected at distance 14.68 km
#> Point 1075 corrected at distance 14.68 km
#> Point 1076 corrected at distance 14.68 km
#> Point 1078 corrected at distance 19.92 km
#> Point 1081 corrected at distance 14.68 km
#> Point 1086 corrected at distance 14.68 km
#> Point 1089 corrected at distance 18.11 km
#> Point 1091 corrected at distance 14.90 km
#> Point 1101 corrected at distance 19.92 km
#> Point 1108 corrected at distance 16.58 km
#> Point 1115 corrected at distance 15.90 km
#> Point 1138 corrected at distance 15.89 km
#> Point 1139 corrected at distance 18.46 km
#> Point 1145 corrected at distance 23.91 km
#> Point 1152 corrected at distance 21.15 km
#> Point 1154 corrected at distance 14.86 km
#> Point 1155 corrected at distance 18.82 km
#> Point 1161 corrected at distance 18.47 km
#> Point 1168 corrected at distance 15.89 km
#> Point 1171 corrected at distance 18.29 km
#> Point 1173 corrected at distance 18.82 km
#> Point 1177 corrected at distance 16.73 km
#> Point 1190 corrected at distance 23.91 km
#> Point 1201 corrected at distance 18.97 km
#> Point 1205 corrected at distance 14.04 km
#> Point 1210 corrected at distance 14.86 km
#> Point 1221 corrected at distance 18.47 km
#> Point 1230 corrected at distance 18.82 km
#> Point 1232 corrected at distance 16.20 km
#> Point 1234 corrected at distance 18.46 km
#> Point 1235 corrected at distance 18.29 km
#> Point 1251 corrected at distance 15.60 km
#> Point 1259 corrected at distance 18.29 km
#> Point 1265 corrected at distance 18.97 km
#> Point 1278 corrected at distance 18.29 km
#> Point 1281 corrected at distance 15.89 km
```

```
#> Point 1282 corrected at distance 18.29 km
#> Point 1286 corrected at distance 21.15 km
#> Point 1287 corrected at distance 18.46 km
#> Point 1291 corrected at distance 17.68 km
#> Point 1293 corrected at distance 18.46 km
#> Point 1294 corrected at distance 18.29 km
#> Point 1297 corrected at distance 16.83 km
#> Point 1303 corrected at distance 18.82 km
#> Point 1308 corrected at distance 18.29 km
#> Point 1314 corrected at distance 23.91 km
#> Point 1337 corrected at distance 15.89 km
#> Point 1349 corrected at distance 18.51 km
#> Point 1353 corrected at distance 20.27 km
#> Point 1355 corrected at distance 18.14 km
#> Point 1356 corrected at distance 18.82 km
#> Point 1373 corrected at distance 24.82 km
#> Point 1376 corrected at distance 24.82 km
#> Point 1381 corrected at distance 15.89 km
#> Point 1383 corrected at distance 14.68 km
#> Point 1386 corrected at distance 19.92 km
#> Point 1390 corrected at distance 19.92 km
#> Point 1397 corrected at distance 21.29 km
#> Point 1399 corrected at distance 19.92 km
#> Point 1403 corrected at distance 19.92 km
#> Point 1408 corrected at distance 19.92 km
#> Point 1412 corrected at distance 15.89 km
#> Point 1418 corrected at distance 19.92 km
#> Point 1419 corrected at distance 15.89 km
#> Point 1432 corrected at distance 19.92 km
#> Point 1439 corrected at distance 16.79 km
#> Point 1441 corrected at distance 16.18 km
#> Point 1443 corrected at distance 42.39 km
#> Point 1444 corrected at distance 14.68 km
#> Point 1457 corrected at distance 14.76 km
#> Point 1459 corrected at distance 24.68 km
#> Point 1461 corrected at distance 14.68 km
#> Point 1463 corrected at distance 16.61 km
#> Point 1465 corrected at distance 19.92 km
#> Point 1471 corrected at distance 19.92 km
#> Point 1481 corrected at distance 16.61 km
#> Point 1485 corrected at distance 18.44 km
#> Point 1486 corrected at distance 24.82 km
#> Point 1489 corrected at distance 16.61 km
#> Point 1491 corrected at distance 19.99 km
#> Point 1492 corrected at distance 19.15 km
#> Point 1493 corrected at distance 14.76 km
#> Point 1498 corrected at distance 15.39 km
#> Point 1502 corrected at distance 19.99 km
#> Point 1508 corrected at distance 14.76 km
#> Point 1513 corrected at distance 21.29 km
#> Point 1532 corrected at distance 19.99 km
#> Point 1545 corrected at distance 19.99 km
```

```
#> Point 1546 corrected at distance 33.92 km
#> Point 1547 corrected at distance 30.99 km
#> Point 1548 corrected at distance 20.46 km
#> Point 1553 corrected at distance 17.17 km
#> Point 1559 corrected at distance 19.99 km
#> Point 1562 corrected at distance 19.99 km
#> Point 1600 corrected at distance 14.76 km
#> Point 1625 removed: nearest cell at 85.59 km (exceeds max 56.0 km)
#> Point 1627 corrected at distance 16.39 km
#> Point 1729 corrected at distance 18.88 km
#> Point 1740 corrected at distance 25.38 km
#> Point 1816 corrected at distance 45.53 km
#> Point 1829 corrected at distance 18.82 km
#> Point 1830 corrected at distance 14.86 km
#> Point 1838 corrected at distance 21.15 km
#> Point 1842 corrected at distance 15.89 km
#> Point 1844 corrected at distance 15.89 km
#> Point 1848 corrected at distance 18.82 km
#> Point 1871 corrected at distance 19.99 km
#> Point 1914 corrected at distance 18.47 km
#> Point 1917 corrected at distance 19.39 km
#> Point 1934 corrected at distance 47.57 km
#> Point 1948 corrected at distance 31.10 km
#> Point 1952 corrected at distance 22.16 km
#> Point 1957 corrected at distance 22.16 km
#> Point 1959 corrected at distance 20.45 km
#> Point 1962 corrected at distance 22.16 km
#> Point 1964 corrected at distance 17.15 km
#> Point 1965 corrected at distance 16.18 km
#> Point 1966 corrected at distance 22.16 km
#> Point 1967 corrected at distance 22.16 km
#> Point 1968 corrected at distance 15.60 km
#> Point 1969 corrected at distance 18.14 km
#> Point 1970 corrected at distance 17.98 km
#> Point 1972 corrected at distance 18.27 km
#> Point 1989 corrected at distance 22.16 km
#> Point 1992 corrected at distance 22.16 km
#> Point 2010 corrected at distance 26.43 km
#> Point 2011 corrected at distance 18.40 km
#> Point 2025 corrected at distance 17.27 km
#> Point 2046 corrected at distance 32.41 km
#> Point 2057 corrected at distance 18.38 km
#> Point 2058 corrected at distance 18.38 km
#> Point 2059 corrected at distance 18.38 km
#> Point 2075 corrected at distance 18.38 km
#> Point 2076 corrected at distance 18.38 km
#> Point 2077 removed: nearest cell at 57.66 km (exceeds max 56.0 km)
#> Point 2078 corrected at distance 18.38 km
#> Point 2113 corrected at distance 19.99 km
#> Point 2120 corrected at distance 19.99 km
#> Point 2126 corrected at distance 19.99 km
#> Point 2127 corrected at distance 14.76 km
```

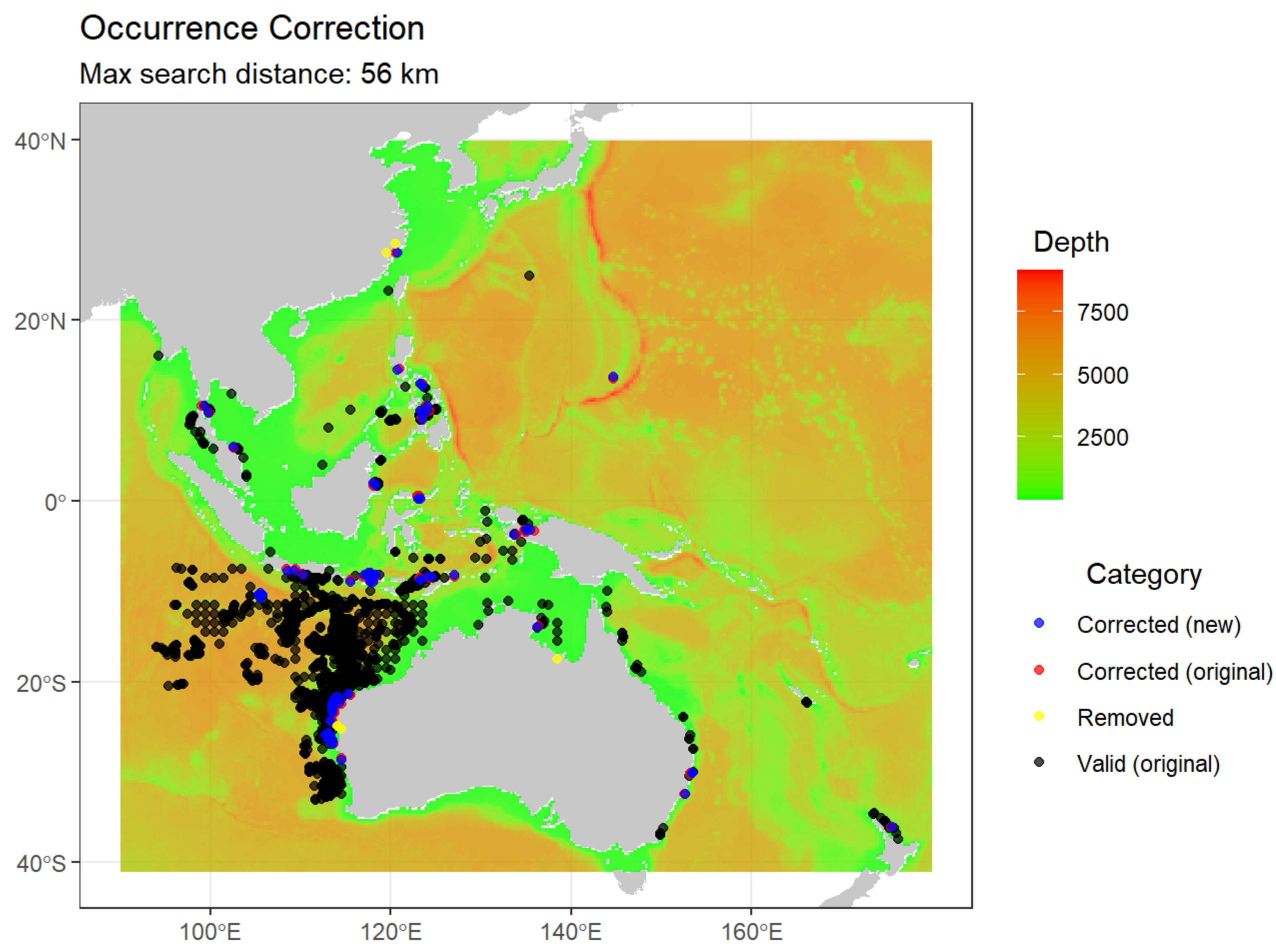
```
#> Point 2131 corrected at distance 19.99 km
#> Point 2148 corrected at distance 18.24 km
#> Point 2162 corrected at distance 19.99 km
#> Point 2165 corrected at distance 19.99 km
#> Point 2179 corrected at distance 19.99 km
#> Point 2213 corrected at distance 19.99 km
#> Point 2218 corrected at distance 18.01 km
#> Point 2239 corrected at distance 16.58 km
#> Point 2283 corrected at distance 17.42 km
#> Point 2317 corrected at distance 16.58 km
#> Point 2400 corrected at distance 19.99 km
#> Point 2543 corrected at distance 18.14 km
#> Point 2546 corrected at distance 15.60 km
#> Point 2550 corrected at distance 14.66 km
#> Point 2562 corrected at distance 16.18 km
#> Point 2567 corrected at distance 17.15 km
#> Point 2575 corrected at distance 19.99 km
#> Point 2576 corrected at distance 21.29 km
#> Point 2579 corrected at distance 16.58 km
#> Point 2580 corrected at distance 16.58 km
#> Point 2590 corrected at distance 14.76 km
#> Point 2614 corrected at distance 19.99 km
#> Point 2619 corrected at distance 21.29 km
#> Point 2627 corrected at distance 14.76 km
#> Point 2654 corrected at distance 14.76 km
#> Point 2684 corrected at distance 24.68 km
#> Point 2698 corrected at distance 19.99 km
#> Point 2701 corrected at distance 16.58 km
#> Point 2743 corrected at distance 19.99 km
#> Point 2754 corrected at distance 19.99 km
#> Point 2755 corrected at distance 21.29 km
#> Point 2762 corrected at distance 15.60 km
#> Point 2763 corrected at distance 17.15 km
#> Point 2764 corrected at distance 16.18 km
#> Point 2765 corrected at distance 17.15 km
#> Point 2766 corrected at distance 16.18 km
#> Point 2771 corrected at distance 14.68 km
#> Point 2780 corrected at distance 19.92 km
#> Point 2784 corrected at distance 15.66 km
#> Point 2786 corrected at distance 19.92 km
#> Point 2802 corrected at distance 19.92 km
#> Point 2805 corrected at distance 24.82 km
#> Point 2818 corrected at distance 13.96 km
#> Point 2819 corrected at distance 19.92 km
#> Point 2859 corrected at distance 14.68 km
#> Point 2860 corrected at distance 28.26 km
#> Point 2884 corrected at distance 14.68 km
#> Point 2896 corrected at distance 14.68 km
#> Point 2904 corrected at distance 19.92 km
#> Point 2906 corrected at distance 26.01 km
#> Point 2911 corrected at distance 14.68 km
#> Point 2915 corrected at distance 14.68 km
```

```
#> Point 2922 corrected at distance 14.39 km
#> Point 2942 corrected at distance 15.60 km
#> Point 3063 corrected at distance 22.95 km
#> Point 3073 corrected at distance 18.82 km
#> Point 3111 corrected at distance 19.92 km
#> Point 3112 corrected at distance 19.92 km
#> Point 3113 corrected at distance 19.92 km
#> Point 3120 corrected at distance 19.92 km
#> Point 3123 corrected at distance 26.35 km
#> Point 3124 corrected at distance 18.74 km
#> Point 3130 corrected at distance 18.14 km
#> Point 3137 corrected at distance 13.47 km
#> Point 3138 corrected at distance 14.68 km
#> Point 3139 corrected at distance 14.68 km
#> Point 3141 corrected at distance 19.92 km
#> Point 3142 corrected at distance 14.68 km
#> Point 3143 corrected at distance 19.92 km
#> Point 3157 corrected at distance 14.68 km
#> Point 3158 corrected at distance 18.51 km
#> Point 3164 corrected at distance 23.41 km
#> Point 3168 corrected at distance 24.82 km
#> Point 3172 corrected at distance 19.92 km
#> Point 3179 corrected at distance 24.80 km
#> Point 3180 corrected at distance 19.60 km
#> Point 3181 corrected at distance 16.33 km
#> Point 3182 corrected at distance 18.65 km
#> Point 3183 corrected at distance 14.54 km
#> Point 3184 corrected at distance 26.93 km
#> Point 3185 corrected at distance 14.74 km
#> Point 3188 corrected at distance 45.28 km
#> Point 3210 corrected at distance 19.29 km
#> Point 3219 corrected at distance 17.85 km
#> Point 3589 corrected at distance 19.99 km
#> Point 3631 corrected at distance 14.76 km
#> Point 3632 corrected at distance 19.92 km
#> Point 3635 corrected at distance 19.92 km
#> Point 3638 corrected at distance 14.68 km
#> Point 3642 corrected at distance 16.58 km
#> Point 3645 corrected at distance 24.68 km
#> Point 3682 corrected at distance 15.60 km
#> Point 3684 corrected at distance 14.66 km
#> Point 3691 corrected at distance 19.92 km
#> Point 3696 corrected at distance 17.15 km
#> Point 3697 corrected at distance 16.18 km
#> Point 3718 corrected at distance 19.29 km
#> Point 3721 corrected at distance 18.76 km
#> Point 3726 corrected at distance 19.68 km
#> Point 3739 corrected at distance 29.26 km
#> Point 3748 corrected at distance 21.70 km
#> Point 3755 corrected at distance 19.99 km
#> Point 4219 corrected at distance 15.72 km
#> Point 4222 corrected at distance 16.18 km
```

```
#> Point 4226 corrected at distance 17.15 km
#> Point 4227 corrected at distance 18.14 km
#> Point 4228 corrected at distance 17.15 km
#> Point 4229 corrected at distance 19.92 km
#> Point 4251 corrected at distance 14.68 km
#> Point 4275 corrected at distance 14.68 km
#> Point 4295 corrected at distance 16.61 km
#> Point 4296 corrected at distance 15.00 km
#> Point 4304 corrected at distance 14.68 km
#> Point 4342 corrected at distance 16.58 km
#> Point 4401 corrected at distance 16.61 km
#> Point 4406 corrected at distance 25.97 km
#> Point 4425 corrected at distance 24.82 km
#> Point 4429 corrected at distance 14.45 km
#> Point 4437 corrected at distance 24.82 km
#> Point 4441 corrected at distance 14.68 km
#> Point 4452 corrected at distance 18.14 km
#> Point 4482 corrected at distance 26.01 km
#> Point 4483 corrected at distance 24.82 km
#> Point 4490 corrected at distance 14.66 km
#> Point 4519 corrected at distance 16.61 km
#> Point 4529 corrected at distance 24.78 km
#> Point 4536 corrected at distance 19.92 km
#> Point 4553 corrected at distance 16.58 km
#> Point 4579 corrected at distance 17.15 km
#> Point 4580 corrected at distance 18.14 km
#> Point 4581 corrected at distance 16.18 km
#> Point 4582 corrected at distance 18.14 km
#> Point 4591 corrected at distance 16.58 km
#> Point 4592 corrected at distance 28.15 km
#> Point 4596 corrected at distance 17.89 km
#> Point 4597 corrected at distance 14.68 km
#> Point 4599 corrected at distance 14.68 km
#> Point 4603 corrected at distance 16.58 km
#> Point 4605 corrected at distance 13.65 km
#> Point 4606 corrected at distance 14.68 km
#> Point 4614 corrected at distance 14.45 km
#> Point 4619 corrected at distance 16.61 km
#> Point 4623 corrected at distance 14.68 km
#> Point 4632 corrected at distance 16.58 km
#> Point 4635 corrected at distance 12.95 km
#> Point 4639 corrected at distance 25.97 km
#> Point 4640 corrected at distance 14.68 km
#> Point 4644 corrected at distance 14.68 km
#> Point 4645 corrected at distance 24.82 km
#> Point 4648 corrected at distance 16.58 km
#> Point 4659 corrected at distance 16.58 km
#> Point 4662 corrected at distance 24.82 km
#> Point 4673 corrected at distance 14.68 km
#> Point 4682 corrected at distance 14.20 km
#> Point 4687 corrected at distance 24.82 km
#> Point 4693 corrected at distance 14.68 km
```

```
#> Point 4694 corrected at distance 14.68 km
#> Point 4698 corrected at distance 14.68 km
#> Point 4709 corrected at distance 14.68 km
#> Point 4719 corrected at distance 16.61 km
#> Point 4721 corrected at distance 14.24 km
#> Point 4730 corrected at distance 14.68 km
#> Point 4741 corrected at distance 14.76 km
#> Point 4743 corrected at distance 14.68 km
#> Point 4766 corrected at distance 16.29 km
#> Point 4769 corrected at distance 16.61 km
#> Point 4783 corrected at distance 24.82 km
#> Point 4787 corrected at distance 14.68 km
#> Point 4823 corrected at distance 19.92 km
#> Point 4838 corrected at distance 14.68 km
#> Point 4852 corrected at distance 26.01 km
#> Point 4871 corrected at distance 24.82 km
#> Point 4886 corrected at distance 16.61 km
#> Point 4894 corrected at distance 24.82 km
#> Point 4898 corrected at distance 16.97 km
#> Point 4908 corrected at distance 24.82 km
#> Point 4918 corrected at distance 14.05 km
#> Point 4921 corrected at distance 16.58 km
#> Point 4922 corrected at distance 13.90 km
#> Point 4923 corrected at distance 19.92 km
#> Point 4928 corrected at distance 24.82 km
#> Point 4936 corrected at distance 14.68 km
#> Point 4953 corrected at distance 13.94 km
#> Point 4965 corrected at distance 14.54 km
#> Point 4988 corrected at distance 16.61 km
#> Point 4989 corrected at distance 16.64 km
#> Point 4998 corrected at distance 17.15 km
#> Point 4999 corrected at distance 16.18 km
#> Point 5017 corrected at distance 16.82 km
#> Point 5018 corrected at distance 19.92 km
#> Point 5025 corrected at distance 14.76 km
#> Point 5032 corrected at distance 16.58 km
#> Point 5075 corrected at distance 24.68 km
#> Point 5081 corrected at distance 14.68 km
#> Point 5090 corrected at distance 14.68 km
#> Point 5097 corrected at distance 14.68 km
#> Point 5110 corrected at distance 24.82 km
#> Point 5116 corrected at distance 24.82 km
#> Point 5117 corrected at distance 16.61 km
#> Point 5124 corrected at distance 19.92 km
#> Point 5129 corrected at distance 14.68 km
#> Point 5136 corrected at distance 16.36 km
#> Point 5152 corrected at distance 13.69 km
#> Point 5160 corrected at distance 24.82 km
#> Point 5166 corrected at distance 14.68 km
#> Point 5168 corrected at distance 16.58 km
#> Point 5187 corrected at distance 21.70 km
#> Point 5202 corrected at distance 19.68 km
```

```
#> Point 5203 corrected at distance 18.38 km
#> Point 5205 corrected at distance 18.38 km
#> Point 5206 corrected at distance 19.29 km
#> Point 5210 corrected at distance 18.38 km
#> Point 5212 removed: nearest cell at 57.66 km (exceeds max 56.0 km)
#> Point 5214 corrected at distance 19.29 km
#> Point 5218 corrected at distance 18.38 km
#> Point 5220 corrected at distance 21.42 km
#> Point 5222 corrected at distance 21.13 km
#> Point 5223 corrected at distance 17.85 km
#> Point 5225 corrected at distance 18.76 km
#> Point 5232 corrected at distance 18.38 km
#> Point 5249 corrected at distance 18.38 km
#> Point 5252 removed: nearest cell at 94.63 km (exceeds max 56.0 km)
#> Point 5253 corrected at distance 29.51 km
#> Point 5254 removed: nearest cell at 82.09 km (exceeds max 56.0 km)
#> Point 5265 corrected at distance 15.43 km
```



```
#> Original records: 5267
#> Records removed: 6
#> Records corrected: 827
#> Final records: 5261
#> Output saved to: F:\whaleshark_sdm\occ\R_typus_adj.csv
summary(corrected_ob)
#>    species          decimalLatitude  decimalLongitude      date           year          month
#>  Length:5261        Min.   :-37.44   Min.   : 94.01   Min.   :16440   Min.   :2015   Min.   :
1.000
#>  Class :character   1st Qu.:-22.60   1st Qu.:112.90   1st Qu.:16740   1st Qu.:2015   1st Qu.:
5.000
#>  Mode  :character   Median :-20.29   Median :113.71   Median :17394   Median :2017   Median :
8.000
#>                     Mean   :-17.92   Mean   :113.85   Mean   :17557   Mean   :2018   Mean   :
7.278
#>                     3rd Qu.:-13.29   3rd Qu.:115.06   3rd Qu.:17777   3rd Qu.:2018   3rd Qu.:1
0.000
#>                     Max.   : 27.43   Max.   :176.26   Max.   :20085   Max.   :2024   Max.   :1
2.000
#>
#>       day          original_x       original_y     correction_status  correction_distance_km  c
orrection_x
#>  Min.   : 1.00   Min.   : 94.01   Min.   :-37.44   Length:5261        Min.   : 0.00          Mi
n.   : 99.35
#>  1st Qu.: 8.00   1st Qu.:112.90   1st Qu.:-22.60   Class :character   1st Qu.: 0.00          1s
t Qu.:105.58
#>  Median :15.00   Median :113.76   Median :-20.29   Mode  :character   Median : 0.00          Me
dian :113.56
#>  Mean   :15.51   Mean   :113.86   Mean   :-17.92                      Mean   : 3.20          Me
an   :111.36
#>  3rd Qu.:23.00   3rd Qu.:115.06   3rd Qu.:-13.29                      3rd Qu.: 0.00          3r
d Qu.:113.56
#>  Max.   :31.00   Max.   :176.26   Max.   : 27.50                      Max.   :53.82          Ma
x.   :175.64
#>  NA's   :27                                                                                  N
A's   :4434
#>   correction_y         env
#>  Min.   :-36.14   Min.   :   2.52
#>  1st Qu.:-22.93   1st Qu.: 129.11
#>  Median :-21.68   Median : 402.42
#>  Mean   :-16.18   Mean   :1656.59
#>  3rd Qu.:-10.71   3rd Qu.:3130.92
#>  Max.   : 27.43   Max.   :6257.69
#>  NA's   :4434
```

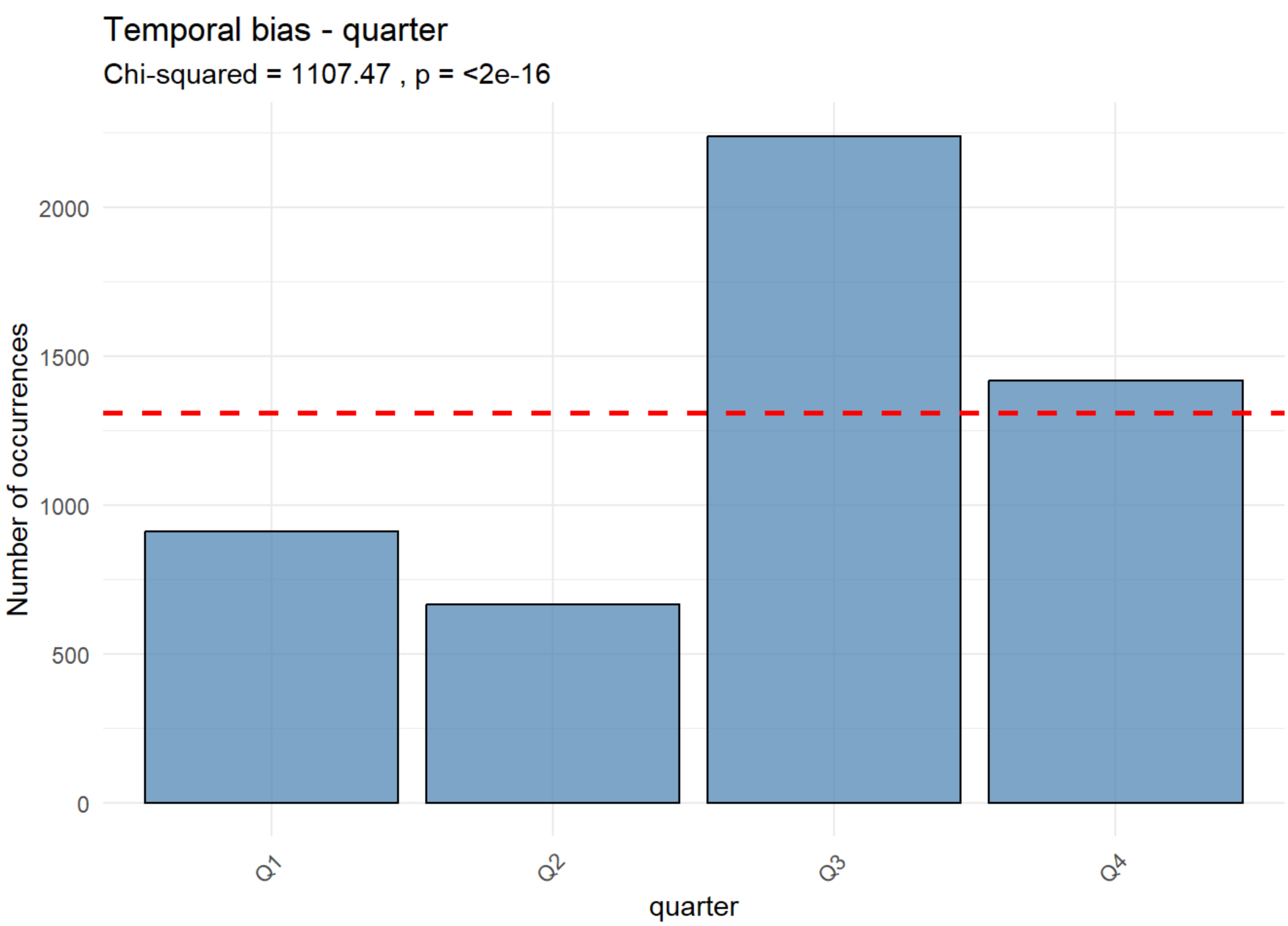
1.5 Check and remove spatio-temporal bias with bias_check() and bias_reduce()

Spatio-temporal sampling bias is a common issue in global biodiversity data and it degrades model robustness. Here, bias_check() and bias_reduce() are two functions to examine and reduce sampling bias (see details in Supplementary Information), respectively, adapted from the dynamicSDM (Dobson et al., 2023).

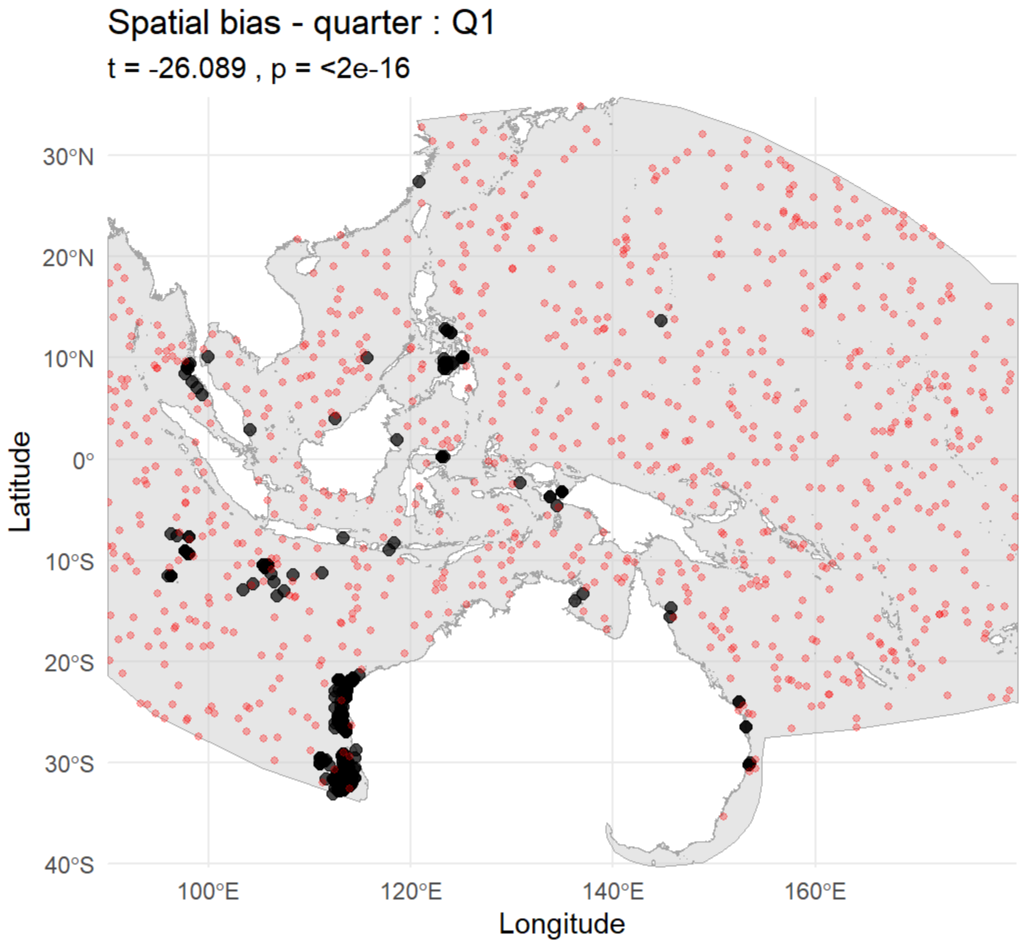
Although designed for highly-mobile marine fish, these data-preparation functions can apply broadly to other marine species.

```
library(dplyr)
## Read input data or skip this line if you have run the section 1.4
corrected_ob<-read.csv(system.file("extdata", "R_typus_adj.csv", package = "OceanSDM"))
## Read input data or skip this line if you have run the section 1.4
range_path <-system.file("extdata", "R_typus_wip2.shp", package = "OceanSDM")
colnames(corrected_ob)
#>  [1] "species"               "decimalLatitude"       "decimalLongitude"       "date"
#>  [5] "year"                  "month"                 "day"                    "original_x"
#>  [9] "original_y"            "correction_status"     "correction_distance_km" "correction_x"
#> [13] "correction_y"          "env"
occ_ob<-corrected_ob[,c(1:7)] # removed unuseful columns
colnames(occ_ob)[2:3]<-c("y","x") # rename latitude and longitude columns
#### For the quarterly SDMs
occ_ob2 <- occ_ob %>%
  mutate(
       quarter = dplyr::case_when(
         month %in% 1:3   ~ "Q1",
         month %in% 4:6   ~ "Q2",
         month %in% 7:9  ~ "Q3",
         month %in% 10:12 ~ "Q4",
         TRUE ~ NA_character_
       ))%>% dplyr::filter(!is.na(quarter))
table(occ_ob2$quarter)
#>
#>   Q1   Q2   Q3   Q4
#>  915  681 2245 1420

#### For the seasonal SDMs
# occ_ob3 <- occ_ob2 %>%
#  mutate(date = make_date(year, month, day),
#         adjusted_date = date - days(73),
# # Parse date information and adjust for sea temperature lag (73 days)
#         adjusted_month = month(adjusted_date),
#         hemisphere = ifelse(y >= 0, "Northern", "Southern"),
#         season = case_when(
#           hemisphere == "Northern" & adjusted_month %in% 3:5    ~ "Spring",
#           hemisphere == "Northern" & adjusted_month %in% 6:8    ~ "Summer",
#           hemisphere == "Northern" & adjusted_month %in% 9:11   ~ "Autumn",
 #          hemisphere == "Northern" & adjusted_month %in% c(12,1,2) ~ "Winter",
  #         hemisphere == "Southern" & adjusted_month %in% 3:5    ~ "Autumn",
  #         hemisphere == "Southern" & adjusted_month %in% 6:8    ~ "Winter",
  #         hemisphere == "Southern" & adjusted_month %in% 9:11   ~ "Spring",
#           hemisphere == "Southern" & adjusted_month %in% c(12,1,2) ~ "Summer",
#           TRUE ~ NA_character_
#         ))
# occ_ob2 <- occ_ob3 %>% filter(!is.na(season))
#range_path<-"F:/whaleshark_sdm/range_map/R_typus_wip2.shp"
## check sampling bias
bias_test<-bias_check(occ_data=occ_ob2,
                        temporal_level="quarter",
                        spatial_method="range_map",
                        range_path=range_path,
```

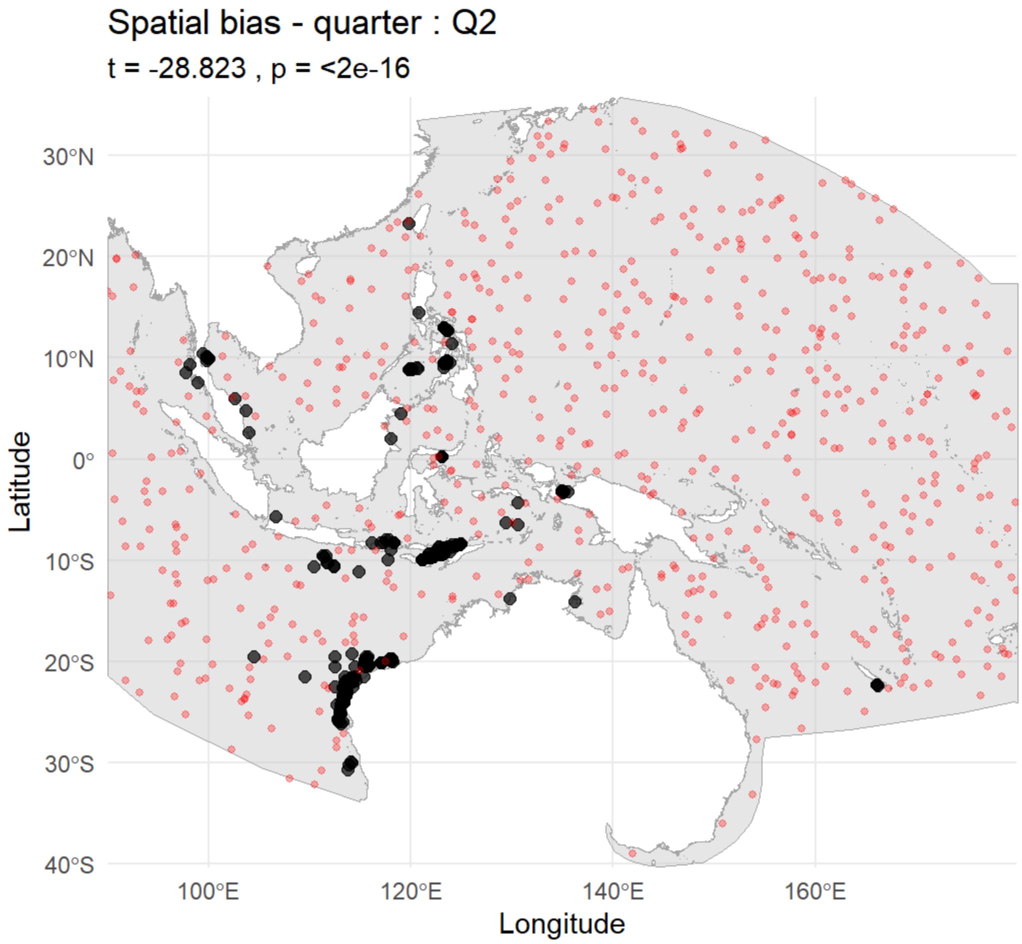
```
                    prj = "+proj=longlat +datum=WGS84")
#> Warning: attribute variables are assumed to be spatially constant throughout all geometries
#> Time period Q1 : some points fall outside the defined area. They will be excluded from spatial
test.
#> Warning: attribute variables are assumed to be spatially constant throughout all geometries
#> Time period Q2 : some points fall outside the defined area. They will be excluded from spatial
test.
#> Warning: attribute variables are assumed to be spatially constant throughout all geometries
#> Time period Q3 : some points fall outside the defined area. They will be excluded from spatial
test.
#> Warning: attribute variables are assumed to be spatially constant throughout all geometries
#> Time period Q4 : some points fall outside the defined area. They will be excluded from spatial
test.
bias_test$Plots
#> $Temporal_bias_plot
```



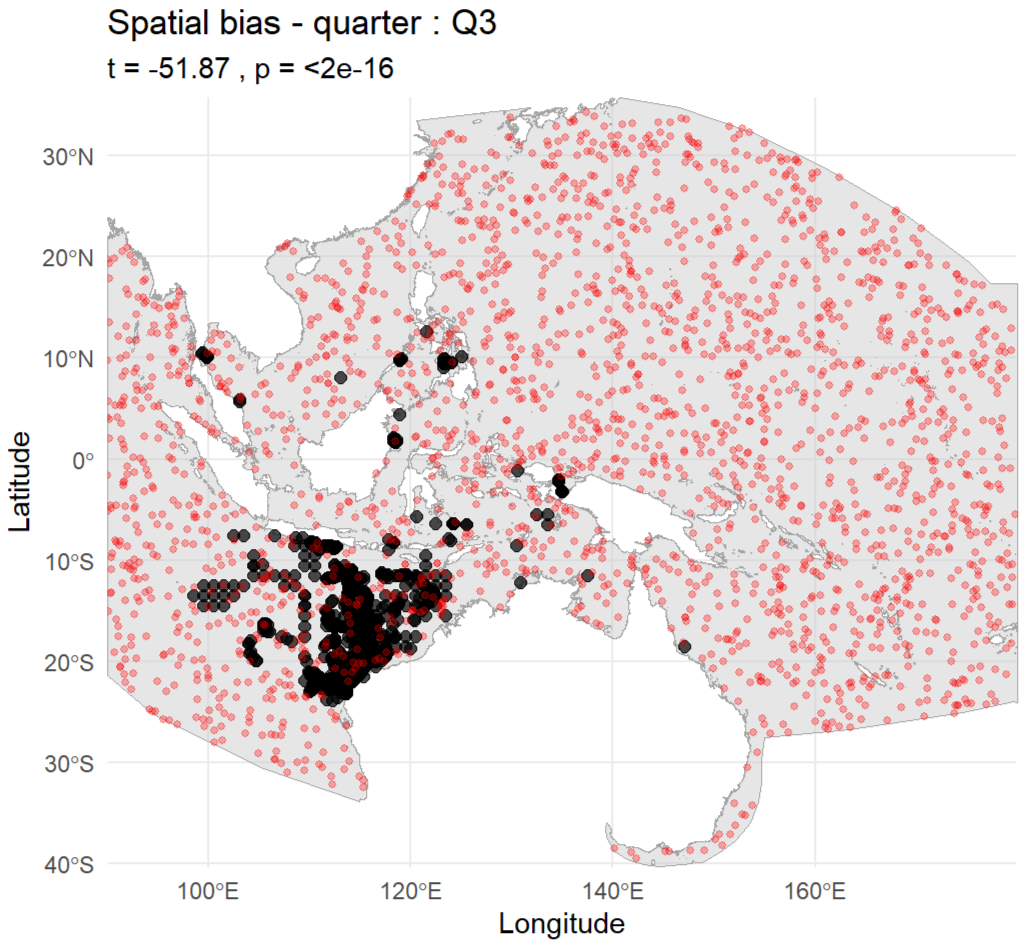

```
#>
#> $Spatial_bias_plot_quarter_Q1
```



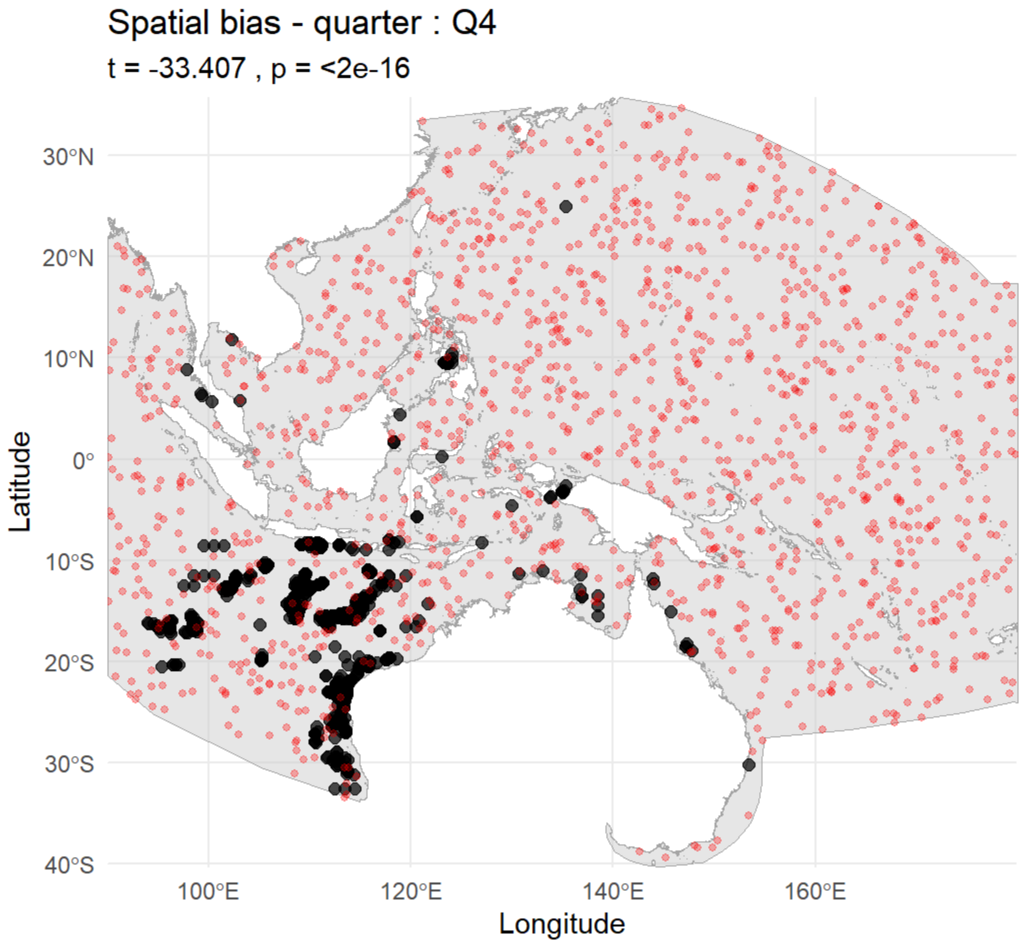

```
#>
#> $Spatial_bias_plot_quarter_Q2
```




```
#>
#> $Spatial_bias_plot_quarter_Q3
```



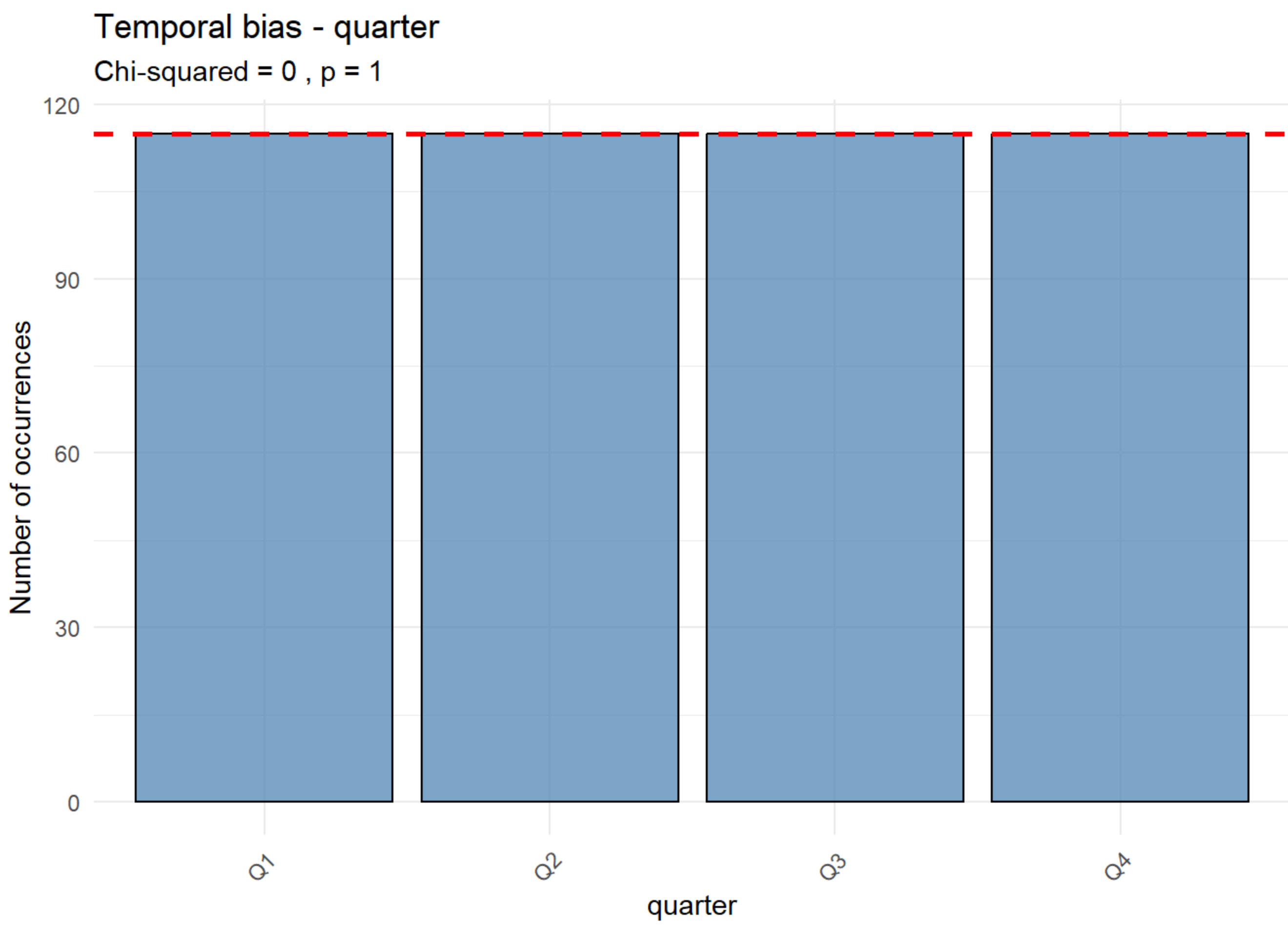

```
#>
#> $Spatial_bias_plot_quarter_Q4
```



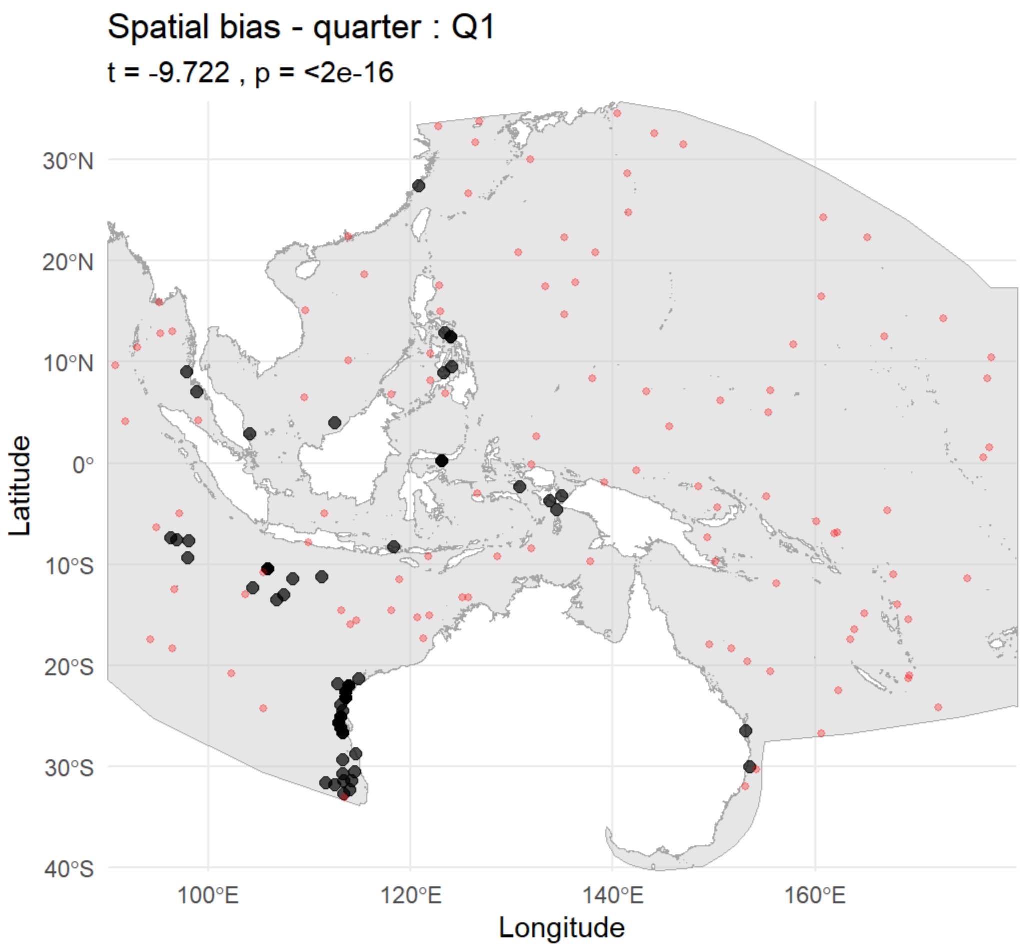
Spatial bias - quarter : Q4
t = -33.407 , p = <2e-16
30°N
20°N
10°N
0°
10°S
20°S
30°S
40°S
Latitude
100°E
120°E
140°E
160°E
Longitude

```
occ_thinned<-bias_reduce(occ_data=occ_ob2,
                          bias_result=bias_test,
                  min_dist_km = 56,
                         thin_reps = 50,
                         temporal_level = "quarter",
                         seed = 123)
#> Temporal bias test p-value: 0
#> 
#> === Step 1: Spatial thinning ===
#> Quarter Q1 spatial bias p-value: 0
#> Quarter Q1 has spatial bias, performing thinning...
#> Quarter Q1 thinning completed. Original points: 915
#> Points after thinning: 204
#> Quarter Q2 spatial bias p-value: 0
#> Quarter Q2 has spatial bias, performing thinning...
#> Quarter Q2 thinning completed. Original points: 681
#> Points after thinning: 115
#> Quarter Q3 spatial bias p-value: 0
#> Quarter Q3 has spatial bias, performing thinning...
#> Quarter Q3 thinning completed. Original points: 2245
#> Points after thinning: 317
#> Quarter Q4 spatial bias p-value: 0
#> Quarter Q4 has spatial bias, performing thinning...
#> Quarter Q4 thinning completed. Original points: 1420
#> Points after thinning: 266
#> 
#> === Step 2: Temporal balancing ===
#> Temporal bias detected, performing temporal balancing...
#> Quarter with smallest sample size: Q2 sample size: 115
#> Quarter Q1 sampled from 204 points to 115 points
#> Quarter Q2 keeping 115 points
#> Quarter Q3 sampled from 317 points to 115 points
#> Quarter Q4 sampled from 266 points to 115 points
#> 
#> Sample sizes per quarter after balancing:
#> Quarter Q1 : 115
#> Quarter Q2 : 115
#> Quarter Q3 : 115
#> Quarter Q4 : 115
#> 
#> === Thinning summary ===
#> Original total sample size: 5261
#> Sample size after spatial thinning: 902
#> Final sample size: 460
#> Retention ratio: 8.74 %
#> 
#> Retention ratio per quarter:
#> Quarter Q1 :  915  ->  115  ( 12.57 %)
#> Quarter Q2 :  681  ->  115  ( 16.89 %)
#> Quarter Q3 :  2245  ->  115  ( 5.12 %)
#> Quarter Q4 :  1420  ->  115  ( 8.1 %)
bias_test<-bias_check(occ_data=occ_thinned,
```

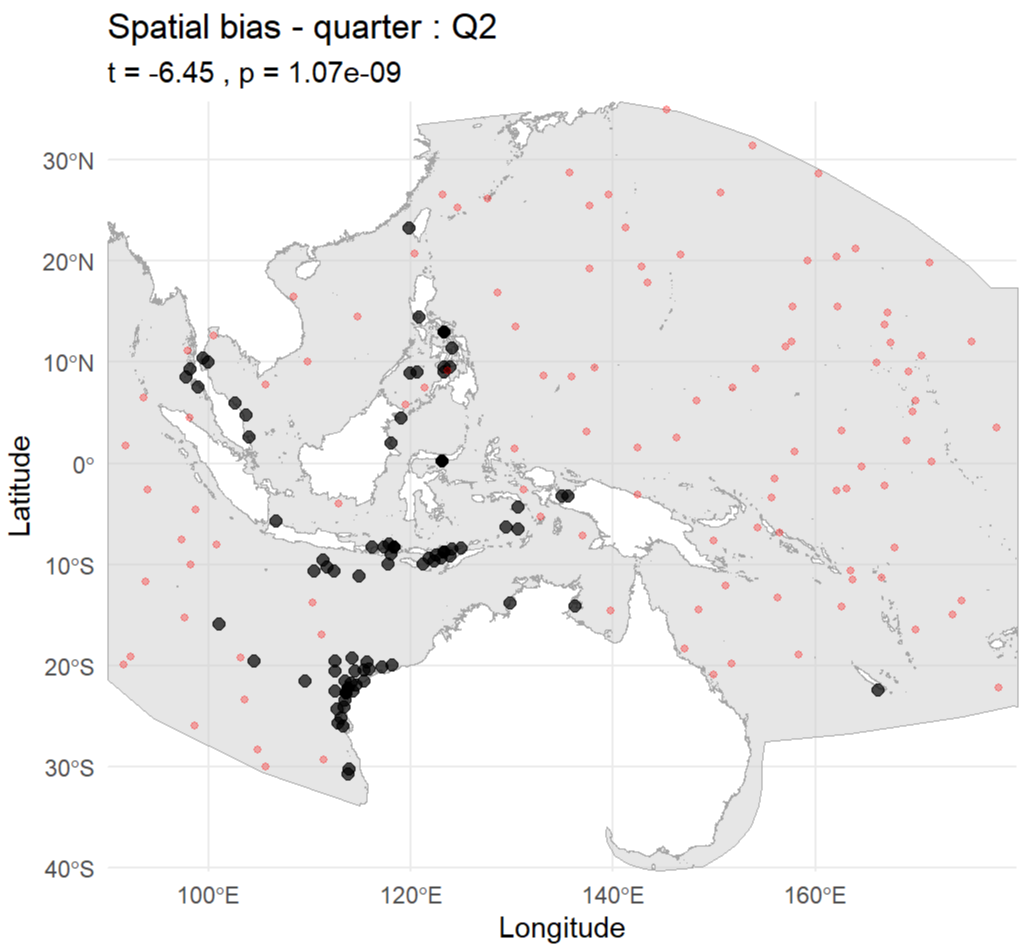
```
                        temporal_level="quarter",
                        spatial_method="range_map",
                        range_path=range_path,
                        prj = "+proj=longlat +datum=WGS84")
#> Warning: attribute variables are assumed to be spatially constant throughout all geometries
#> Time period Q1 : some points fall outside the defined area. They will be excluded from spatial
test.
#> Warning: attribute variables are assumed to be spatially constant throughout all geometries
#> Time period Q2 : some points fall outside the defined area. They will be excluded from spatial
test.
#> Warning: attribute variables are assumed to be spatially constant throughout all geometries
#> Time period Q3 : some points fall outside the defined area. They will be excluded from spatial
test.
#> Warning: attribute variables are assumed to be spatially constant throughout all geometries
#> Time period Q4 : some points fall outside the defined area. They will be excluded from spatial
test.
bias_test$Plots
#> $Temporal_bias_plot
```



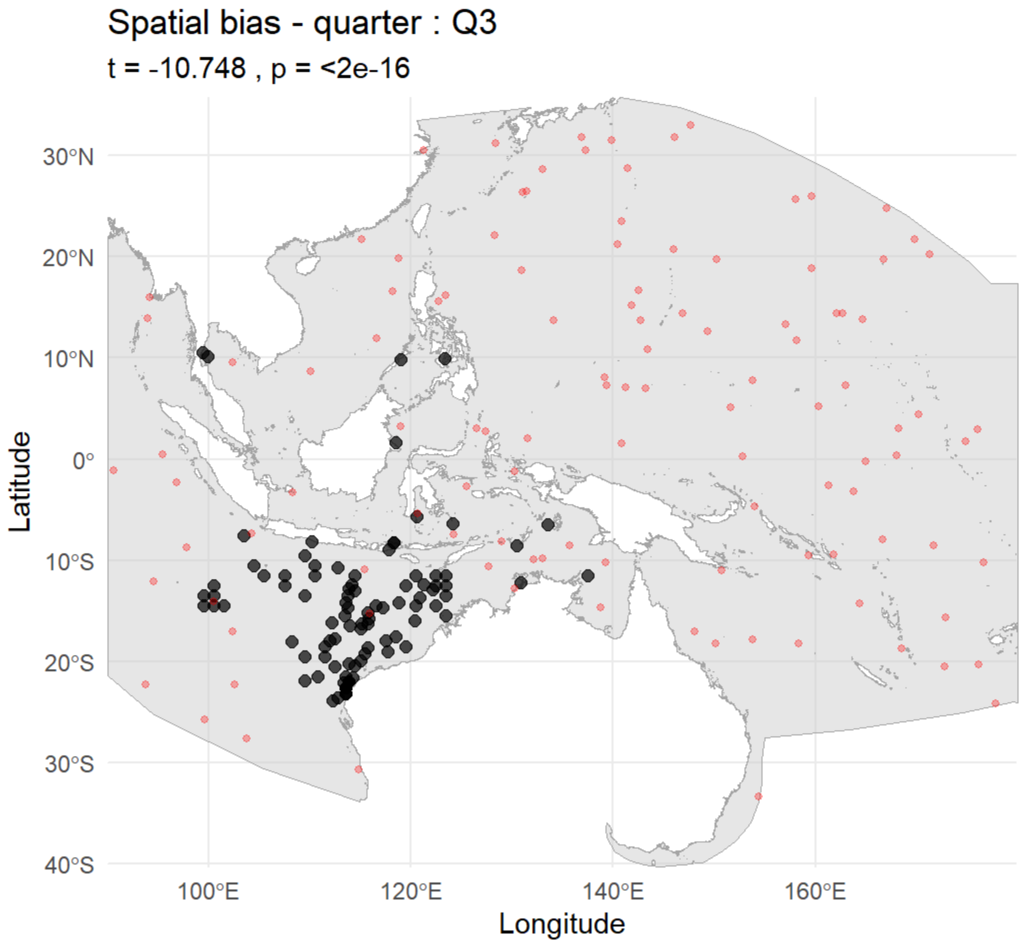

```
#>
#> $Spatial_bias_plot_quarter_Q1
```



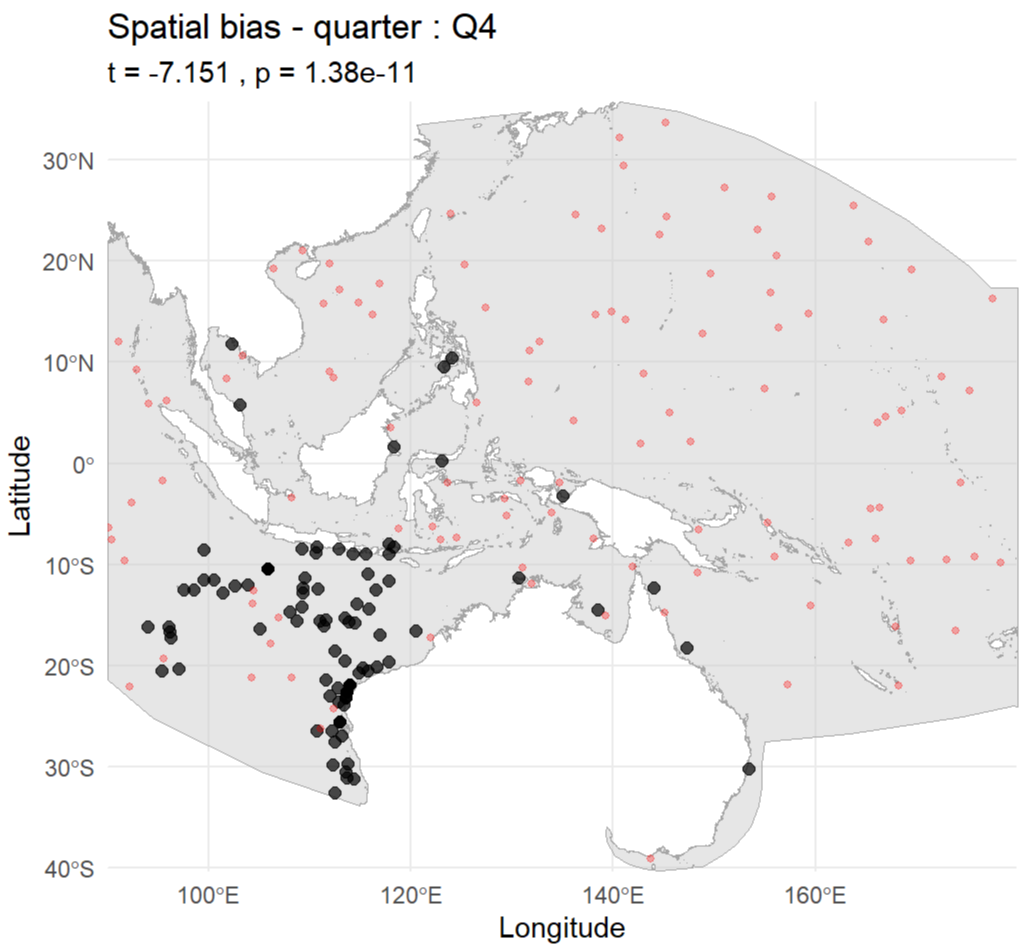

```
#>
#> $Spatial_bias_plot_quarter_Q2
```




```
#>
#> $Spatial_bias_plot_quarter_Q3
```




```
#>
#> $Spatial_bias_plot_quarter_Q4
```




```
#### save the thinned occurrences
write.table(occ_thinned, "F:/whaleshark_sdm/occ/R_typus_qs.txt",
   append = FALSE, sep = "\t", dec = ".",row.names = FALSE, col.names = TRUE)
occ_q1<-occ_thinned[which(occ_thinned$quarter=="Q1"),]
write.table(occ_q1, "F:/whaleshark_sdm/occ/R_typus_q1.txt",
 append = FALSE, sep = "\t", dec = ".",row.names = FALSE, col.names = TRUE)
occ_q2<-occ_thinned[which(occ_thinned$quarter=="Q2"),]
write.table(occ_q2, "F:/whaleshark_sdm/occ/R_typus_q2.txt",
 append = FALSE, sep = "\t", dec = ".",row.names = FALSE, col.names = TRUE)
occ_q3<-occ_thinned[which(occ_thinned$quarter=="Q3"),]
write.table(occ_q3, "F:/whaleshark_sdm/occ/R_typus_q3.txt",
   append = FALSE, sep = "\t", dec = ".",row.names = FALSE, col.names = TRUE)
occ_q4<-occ_thinned[which(occ_thinned$quarter=="Q4"),]
write.table(occ_q4, "F:/whaleshark_sdm/occ/R_typus_q4.txt",
   append = FALSE, sep = "\t", dec = ".", row.names = FALSE, col.names = TRUE)
```

Please note that the spatial bias still exist but have been largely reduced. A certain level of spatial autocorrelation of occurrences are ecological meaningful given species are rarely uniformly distributed in the ocean (due to the clumping nature of resources and sometimes species natural schooling or aggregation habits), and such autocorrelation is also allowed for machine-learning algorithms.

2. Build 2D SDMs for each temporal bin and generate related figures

2.1 SDM construction for each quarter with TB_sdm()

To predict horizontal geographic distributions within each temporal bin, we created a single function TB_sdm(), which leveraged some key functions from the packages sdm (Naimi & Araújo, 2016), blockCV (Valavi et al., 2018), sf (Pebesma, 2018), terra (Hijmans et al., 2022), and Metrics (Hamner & Frasco, 2012). This function streamlines multiple modelling steps: (i) reading occurrences and predictors, (ii) creating folds for spatial block cross-validation, (iii) pseudo absence generation, (iv) collinearity diagnostic and selection of predictors (in an interactive manner), (v) model calibration and evaluation, (vi) ensemble construction and final prediction, and (vii) variable importance and partial dependence data extraction. It constructs models based on the sdm package, which supports classical statistical methods, machine learning algorithms, envelope methods, as well as their ensembles (see details in Naimi & Araújo, 2016). Here I only run the code for the 1st quarter as an example to create this vignette webpage.

```
#### Prior to this step, please rename the environmental predictors to make the
#### the names of variables of each quarterly bin the same as follows:
#### Depth, Mixed_layer_depth, Dissolved_oxygen, Temperature, Salinity,
####      Net_primary_productivity, Sea_surface_height.
## Q1
rm()
gc()
#>            used  (Mb) gc trigger  (Mb) max used  (Mb)
#> Ncells  6552547 350.0    9980527 533.1  9756850 521.1
#> Vcells 14578716 111.3   47082104 359.3 58852608 449.1
# You can get access to the Q1 occurrences of whale sharks in the system file folder extdata.
d_occ <- system.file("extdata", "R_typus_q1.txt", package = "OceanSDM")
d_env <- system.file("extdata", "Q1", package = "OceanSDM")
d_prj<-system.file("extdata", "proj", package = "OceanSDM")  # Here I set up a folder where predic
tors of the 2nd quarter were stored, and was used as a faked future climate scenarios to demonstra
te how future projections can be done.
d_res<-"F:/whaleshark_sdm/result_sdm/Q1" # Change to your own directory
#
# library(sdm)
sdm_q1<- TB_sdm(env_path=d_env, occ_path=d_occ, res_path=d_res,
                species="Species", x="x", y="y",
                algorithms = c("brt", "rf", "svm"),
                cv_method = "random", n_folds =5,
                pre_abs_ratio = 0.3, block_size = 500000,
                var.importance = TRUE, response.curve = TRUE,
                ensemble.method = "weighted", ensemble.stat = "TSS",
                ncore = 1, seed = 123,
                proj_path = d_prj,
                collinearity_thresh = 0.75, remove_collinear = F)
#> Reading environmental rasters...
#> Reading occurrence data...
#> Number of unique presence records: 54
#> Generating pseudo-absences using sdm::background with method = 'eDist'...
#> Total points (presence + pseudo-absence): 234 (presence: 54, pseudo-absence: 180)
#> Performing collinearity diagnostics...
#> Found 1 variable pairs with |cor| > 0.75.
#> Variables are kept unchanged for modeling (remove_collinear = FALSE).
#> Creating full sdmData object...
#> Creating spatial folds using blockCV...
#>
#>   train_0 train_1 test_0 test_1
#> 1     144      40     36     14
#> 2     145      44     35     10
#> 3     140      44     40     10
#> 4     142      46     38      8
#> 5     149      42     31     12
#> Spatial folds created. Fold sizes: 50, 45, 50, 46, 43
#> Starting manual cross-validation...
#>  Processing fold 1/5
#>  Processing fold 2/5
#>  Processing fold 3/5
#>  Processing fold 4/5
```

```
#>  Processing fold 5/5
#> Fitting final model on all data...
#> Building ensemble prediction...
#> Best threshold based on MAX_TSS method = 0.23855
#> Evaluate ensemble model ...
#> Processing fold  1 / 5
#> Prepare training data...
#> Create sdmdata to build the model for fold =  1
#> Prepare test data...
#> Processing fold  2 / 5
#> Prepare training data...
#> Create sdmdata to build the model for fold =  2
#> Prepare test data...
#> Processing fold  3 / 5
#> Prepare training data...
#> Create sdmdata to build the model for fold =  3
#> Prepare test data...
#> Processing fold  4 / 5
#> Prepare training data...
#> Create sdmdata to build the model for fold =  4
#> Prepare test data...
#> Processing fold  5 / 5
#> Prepare training data...
#> Create sdmdata to build the model for fold =  5
#> Prepare test data...
#> Evaluation of the models ...
#>   methodID  AUC_mean     AUC_sd  TSS_mean     TSS_sd Boyce_mean   Boyce_sd
#> 1      brt 0.8527763 0.05888217 0.6796761 0.08314130  0.3907319 0.24871505
#> 2       rf 0.8809475 0.06155826 0.7284271 0.08321856  0.6481045 0.12126955
#> 3      svm 0.9196268 0.05330382 0.7414911 0.12677391  0.4632004 0.26484996
#> 4 Ensemble 0.8989715 0.06173640 0.7687364 0.09768171  0.7346307 0.09088736
#> Variable importance:...
#>                                           variables corTest AUCtest
#> Depth                                         Depth  0.1091  0.0588
#> Dissolved_oxygen                   Dissolved_oxygen  0.1041  0.0448
#> Mixed_layer_depth                 Mixed_layer_depth  0.1419  0.0828
#> Net_primary_productivity Net_primary_productivity  0.0302  0.0144
#> Salinity                                   Salinity  0.0489  0.0276
#> Sea_surface_height               Sea_surface_height  0.2967  0.1756
#> Temperature                             Temperature  0.0431  0.0172
#> Generating response curves...
#> The id argument is not specified; The modelIDs of 3 successfully fitted models are assigned to
id...!
#> Done...response curves were successfully created!
#> Projecting to new environmental conditions...
## Q2
d_occ <- system.file("extdata", "R_typus_q2.txt", package = "OceanSDM")
d_env <- system.file("extdata", "Q2", package = "OceanSDM")
d_res<-"F:/whaleshark_sdm/result_sdm/Q2"
sdm_q2<- TB_sdm(env_path=d_env, occ_path=d_occ, res_path=d_res,
                species="Species", x="x", y="y",
                algorithms = c("brt", "rf", "svm"),
```

```
                 cv_method = "random", n_folds =5,
                 pre_abs_ratio = 0.3, block_size = 500000,
                 var.importance = TRUE, response.curve = TRUE,
                 ensemble.method = "weighted", ensemble.stat = "TSS",
                 ncore = 1, seed = 123,
                 collinearity_thresh = 0.75, remove_collinear = F)
#> Reading environmental rasters...
#> Reading occurrence data...
#> Number of unique presence records: 82
#> Generating pseudo-absences using sdm::background with method = 'eDist'...
#> Total points (presence + pseudo-absence): 355 (presence: 82, pseudo-absence: 273)
#> Performing collinearity diagnostics...
#> Found 2 variable pairs with |cor| > 0.75.
#> Variables are kept unchanged for modeling (remove_collinear = FALSE).
#> Creating full sdmData object...
#> Creating spatial folds using blockCV...
#> 
#>   train_0 train_1 test_0 test_1
#> 1     223      67     50     15
#> 2     220      66     53     16
#> 3     218      60     55     22
#> 4     207      69     66     13
#> 5     224      66     49     16
#> Spatial folds created. Fold sizes: 65, 69, 77, 79, 65
#> Starting manual cross-validation...
#>  Processing fold 1/5
#>  Processing fold 2/5
#>  Processing fold 3/5
#>  Processing fold 4/5
#>  Processing fold 5/5
#> Fitting final model on all data...
#> Building ensemble prediction...
#> Best threshold based on MAX_TSS method = 0.29232
#> Evaluate ensemble model ...
#> Processing fold  1 / 5
#> Prepare training data...
#> Create sdmdata to build the model for fold =  1
#> Prepare test data...
#> Processing fold  2 / 5
#> Prepare training data...
#> Create sdmdata to build the model for fold =  2
#> Prepare test data...
#> Processing fold  3 / 5
#> Prepare training data...
#> Create sdmdata to build the model for fold =  3
#> Prepare test data...
#> Processing fold  4 / 5
#> Prepare training data...
#> Create sdmdata to build the model for fold =  4
#> Prepare test data...
#> Processing fold  5 / 5
#> Prepare training data...
```

```
#> Create sdmdata to build the model for fold =  5
#> Prepare test data...
#> Evaluation of the models ...
#>   methodID  AUC_mean     AUC_sd  TSS_mean     TSS_sd Boyce_mean  Boyce_sd
#> 1      brt 0.9245592 0.03639817 0.7763667 0.13093364  0.7187464 0.1324599
#> 2       rf 0.9197650 0.03715363 0.7734540 0.09753788  0.5952595 0.3064057
#> 3      svm 0.8429604 0.05340640 0.7232768 0.09856177  0.4752345 0.2121778
#> 4 Ensemble 0.8988553 0.06239008 0.7623221 0.12983450  0.6833981 0.1542716
#> Variable importance:...
#>                                           variables corTest AUCtest
#> Depth                                         Depth  0.2936  0.1456
#> Dissolved_oxygen                   Dissolved_oxygen  0.2625  0.1540
#> Mixed_layer_depth                 Mixed_layer_depth  0.0112  0.0104
#> Net_primary_productivity Net_primary_productivity  0.0148  0.0184
#> Salinity                                   Salinity  0.0868  0.0584
#> Sea_surface_height               Sea_surface_height  0.0458  0.0196
#> Temperature                             Temperature  0.0589  0.0176
#> Generating response curves...
#> The id argument is not specified; The modelIDs of 3 successfully fitted models are assigned to
id...!
#> Done...response curves were successfully created!
# # Q3
d_occ <- system.file("extdata", "R_typus_q3.txt", package = "OceanSDM")
d_env <- system.file("extdata", "Q3", package = "OceanSDM")
d_res<-"F:/whaleshark_sdm/result_sdm/Q3"
sdm_q3<- TB_sdm(env_path=d_env, occ_path=d_occ, res_path=d_res,
                species="Species", x="x", y="y",
                algorithms = c("brt", "rf", "svm"),
                cv_method = "random", n_folds =5,
                pre_abs_ratio = 0.3, block_size = 500000,
                var.importance = TRUE, response.curve = TRUE,
                ensemble.method = "weighted", ensemble.stat = "TSS",
                ncore = 1, seed = 123,
                collinearity_thresh = 0.75, remove_collinear = F)
#> Reading environmental rasters...
#> Reading occurrence data...
#> Number of unique presence records: 89
#> Generating pseudo-absences using sdm::background with method = 'eDist'...
#> Total points (presence + pseudo-absence): 385 (presence: 89, pseudo-absence: 296)
#> Performing collinearity diagnostics...
#> Found 3 variable pairs with |cor| > 0.75.
#> Variables are kept unchanged for modeling (remove_collinear = FALSE).
#> Creating full sdmData object...
#> Creating spatial folds using blockCV...
#>
#>   train_0 train_1 test_0 test_1
#> 1     226      75     70     14
#> 2     240      68     56     21
#> 3     236      75     60     14
#> 4     240      68     56     21
#> 5     242      70     54     19
#> Spatial folds created. Fold sizes: 84, 77, 74, 77, 73
```

```
#> Starting manual cross-validation...
#>  Processing fold 1/5
#>  Processing fold 2/5
#>  Processing fold 3/5
#>  Processing fold 4/5
#>  Processing fold 5/5
#> Fitting final model on all data...
#> Building ensemble prediction...
#> Best threshold based on MAX_TSS method = 0.25483
#> Evaluate ensemble model ...
#> Processing fold  1 / 5
#> Prepare training data...
#> Create sdmdata to build the model for fold =  1
#> Prepare test data...
#> Processing fold  2 / 5
#> Prepare training data...
#> Create sdmdata to build the model for fold =  2
#> Prepare test data...
#> Processing fold  3 / 5
#> Prepare training data...
#> Create sdmdata to build the model for fold =  3
#> Prepare test data...
#> Processing fold  4 / 5
#> Prepare training data...
#> Create sdmdata to build the model for fold =  4
#> Prepare test data...
#> Processing fold  5 / 5
#> Prepare training data...
#> Create sdmdata to build the model for fold =  5
#> Prepare test data...
#> Evaluation of the models ...
#>   methodID  AUC_mean     AUC_sd  TSS_mean     TSS_sd Boyce_mean  Boyce_sd
#> 1      brt 0.9519133 0.01197501 0.8397222 0.06666230  0.5098315 0.2003312
#> 2       rf 0.9624735 0.01400719 0.8838228 0.05944155  0.6913190 0.1916692
#> 3      svm 0.9564768 0.01093386 0.8649206 0.06475711  0.5649816 0.2106181
#> 4 Ensemble 0.9640127 0.01112376 0.8887434 0.04196996  0.6812705 0.1358886
#> Variable importance:...
#>                                          variables corTest AUCtest
#> Depth                                        Depth  0.0143  0.0140
#> Dissolved_oxygen                  Dissolved_oxygen  0.2708  0.2056
#> Mixed_layer_depth                Mixed_layer_depth  0.1689  0.0592
#> Net_primary_productivity Net_primary_productivity  0.0065  0.0084
#> Salinity                                  Salinity  0.0193  0.0132
#> Sea_surface_height              Sea_surface_height  0.0651  0.0260
#> Temperature                            Temperature  0.1869  0.0864
#> Generating response curves...
#> The id argument is not specified; The modelIDs of 3 successfully fitted models are assigned to id...!
#> Done...response curves were successfully created!
# # Q4
d_occ <- system.file("extdata", "R_typus_q4.txt", package = "OceanSDM")
d_env <- system.file("extdata", "Q4", package = "OceanSDM")
```

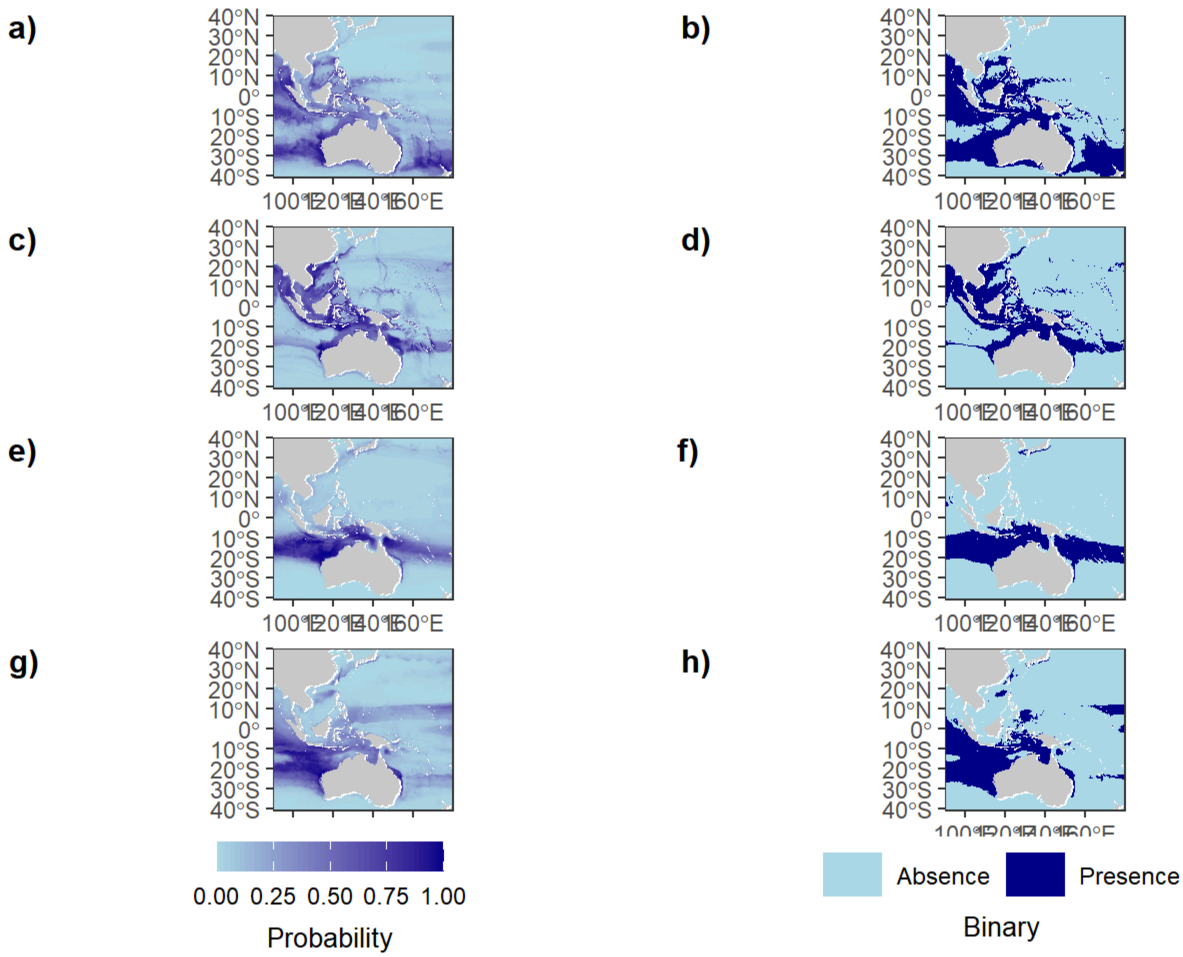
```
d_res<-"F:/whaleshark_sdm/result_sdm/Q4"
sdm_q4<- TB_sdm(env_path=d_env, occ_path=d_occ, res_path=d_res,
                species="Species", x="x", y="y",
                algorithms = c("brt", "rf", "svm"),
                cv_method = "random", n_folds =5,
                pre_abs_ratio = 0.3, block_size = 500000,
                var.importance = TRUE, response.curve = TRUE,
                ensemble.method = "weighted", ensemble.stat = "TSS",
                ncore = 1, seed = 123,
                collinearity_thresh = 0.75, remove_collinear = F)
#> Reading environmental rasters...
#> Reading occurrence data...
#> Number of unique presence records: 86
#> Generating pseudo-absences using sdm::background with method = 'eDist'...
#> Total points (presence + pseudo-absence): 372 (presence: 86, pseudo-absence: 286)
#> Performing collinearity diagnostics...
#> Found 3 variable pairs with |cor| > 0.75.
#> Variables are kept unchanged for modeling (remove_collinear = FALSE).
#> Creating full sdmData object...
#> Creating spatial folds using blockCV...
#>
#>   train_0 train_1 test_0 test_1
#> 1     214      70     72     16
#> 2     227      72     59     14
#> 3     238      73     48     13
#> 4     232      65     54     21
#> 5     233      64     53     22
#> Spatial folds created. Fold sizes: 88, 73, 61, 75, 75
#> Starting manual cross-validation...
#>  Processing fold 1/5
#>  Processing fold 2/5
#>  Processing fold 3/5
#>  Processing fold 4/5
#>  Processing fold 5/5
#> Fitting final model on all data...
#> Building ensemble prediction...
#> Best threshold based on MAX_TSS method = 0.27649
#> Evaluate ensemble model ...
#> Processing fold  1 / 5
#> Prepare training data...
#> Create sdmdata to build the model for fold =  1
#> Prepare test data...
#> Processing fold  2 / 5
#> Prepare training data...
#> Create sdmdata to build the model for fold =  2
#> Prepare test data...
#> Processing fold  3 / 5
#> Prepare training data...
#> Create sdmdata to build the model for fold =  3
#> Prepare test data...
#> Processing fold  4 / 5
#> Prepare training data...
```

```
#> Create sdmdata to build the model for fold =  4
#> Prepare test data...
#> Processing fold  5 / 5
#> Prepare training data...
#> Create sdmdata to build the model for fold =  5
#> Prepare test data...
#> Evaluation of the models ...
#>   methodID  AUC_mean     AUC_sd  TSS_mean     TSS_sd Boyce_mean  Boyce_sd
#> 1      brt 0.8687912 0.06648495 0.6484074 0.10901919  0.5853977 0.2817642
#> 2       rf 0.9148589 0.05096988 0.7562889 0.13084112  0.8279228 0.1156046
#> 3      svm 0.9086926 0.02127888 0.7661640 0.04751780  0.6623615 0.2110698
#> 4 Ensemble 0.9258699 0.02675036 0.7983600 0.03604724  0.7113129 0.1124220
#> Variable importance:...
#>                                          variables corTest AUCtest
#> Depth                                        Depth  0.0083  0.0116
#> Dissolved_oxygen                  Dissolved_oxygen  0.1433  0.0968
#> Mixed_layer_depth                Mixed_layer_depth  0.0914  0.0532
#> Net_primary_productivity Net_primary_productivity  0.0233  0.0156
#> Salinity                                  Salinity  0.1155  0.0724
#> Sea_surface_height              Sea_surface_height  0.4474  0.3180
#> Temperature                            Temperature  0.1024  0.0520
#> Generating response curves...
#> The id argument is not specified; The modelIDs of 3 successfully fitted models are assigned to id...!
#> Done...response curves were successfully created!
## save data
# save(sdm_q1,sdm_q2,sdm_q3,sdm_q4, file = "F:/whaleshark_sdm/example.RData")
```

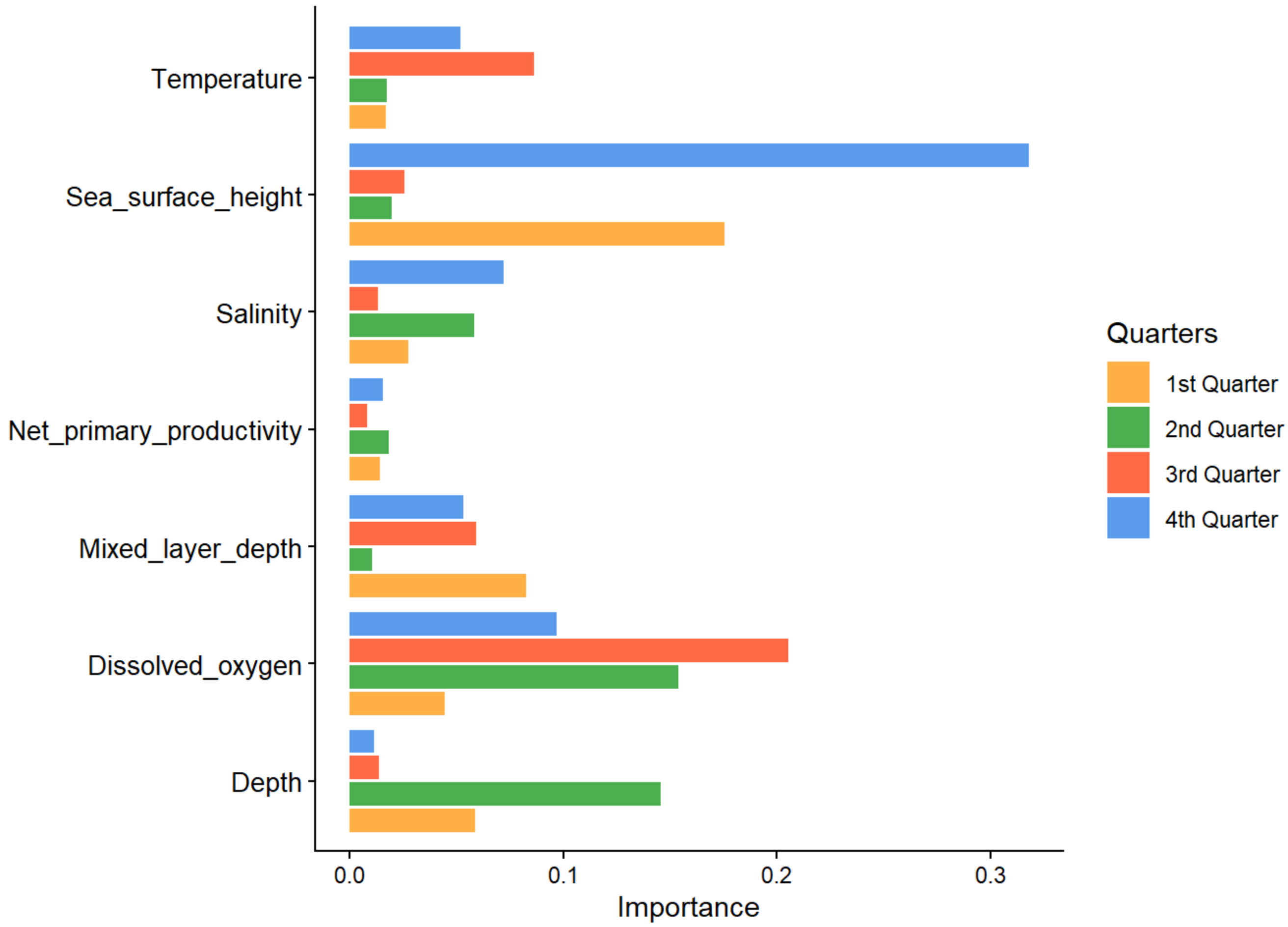
2.2 Plot predictive maps, variable importance and species response curves with TB_map(), TB_varimp(), and TB_rescur()

After successful 2D-SDM construction, TB_map(), TB_varimp(), and TB_rescur() automatically produces related figures (i.e., probability and binary maps, variable importance, and species response curves) required for publication.

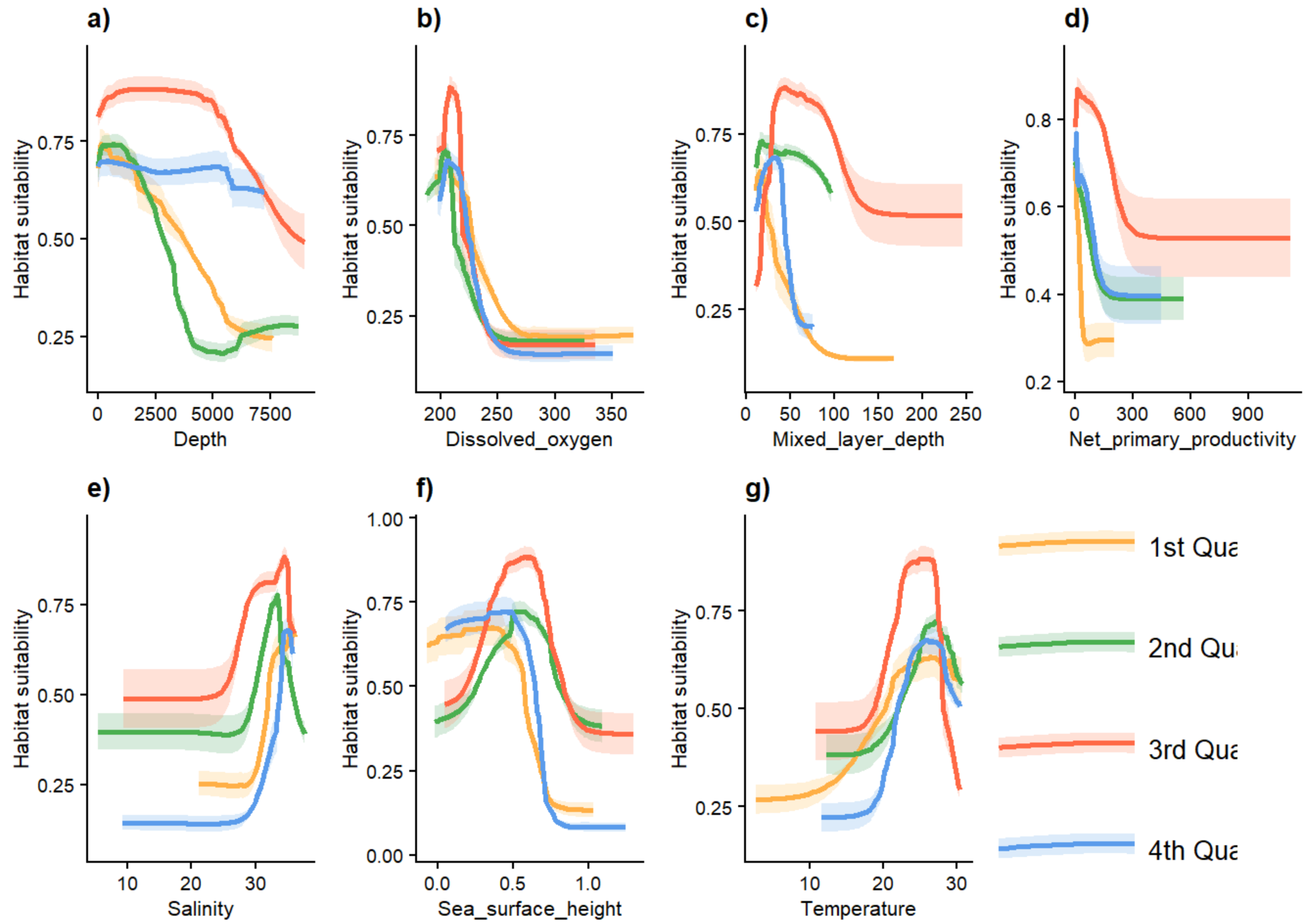
```
## Clean the R environment
rm()
gc()
#>            used (Mb) gc trigger  (Mb) max used  (Mb)
#> Ncells  6814759  364   12737551 680.3 12737551 680.3
#> Vcells 16513708  126   47291705 360.9 58852608 449.1
## Plot predictive maps
map_plot<-TB_map(root_path = "F:/whaleshark_sdm/result_sdm",
  time_type = "quarter",
  output_width = 8,
  output_height = 12,
  output_filename_base = "SDM_quarterly")
#> Land background prepared.
#> Plot saved to: SDM_quarterly.tif
print(map_plot)
```




```
# Name each temporal bin
time_names <- c("1st Quarter", "2nd Quarter", "3rd Quarter", "4th Quarter")

#  Plot variable importance
imp_plot <- TB_varimp(root_path = "F:/whaleshark_sdm/result_sdm",
                      time_type = "quarter",
                      file_name = "VarImportance_Table.csv",
                      time_names = time_names,
                      method = "AUCtest",
                      colors = c("1st Quarter" = "#FFB347",
                                 "2nd Quarter" = "#4CAF50",
                                 "3rd Quarter" = "#FF6B4A",
                                 "4th Quarter" = "#5D9BEC"),
                      title = NULL,
                      x_label = "Importance",
                      y_label = NULL,
                      group_label= "Quarters",
                      width = 8,
                      height = 6,
                      save_path = "F:/whaleshark_sdm/result_sdm/variable_importance.tif")
print(imp_plot)
```




```
#  Plot response curves panel (legend at the bottom)
res_plot <- TB_rescur(root_path = "F:/whaleshark_sdm/result_sdm",
                      time_type = "quarter",
                      file_name = "rcurve_Table.csv",
                      colors = c("1st Quarter" = "#FFB347",
                                 "2nd Quarter" = "#4CAF50",
                                 "3rd Quarter" = "#FF6B4A",
                                 "4th Quarter" = "#5D9BEC"),
                      time_names = time_names,
                      time_folders = NULL,
                      legend_position = "vertical",
                      ncol=4, nrow =NULL,
                      width = 10,
                      height = 4,
                      save_path = "F:/whaleshark_sdm/result_sdm/response_curves2.tif")
print(res_plot)
```



3. Estimate suitable/preferred depth and map the 3D habitat for each temporal bin

3.1 Estimate climatic niche (temperature) with niche_tag() or niche_occ()

A key contribution of this package is providing tools to estimate species realized climatic niche based on either tagging studies or species traits about water-column preference. I correspondingly developed two functions, niche_tag() and niche_occ(). Both functions will save a dataframe “summary_niche_fit.csv”, which stored the estimated optimal temperature, preferred temperature range, and tolerable temperature range. It will be used as an input for TB_depth(). The niche fitting plot was also saved.

3.1.1 Use niche_tag to estimate thermal niche based on temperature utilization data from tagging studies

Many studies have indicated that the observed depths of highly-mobile marine fishes are primarily determined by physiological constraints (especially thermal niche and oxygen demand) (Arrowsmith et al., 2021; Coffey et al., 2017; Dahms & Killen, 2023; Valle et al., 2024), prey availability (Braun et al., 2023), and life history needs (e.g., reproduction, nursery, and wintering) (Andrzejaczek et al., 2019). While prey availability and life history needs are vital, these factors are currently difficult to monitor or correlate with depths. Nevertheless, species realized climatic niche (e.g., the observed thermal niche) is a result of the integrated impacts of such physical and biological factors and can be correlated with depths. The major challenge is thus how to quantify species realized climatic niche.

An optimal approach to estimate species realized climatic niche is using tag-based frequency of occupancy across in-situ ocean climate bins (e.g., temperature bins), which can be readily extracted from figures (using WebPlotDigitizer) in published tagging studies. niche_tag() estimates tolerable climate range (TCR) and preferred climate range (PCR) from such data. It first combines frequency data across studies using a weighted average approach (see the equation in the paper).

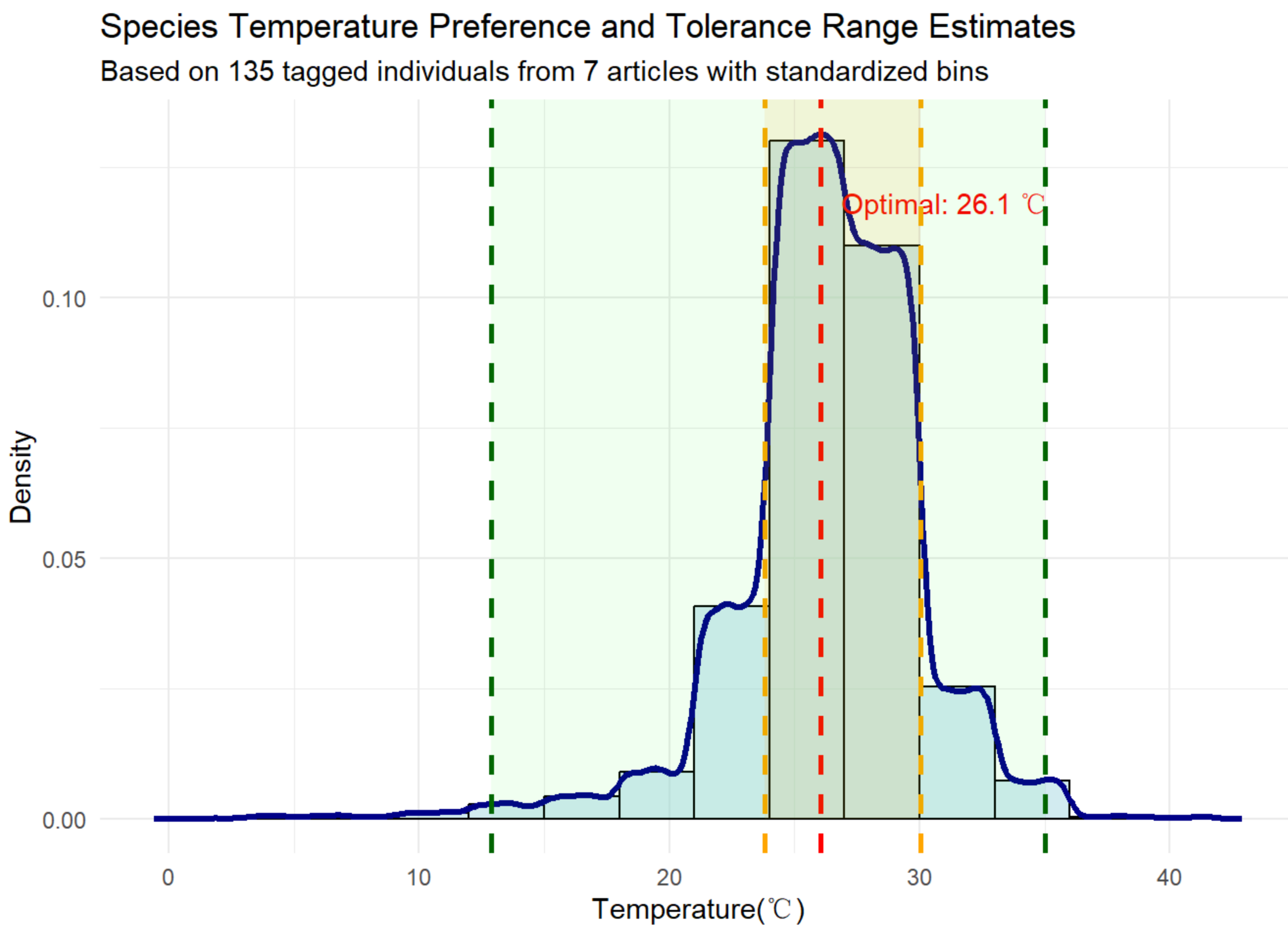
The function then allows users to choose suitable approaches to estimate TCR and PCR. For TCR, users can apply specific percentiles (default, 1st and 99th) of the temperature data, so as for PCR (default, 25th and 75th) (Hertz et al., 1993). The function also provides another option—the full width at half maximum (FWHM) from Gaussian kernel density estimation—to estimate PCR.

When tag-based temperature data are unavailable, niche_occ() implements an alternative, trait based approach. It utilizes bias-reduced occurrences to extract real-time temperatures from the depth layer preferred by species (e.g., surface for pelagic fish) rather than depths recorded in biodiversity databases as in the voluModel package. It then allows users to select similar methods (quantiles or FWHM) to estimate TCR and PCR as mentioned for niche_tag().

```
## Article 1 data (used as the standard bin given the no. of tagged individuals is the highest)
article1_df <- data.frame(
  bin_lower = c(0, 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39),
  bin_upper = c(3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42),
  percentage = c(0,0,0,0,0,0,0.3, 11, 29, 37, 16, 6, 0.4, 0.3) / 100,
  article_weight = 50  # Sample size: 50 individuals
)

## Article 2 data (different binning scheme)
article2_df <- data.frame(
  bin_lower = c(10, 12.5, 15, 17.5, 20, 22.5, 25, 27.5, 30),
  bin_upper = c(12.5, 15, 17.5, 20, 22.5, 25, 27.5, 30, 32.5),
  percentage =c(0.8856, 0.8856, 0.6642, 0.4428, 0.4428, 4.6494, 29.4465, 59.3358, 2.8782)/100 ,
  article_weight = 8  # Sample size: 8 individuals
)

## Article 3 data (another different binning scheme)
article3_df <- data.frame(
  bin_lower = c(10, 12, 14, 16, 18, 20, 22, 24, 26, 28),
  bin_upper = c(12, 14, 16, 18, 20, 22, 24, 26, 28, 30),
  percentage = c(0.65445, 3.02054, 4.36468, 4.36468, 4.36468, 6.207209, 12.58558, 25.50846, 26.011
88, 12.91784)/100 ,
  article_weight = 1  # Sample size: 1 individuals
)

## Article 4 data (another different binning scheme)
article4_df <- data.frame(
  bin_lower = c(0, 2, 5, 10, 15, 17.5, 20, 22, 25, 27, 29,31),
  bin_upper = c(2, 5, 10, 15, 17.5, 20, 22, 25, 27, 29,31,33),
  percentage =  c(0.039048, 0.002092, 0.055783, 2.545097, 4.554678, 8.029258, 18.46834, 44.00787,
12.77708, 8.686102, 0.594088, 0.240564)/100,
  article_weight = 15  # Sample size: 15 individuals
)

## Article 5 data (another different binning scheme)
article5_df <- data.frame(
  bin_lower = c(0, 5, 10, 15, 18, 20, 22, 24, 26, 28, 30, 32, 34),
  bin_upper = c(5, 10, 15, 18, 20, 22, 24, 26, 28, 30, 32, 34,36),
  percentage =  c(0.002896, 0.004234, 0.013009, 0.012676, 0.011109, 0.018206, 0.055804, 0.443287,
0.41829, 0.020489,0,0,0),
  article_weight = 27  # Sample size: 15 individuals
)
## Article 6 data (another different binning scheme)
article6_df <- data.frame(
  bin_lower = c(4,8,12,16,20,22,24,26,28),
  bin_upper = c(8,12,16,20,22,24,26,28,30),
  percentage =  c(0.0188, 0.0075, 0.0063, 0.0088, 0.0151, 0.1217, 0.3111, 0.4027, 0.1116),
  article_weight = 17  # Sample size: 15 individuals
)
## Article 7 data (another different binning scheme)
article7_df <- data.frame(
  bin_lower = c(10,12.5,15,17.5,20,22.5,25,27.5,30,32.5),
```

```
  bin_upper = c(12.5,15,17.5,20,22.5,25,27.5,30,32.5,35),
  percentage =  c(0, 0.029412, 0.011765, 0.011765, 0.019608, 0.060785, 0.372549, 0.37647, 0.115686, 0.001961),
  article_weight = 17  # Sample size: 15 individuals
)

## Create list of articles
data_list <- list(article1_df, article2_df, article3_df,article4_df,article5_df,article6_df,article7_df)

## Define unified global binning scheme (follow article1_df)
global_bins <- article1_df[,1:2]
## Run estimation
temp_fit<-niche_tag(
    data_list,
    global_bins,
    samples_per_article = 10000,
    toler_quantiles = c(0.01, 0.99),
    pref_method = "FWHM",
    pref_quantiles =NULL,
    variable_name = "Temperature",
    unit = "℃",
    seed = 123,
    width = 6,
    height = 5,
    save_path = "F:/whaleshark_sdm/temp_fit/tag"
)
#>
#> Processing Article 1 ...
#>   Simulated 10000 raw values from original bins
#>
#> Processing Article 2 ...
#>   Simulated 9964 raw values from original bins
#>
#> Processing Article 3 ...
#>   Simulated 9999 raw values from original bins
#>
#> Processing Article 4 ...
#>   Simulated 10001 raw values from original bins
#>
#> Processing Article 5 ...
#>   Simulated 10000 raw values from original bins
#>
#> Processing Article 6 ...
#>   Simulated 10036 raw values from original bins
#>
#> Processing Article 7 ...
#>   Simulated 10001 raw values from original bins
#>
#> === Combining articles with weighted average ===
#> Total number of individuals = 135
#> Normalized article weights: 0.37, 0.059, 0.007, 0.111, 0.2, 0.126, 0.126
```

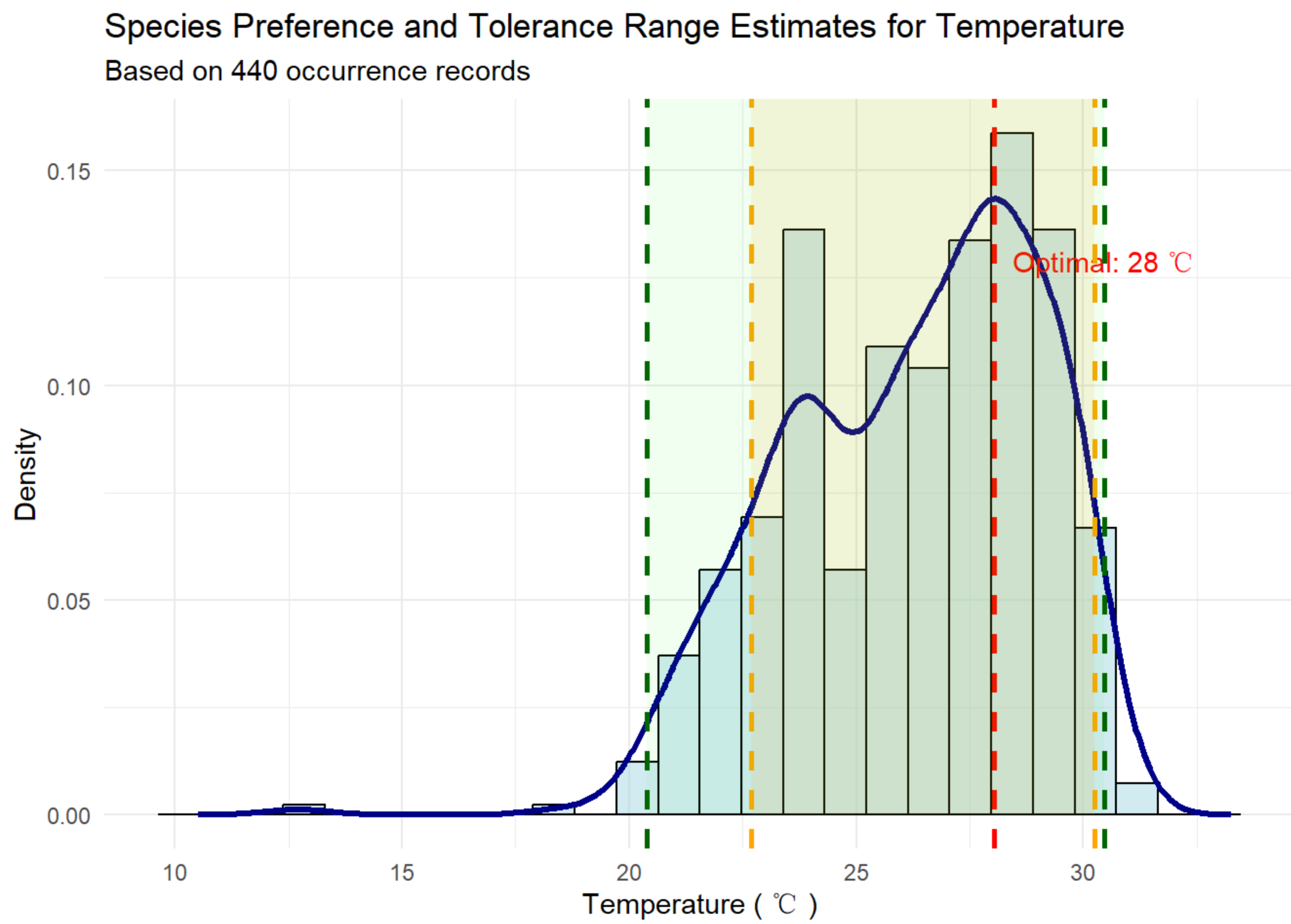
```
#>
#> Final standardized binned data (weighted average):
#>    bin_lower bin_upper  percentage      density
#> 1          0         3 0.000444440 0.0001481467
#> 2          3         6 0.001647290 0.0005490965
#> 3          6         9 0.001731730 0.0005772432
#> 4          9        12 0.003665289 0.0012217631
#> 5         12        15 0.008542011 0.0028473370
#> 6         15        18 0.013076554 0.0043588512
#> 7         18        21 0.027417876 0.0091392920
#> 8         21        24 0.122737990 0.0409126633
#> 9         24        27 0.389760985 0.1299203283
#> 10        27        30 0.329802320 0.1099341066
#> 11        30        33 0.076169832 0.0253899439
#> 12        33        36 0.022411092 0.0074703641
#> 13        36        39 0.001481481 0.0004938272
#> 14        39        42 0.001111111 0.0003703704
#>
#> === Generating final simulated data from standardized bins ===
#> Generated 50000 simulated values from standardized bins
#>
#> === Species Temperature Preference and Tolerance Range Estimates ===
#> Method: Standardized bins + weighted average across articles
#> Preference range method: FWHM
#> Final sample size for KDE: 50000
#> Total individuals across articles: 135
#> Number of articles combined: 7
#>
#> Optimal Temperature :
#>   Value: 26.05 ℃
#>
#> Preferred Range:
#>   Lower: 23.82 ℃
#>   Upper: 30.04 ℃
#>
#> Tolerance Range [0.01-0.99 percentile]:
#>   Lower: 12.87 ℃
#>   Upper: 35.03 ℃
```



3.1.2 Extract temperature data from thinned occurrences to estimate climatic niche

Here, based on the pelagic trait of the focal species, I used niche_occ() to extract the temperature at the depth of 1 m at the thinned occurrences to estimate thermal niche.

```r
occ_path<-system.file("extdata", "R_typus_qs.txt", package = "OceanSDM")

var_name<-"thetao_mean"
# ocean temperature at the surface layer (depth = 1m)
temp_path<-system.file("extdata", "cms/IndoWPac_temp_1m_y2015_y2024.nc", package = "OceanSDM")
save_path <- "F:/whaleshark_sdm/temp_fit/habitat"
min_year<-2015
lon_range<-c(90,180)
lat_range<-c(-41,40)
width<-6
height<-5
temp_occ<-niche_occ(
    occ_path,
    temp_path,
    toler_quantiles = c(0.01, 0.99),
    pref_method = "FWHM",
    pref_quantiles =NULL,
    save_path,
    min_year,
    var_name = "thetao_mean",
    env_label = "Temperature",
    unit = "℃",
    lon_range,
    lat_range,
    width = 6,
    height = 5
)
#> Summary of extracted thetao_mean values:
#>    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#>   12.71   24.03   26.60   26.24   28.29   31.06
#> Preferred Range (FWHM): 22.69 - 30.25 (Optimal: 28.05)
#> Tolerance Range (1%-99%): 20.39 - 30.46
#> Saving niche plot and data...
```



3.1.3 Extract oxygen data (optional)

Although the depth limits might be best estimated with thermal niche, other physiological constraints such as demand of oxygen can be estimated as well with data in a user-defined depth at species occurrences.

```
# occ_path<-system.file("extdata", "R_typus_qs.txt", package = "OceanSDM")
# var_name<-"o2"
## Download the O2 data with env_download() first
# o2_path<-"F:/cmems/IndoWPac_o2_1m_y2015_y2024.nc" # dissolved oxygen at the surface layer (depth
= 1m)
# save_path <- "F:/whaleshark_sdm/o2_fit/habitat"
#
# oxy_occ<-niche_occ(
#   occ_path,
#   o2_path,
#   toler_quantiles = c(0.01, 0.99),
#   pref_method = "FWHM",
#   pref_quantiles =NULL,
#   save_path,
#   min_year,
#   var_name = "o2",
#   env_label = "Dissolved oxygen",
#   unit = "μmol/L",
#   lon_range,
#   lat_range,
#   width = 8,
#   height = 5
# )
```

3.2 Create temporally-binned temperature for each depth layer

Before predicting vertical distributions across space and time, clim_bin() calculates the average climate value at each depth layer of each predicted suitable pixel in each temporal bin (derived from TB_sdm).

```
## Collect predicted binary maps of each temporal bin (here, quarters)
# quarter_files<-c(
#   Q1="F:/whaleshark_sdm/result_sdm/Q1/predicted_presence.asc",
#   Q2="F:/whaleshark_sdm/result_sdm/Q2/predicted_presence.asc",
#   Q3="F:/whaleshark_sdm/result_sdm/Q3/predicted_presence.asc",
#   Q4="F:/whaleshark_sdm/result_sdm/Q4/predicted_presence.asc")
# presence_asc <- lapply(quarter_files, terra::rast)
# save_path<-"F:/whaleshark_sdm/4D"

## In subsection 1.2.2, you can run the code to downloaded temperature data for three water column
s: 0-200m, 200-400m, 400-1050m. And then you can employ clim_bin() to create temporally-binned pre
dictors for each depth layer based on these three .nc files.
# temp_nc<-"F:/whaleshark_sdm/cmems/IndoWPac_temp_200m_y2015_y2024.nc"
# clim_bin(temp_nc, presence_asc, time_type = "quarter",
#          save_path=save_path, hemisphere = "north", na_value = -9999)
# temp_nc<-"F:/whaleshark_sdm/cmems/IndoWPac_temp_200_400m_y2015_y2024.nc"
# clim_bin(temp_nc, presence_asc, time_type = "quarter",
#          save_path=save_path,  na_value = -9999)
# temp_nc<-"F:/whaleshark_sdm/cmems/IndoWPac_temp_400_1050m_y2015_y2024.nc"
# clim_bin(temp_nc, presence_asc, time_type = "quarter",
#          save_path=save_path, na_value = -9999)
```

3.3 Calculate TD and PD for each temporal bin with TB_depth()

TB_depth() calculates tolerable depths (TD) and preferred depths (PD) in each pixel for each temporal bin by comparing TCR and PCR, respectively, with the climate value along the vertical gradient. It outputs three statistics (minimum, maximum, range) for TD and PD and plots the vertical climate profile for a random sample (default, 100) of grids. However, this process is very time-consuming (lasting > 8 hrs) given the function has to calculate these statistics grid by grid and there are 47 depth layers. Therefore, I did not run the following code when generating this vignette. But I have stored the outputs while producing the paper.

```
##### Example shown with estimates based on tag data
df<-read.csv(system.file("extdata","summary_niche_fit.csv",package = "OceanSDM"))
PTR<-df[2:3,2] # preferred temperature range
TTR<-df[4:5,2] # tolerable temperature range
## Run the whole code in subsection 3.2 to create temporally-binned mean temperature for each dept
h layer. Here I stored all these files in "F:/whaleshark_sdm/4D"
temp_bin_path<-"F:/whaleshark_sdm/4D" # Change to your own directory
save_path_TPD<-"F:/whaleshark_sdm/4D/TPD" ## for storing data of Preferred Depth (PD)
save_path_TTD<-"F:/whaleshark_sdm/4D/TTD" ## for storing data of Tolerable Depth (TD)
## For Tolerable Depth
TTD_quarter<-TB_depth(temp_bin_path, TTR, time_type = "quarter",
                      layer_option="top",
                      save_path_TTD, na_value = -9999,
                      use_makima = TRUE,
                      N_cells = 50)
#> Using makima for interpolation (when data points >= 4)
#> Found 4 time groups: Q1, Q2, Q3, Q4
#>
#> Processing time group Q1...
#>  Reading depth layers into stack...
#>  Extracting values matrix...
#>  Matrix dimensions: 117325 cells x 47 depth layers
#>  Cells with at least one valid depth layer: 29174
#>  Interpolating and calculating depth ranges...
#>   |
|
|   0%  |
|=
|   0%  |
|=
|   1%  |
|==
|   1%  |
|==
|   2%  |
|===
|   2%  |
|===
|   3%  |
|====
|   3%  |
|====
|   4%  |
|=====
|   4%  |
|=====
|   5%  |
|======
|   5%  |
|======
|   6%  |
|=======
```

```
|    6%   |
|=======
|    7%   |
|========
|    7%   |
|========
|    8%   |
|=========
|    8%   |
|=========
|    9%   |
|==========
|    9%   |
|==========
|   10%   |
|===========
|   10%   |
|===========
|   11%   |
|============
|   11%   |
|============
|   12%   |
|=============
|   12%   |
|=============
|   13%   |
|==============
|   13%   |
|==============
|   14%   |
|===============
|   14%   |
|===============
|   15%   |
|================
|   15%   |
|================
|   16%   |
|=================
|   16%   |
|=================
|   17%   |
|==================
|   17%   |
|==================
|   18%   |
|===================
|   18%   |
|===================
|   19%   |
|====================
```

```
/   19%   /
/====================
/   20%   /
/=====================
/   20%   /
/=====================
/   21%   /
/======================
/   21%   /
/======================
/   22%   /
/=======================
/   22%   /
/=======================
/   23%   /
/========================
/   23%   /
/========================
/   24%   /
/=========================
/   24%   /
/=========================
/   25%   /
/==========================
/   25%   /
/==========================
/   26%   /
/===========================
/   26%   /
/===========================
/   27%   /
/============================
/   27%   /
/============================
/   28%   /
/=============================
/   28%   /
/=============================
/   29%   /
/==============================
/   29%   /
/==============================
/   30%   /
/===============================
/   30%   /
/===============================
/   31%   /
/================================
/   31%   /
/================================
/   32%   /
/=================================
```

/ 32% /
/================================
/ 33% /
/=================================
/ 33% /
/=================================
/ 34% /
/==================================
/ 34% /
/==================================
/ 35% /
/===================================
/ 35% /
/===================================
/ 36% /
/====================================
/ 36% /
/====================================
/ 37% /
/=====================================
/ 37% /
/=====================================
/ 38% /
/======================================
/ 38% /
/======================================
/ 39% /
/=======================================
/ 39% /
/=======================================
/ 40% /
/==
/ 40% /
/==
/ 41% /
/===
/ 41% /
/===
/ 42% /
/==
/ 42% /
/==
/ 43% /
/===
/ 43% /
/===
/ 44% /
/==
/ 44% /
/==
/ 45% /
/===

/ 45% /
/===
/ 46% /
/==
/ 46% /
/==
/ 47% /
/===
/ 47% /
/===
/ 48% /
/==
/ 48% /
/==
/ 49% /
/===
/ 49% /
/===
/ 50% /
/==
/ 50% /
/==
/ 51% /
/===
/ 51% /
/===
/ 52% /
/==
/ 52% /
/==
/ 53% /
/===
/ 53% /
/===
/ 54% /
/==
/ 54% /
/==
/ 55% /
/===
/ 55% /
/===
/ 56% /
/==
/ 56% /
/==
/ 57% /
/===
/ 57% /
/===
/ 58% /
/==

/   58%   /
/==
/   59%   /
/===
/   59%   /
/===
/   60%   /
/==
/   60%   /
/==
/   61%   /
/===
/   61%   /
/===
/   62%   /
/==
/   62%   /
/==
/   63%   /
/===
/   63%   /
/===
/   64%   /
/==
/   64%   /
/==
/   65%   /
/===
/   65%   /
/===
/   66%   /
/==
/   66%   /
/==
/   67%   /
/===
/   67%   /
/===
/   68%   /
/==
/   68%   /
/==
/   69%   /
/===
/   69%   /
/===
/   70%   /
/==
/   70%   /
/==
/   71%   /
/===

/ 71% /
/==
/ 72% /
/===
/ 72% /
/===
/ 73% /
/==
/ 73% /
/==
/ 74% /
/===
/ 74% /
/===
/ 75% /
/==
/ 75% /
/==
/ 76% /
/===
/ 76% /
/===
/ 77% /
/==
/ 77% /
/==
/ 78% /
/===
/ 78% /
/===
/ 79% /
/==
/ 79% /
/==
/ 80% /
/===
/ 80% /
/===
/ 81% /
/==
/ 81% /
/==
/ 82% /
/===
/ 82% /
/===
/ 83% /
/==
/ 83% /
/==
/ 84% /
/===

/ 84% /
/===
/ 85% /
/==
/ 85% /
/==
/ 86% /
/===
/ 86% /
/===
/ 87% /
/==
/ 87% /
/==
/ 88% /
/===
/ 88% /
/===
/ 89% /
/==
/ 89% /
/==
/ 90% /
/===
/ 90% /
/===
/ 91% /
/==
/ 91% /
/==
/ 92% /
/===
/ 92% /
/===
/ 93% /
/==
/ 93% /
/==
/ 94% /
/===
/ 94% /
/===
/ 95% /
/==
/ 95% /
/==
/ 96% /
/===
/ 96% /
/===
/ 97% /
/==

```
=   |  97%  |
|==================================================================================================
=   |  98%  |
|==================================================================================================
==  |  98%  |
|==================================================================================================
==  |  99%  |
|==================================================================================================
=== |  99%  |
|==================================================================================================
=== | 100%  |
|==================================================================================================
====| 100%
#>  Group Q1: 29174 cells have valid depth ranges
#>  Group Q1 processing complete, 3 files saved
#>  Saved temperature profile plot: Temperature_profile_Q1.tif
#>
#> Processing time group Q2...
#>  Reading depth layers into stack...
#>  Extracting values matrix...
#>  Matrix dimensions: 117325 cells x 47 depth layers
#>  Cells with at least one valid depth layer: 18815
#>  Interpolating and calculating depth ranges...
#>    |
|
|   0%  |
|=
|   0%  |
|=
|   1%  |
|==
|   1%  |
|==
|   2%  |
|===
|   2%  |
|===
|   3%  |
|====
|   3%  |
|====
|   4%  |
|=====
|   4%  |
|=====
|   5%  |
|======
|   5%  |
|======
|   6%  |
|=======
|   6%  |
```

```
/=======
/    7%    /
/========
/    7%    /
/========
/    8%    /
/=========
/    8%    /
/=========
/    9%    /
/==========
/    9%    /
/==========
/   10%    /
/===========
/   10%    /
/===========
/   11%    /
/============
/   11%    /
/============
/   12%    /
/=============
/   12%    /
/=============
/   13%    /
/==============
/   13%    /
/==============
/   14%    /
/===============
/   14%    /
/===============
/   15%    /
/================
/   15%    /
/================
/   16%    /
/=================
/   16%    /
/=================
/   17%    /
/==================
/   17%    /
/==================
/   18%    /
/===================
/   18%    /
/===================
/   19%    /
/====================
/   19%    /
```

```
/====================
/   20%   /
/=====================
/   20%   /
/=====================
/   21%   /
/======================
/   21%   /
/======================
/   22%   /
/=======================
/   22%   /
/=======================
/   23%   /
/========================
/   23%   /
/========================
/   24%   /
/=========================
/   24%   /
/=========================
/   25%   /
/==========================
/   25%   /
/==========================
/   26%   /
/===========================
/   26%   /
/===========================
/   27%   /
/============================
/   27%   /
/============================
/   28%   /
/=============================
/   28%   /
/=============================
/   29%   /
/==============================
/   29%   /
/==============================
/   30%   /
/===============================
/   30%   /
/===============================
/   31%   /
/================================
/   31%   /
/================================
/   32%   /
/=================================
/   32%   /
```

```
/=================================
/   33%   /
/==================================
/   33%   /
/==================================
/   34%   /
/===================================
/   34%   /
/===================================
/   35%   /
/====================================
/   35%   /
/====================================
/   36%   /
/=====================================
/   36%   /
/=====================================
/   37%   /
/======================================
/   37%   /
/======================================
/   38%   /
/=======================================
/   38%   /
/=======================================
/   39%   /
/========================================
/   39%   /
/========================================
/   40%   /
/=========================================
/   40%   /
/=========================================
/   41%   /
/==========================================
/   41%   /
/==========================================
/   42%   /
/===========================================
/   42%   /
/===========================================
/   43%   /
/============================================
/   43%   /
/============================================
/   44%   /
/=============================================
/   44%   /
/=============================================
/   45%   /
/==============================================
/   45%   /
```

```
/=============================================
/   46%   /
/==============================================
/   46%   /
/==============================================
/   47%   /
/===============================================
/   47%   /
/===============================================
/   48%   /
/================================================
/   48%   /
/================================================
/   49%   /
/=================================================
/   49%   /
/=================================================
/   50%   /
/==================================================
/   50%   /
/==================================================
/   51%   /
/===================================================
/   51%   /
/===================================================
/   52%   /
/====================================================
/   52%   /
/====================================================
/   53%   /
/=====================================================
/   53%   /
/=====================================================
/   54%   /
/======================================================
/   54%   /
/======================================================
/   55%   /
/=======================================================
/   55%   /
/=======================================================
/   56%   /
/========================================================
/   56%   /
/========================================================
/   57%   /
/=========================================================
/   57%   /
/=========================================================
/   58%   /
/==========================================================
/   58%   /
```

/==

/ 59% /

/===

/ 59% /

/===

/ 60% /

/==

/ 60% /

/==

/ 61% /

/===

/ 61% /

/===

/ 62% /

/==

/ 62% /

/==

/ 63% /

/===

/ 63% /

/===

/ 64% /

/==

/ 64% /

/==

/ 65% /

/===

/ 65% /

/===

/ 66% /

/==

/ 66% /

/==

/ 67% /

/===

/ 67% /

/===

/ 68% /

/==

/ 68% /

/==

/ 69% /

/===

/ 69% /

/===

/ 70% /

/==

/ 70% /

/==

/ 71% /

/===

/ 71% /

```
|=======================================================================
|   72%   |
|========================================================================
|   72%   |
|========================================================================
|   73%   |
|=========================================================================
|   73%   |
|=========================================================================
|   74%   |
|==========================================================================
|   74%   |
|==========================================================================
|   75%   |
|===========================================================================
|   75%   |
|===========================================================================
|   76%   |
|============================================================================
|   76%   |
|============================================================================
|   77%   |
|=============================================================================
|   77%   |
|=============================================================================
|   78%   |
|==============================================================================
|   78%   |
|==============================================================================
|   79%   |
|===============================================================================
|   79%   |
|===============================================================================
|   80%   |
|================================================================================
|   80%   |
|================================================================================
|   81%   |
|=================================================================================
|   81%   |
|=================================================================================
|   82%   |
|==================================================================================
|   82%   |
|==================================================================================
|   83%   |
|===================================================================================
|   83%   |
|===================================================================================
|   84%   |
|====================================================================================
|   84%   |
```

```
|=====================================================================================
| 85% |
|======================================================================================
| 85% |
|======================================================================================
| 86% |
|=======================================================================================
| 86% |
|=======================================================================================
| 87% |
|========================================================================================
| 87% |
|========================================================================================
| 88% |
|=========================================================================================
| 88% |
|=========================================================================================
| 89% |
|==========================================================================================
| 89% |
|==========================================================================================
| 90% |
|===========================================================================================
| 90% |
|===========================================================================================
| 91% |
|============================================================================================
| 91% |
|============================================================================================
| 92% |
|=============================================================================================
| 92% |
|=============================================================================================
| 93% |
|==============================================================================================
| 93% |
|==============================================================================================
| 94% |
|===============================================================================================
| 94% |
|===============================================================================================
| 95% |
|================================================================================================
| 95% |
|================================================================================================
| 96% |
|=================================================================================================
| 96% |
|=================================================================================================
| 97% |
|=================================================================================================
=    | 97% |
```

```
|================================================================================================
=   |  98%  |
|================================================================================================
==  |  98%  |
|================================================================================================
==  |  99%  |
|================================================================================================
=== |  99%  |
|================================================================================================
=== | 100%  |
|================================================================================================
====| 100%
#>  Group Q2: 18815 cells have valid depth ranges
#>  Group Q2 processing complete, 3 files saved
#>  Saved temperature profile plot: Temperature_profile_Q2.tif
#>
#> Processing time group Q3...
#>  Reading depth layers into stack...
#>  Extracting values matrix...
#>  Matrix dimensions: 117325 cells x 47 depth layers
#>  Cells with at least one valid depth layer: 15762
#>  Interpolating and calculating depth ranges...
#>    |
|
|   0%  |
|=
|   1%  |
|==
|   1%  |
|==
|   2%  |
|===
|   2%  |
|===
|   3%  |
|====
|   3%  |
|====
|   4%  |
|=====
|   4%  |
|=====
|   5%  |
|======
|   5%  |
|======
|   6%  |
|=======
|   6%  |
|=======
|   7%  |
|========
```

```
/     7%    /
/========
/     8%    /
/=========
/     8%    /
/=========
/     9%    /
/==========
/     9%    /
/==========
/    10%    /
/===========
/    10%    /
/===========
/    11%    /
/============
/    11%    /
/============
/    12%    /
/=============
/    12%    /
/=============
/    13%    /
/==============
/    13%    /
/==============
/    14%    /
/===============
/    14%    /
/===============
/    15%    /
/================
/    15%    /
/================
/    16%    /
/=================
/    16%    /
/=================
/    17%    /
/==================
/    17%    /
/==================
/    18%    /
/===================
/    18%    /
/===================
/    19%    /
/====================
/    19%    /
/====================
/    20%    /
/=====================
```

```
|   20%   |
|=====================
|   21%   |
|======================
|   21%   |
|======================
|   22%   |
|=======================
|   22%   |
|=======================
|   23%   |
|========================
|   23%   |
|========================
|   24%   |
|=========================
|   24%   |
|=========================
|   25%   |
|==========================
|   25%   |
|==========================
|   26%   |
|===========================
|   26%   |
|===========================
|   27%   |
|============================
|   27%   |
|============================
|   28%   |
|=============================
|   28%   |
|=============================
|   29%   |
|==============================
|   29%   |
|==============================
|   30%   |
|===============================
|   30%   |
|===============================
|   31%   |
|================================
|   31%   |
|================================
|   32%   |
|=================================
|   32%   |
|=================================
|   33%   |
|==================================
```

```
/ 33% /
/=================================
/ 34% /
/==================================
/ 34% /
/==================================
/ 35% /
/===================================
/ 35% /
/===================================
/ 36% /
/====================================
/ 36% /
/====================================
/ 37% /
/=====================================
/ 37% /
/=====================================
/ 38% /
/======================================
/ 38% /
/======================================
/ 39% /
/=======================================
/ 39% /
/=======================================
/ 40% /
/========================================
/ 40% /
/========================================
/ 41% /
/=========================================
/ 41% /
/=========================================
/ 42% /
/==========================================
/ 42% /
/==========================================
/ 43% /
/===========================================
/ 43% /
/===========================================
/ 44% /
/============================================
/ 44% /
/============================================
/ 45% /
/=============================================
/ 45% /
/=============================================
/ 46% /
/==============================================
```

/ 46% /
/==
/ 47% /
/===
/ 47% /
/===
/ 48% /
/==
/ 48% /
/==
/ 49% /
/===
/ 49% /
/===
/ 50% /
/==
/ 50% /
/==
/ 51% /
/===
/ 51% /
/===
/ 52% /
/==
/ 52% /
/==
/ 53% /
/===
/ 53% /
/===
/ 54% /
/==
/ 54% /
/==
/ 55% /
/===
/ 55% /
/===
/ 56% /
/==
/ 56% /
/==
/ 57% /
/===
/ 57% /
/===
/ 58% /
/==
/ 58% /
/==
/ 59% /
/===

/   59%   /
/==
/   60%   /
/===
/   60%   /
/===
/   61%   /
/==
/   61%   /
/==
/   62%   /
/===
/   62%   /
/===
/   63%   /
/==
/   63%   /
/==
/   64%   /
/===
/   64%   /
/===
/   65%   /
/==
/   65%   /
/==
/   66%   /
/===
/   66%   /
/===
/   67%   /
/==
/   67%   /
/==
/   68%   /
/===
/   68%   /
/===
/   69%   /
/==
/   69%   /
/==
/   70%   /
/===
/   70%   /
/===
/   71%   /
/==
/   71%   /
/==
/   72%   /
/===

```
|   72%   |
|========================================================================
|   73%   |
|=========================================================================
|   73%   |
|=========================================================================
|   74%   |
|==========================================================================
|   74%   |
|==========================================================================
|   75%   |
|===========================================================================
|   75%   |
|===========================================================================
|   76%   |
|============================================================================
|   76%   |
|============================================================================
|   77%   |
|=============================================================================
|   77%   |
|=============================================================================
|   78%   |
|==============================================================================
|   78%   |
|==============================================================================
|   79%   |
|===============================================================================
|   79%   |
|===============================================================================
|   80%   |
|================================================================================
|   80%   |
|================================================================================
|   81%   |
|=================================================================================
|   81%   |
|=================================================================================
|   82%   |
|==================================================================================
|   82%   |
|==================================================================================
|   83%   |
|===================================================================================
|   83%   |
|===================================================================================
|   84%   |
|====================================================================================
|   84%   |
|====================================================================================
|   85%   |
|=====================================================================================
```

```
/   85%   /
/=====================================================================================
/   86%   /
/======================================================================================
/   86%   /
/======================================================================================
/   87%   /
/=======================================================================================
/   87%   /
/=======================================================================================
/   88%   /
/========================================================================================
/   88%   /
/========================================================================================
/   89%   /
/=========================================================================================
/   89%   /
/=========================================================================================
/   90%   /
/==========================================================================================
/   90%   /
/==========================================================================================
/   91%   /
/===========================================================================================
/   91%   /
/===========================================================================================
/   92%   /
/============================================================================================
/   92%   /
/============================================================================================
/   93%   /
/=============================================================================================
/   93%   /
/=============================================================================================
/   94%   /
/==============================================================================================
/   94%   /
/==============================================================================================
/   95%   /
/===============================================================================================
/   95%   /
/===============================================================================================
/   96%   /
/================================================================================================
/   96%   /
/================================================================================================
/   97%   /
/================================================================================================
=    /   97%   /
/================================================================================================
=    /   98%   /
/================================================================================================
```

```
== |  98%  |
|==================================================================================================
== |  99%  |
|==================================================================================================
=== |  99%  |
|==================================================================================================
====| 100%
#>  Group Q3: 15762 cells have valid depth ranges
#>  Group Q3 processing complete, 3 files saved
#>  Saved temperature profile plot: Temperature_profile_Q3.tif
#>
#> Processing time group Q4...
#>  Reading depth layers into stack...
#>  Extracting values matrix...
#>  Matrix dimensions: 117325 cells x 47 depth layers
#>  Cells with at least one valid depth layer: 17602
#>  Interpolating and calculating depth ranges...
#>    |
|
|   0%  |
|=
|   0%  |
|=
|   1%  |
|==
|   1%  |
|==
|   2%  |
|===
|   2%  |
|===
|   3%  |
|====
|   3%  |
|====
|   4%  |
|=====
|   4%  |
|=====
|   5%  |
|======
|   5%  |
|======
|   6%  |
|=======
|   6%  |
|=======
|   7%  |
|========
|   7%  |
|========
|   8%  |
```

```
/=========
/     8%    /
/=========
/     9%    /
/==========
/     9%    /
/==========
/    10%    /
/===========
/    10%    /
/===========
/    11%    /
/============
/    11%    /
/============
/    12%    /
/=============
/    12%    /
/=============
/    13%    /
/==============
/    13%    /
/==============
/    14%    /
/===============
/    14%    /
/===============
/    15%    /
/================
/    15%    /
/================
/    16%    /
/=================
/    16%    /
/=================
/    17%    /
/==================
/    17%    /
/==================
/    18%    /
/===================
/    18%    /
/===================
/    19%    /
/====================
/    19%    /
/====================
/    20%    /
/=====================
/    20%    /
/=====================
/    21%    /
```

```
/======================
/   21%   /
/======================
/   22%   /
/=======================
/   22%   /
/=======================
/   23%   /
/========================
/   23%   /
/========================
/   24%   /
/=========================
/   24%   /
/=========================
/   25%   /
/==========================
/   25%   /
/==========================
/   26%   /
/===========================
/   26%   /
/===========================
/   27%   /
/============================
/   27%   /
/============================
/   28%   /
/=============================
/   28%   /
/=============================
/   29%   /
/==============================
/   29%   /
/==============================
/   30%   /
/===============================
/   30%   /
/===============================
/   31%   /
/================================
/   31%   /
/================================
/   32%   /
/=================================
/   32%   /
/=================================
/   33%   /
/==================================
/   33%   /
/==================================
/   34%   /
```

/===================================
/ 34% /
/===================================
/ 35% /
/====================================
/ 35% /
/====================================
/ 36% /
/=====================================
/ 36% /
/=====================================
/ 37% /
/======================================
/ 37% /
/======================================
/ 38% /
/=======================================
/ 38% /
/=======================================
/ 39% /
/==
/ 39% /
/==
/ 40% /
/===
/ 40% /
/===
/ 41% /
/==
/ 41% /
/==
/ 42% /
/===
/ 42% /
/===
/ 43% /
/==
/ 43% /
/==
/ 44% /
/===
/ 44% /
/===
/ 45% /
/==
/ 45% /
/==
/ 46% /
/===
/ 46% /
/===
/ 47% /

```
|================================================
|   47%   |
|================================================
|   48%   |
|=================================================
|   48%   |
|=================================================
|   49%   |
|==================================================
|   49%   |
|==================================================
|   50%   |
|===================================================
|   50%   |
|===================================================
|   51%   |
|====================================================
|   51%   |
|====================================================
|   52%   |
|=====================================================
|   52%   |
|=====================================================
|   53%   |
|======================================================
|   53%   |
|======================================================
|   54%   |
|=======================================================
|   54%   |
|=======================================================
|   55%   |
|========================================================
|   55%   |
|========================================================
|   56%   |
|=========================================================
|   56%   |
|=========================================================
|   57%   |
|==========================================================
|   57%   |
|==========================================================
|   58%   |
|===========================================================
|   58%   |
|===========================================================
|   59%   |
|============================================================
|   59%   |
|============================================================
|   60%   |
```

```
/============================================================
/   60%   /
/============================================================
/   61%   /
/=============================================================
/   61%   /
/=============================================================
/   62%   /
/==============================================================
/   62%   /
/==============================================================
/   63%   /
/===============================================================
/   63%   /
/===============================================================
/   64%   /
/================================================================
/   64%   /
/================================================================
/   65%   /
/=================================================================
/   65%   /
/=================================================================
/   66%   /
/==================================================================
/   66%   /
/==================================================================
/   67%   /
/===================================================================
/   67%   /
/===================================================================
/   68%   /
/====================================================================
/   68%   /
/====================================================================
/   69%   /
/=====================================================================
/   69%   /
/=====================================================================
/   70%   /
/======================================================================
/   70%   /
/======================================================================
/   71%   /
/=======================================================================
/   71%   /
/=======================================================================
/   72%   /
/========================================================================
/   72%   /
/========================================================================
/   73%   /
```

/===
/ 73% /
/===
/ 74% /
/==
/ 74% /
/==
/ 75% /
/===
/ 75% /
/===
/ 76% /
/==
/ 76% /
/==
/ 77% /
/===
/ 77% /
/===
/ 78% /
/==
/ 78% /
/==
/ 79% /
/===
/ 79% /
/===
/ 80% /
/==
/ 80% /
/==
/ 81% /
/===
/ 81% /
/===
/ 82% /
/==
/ 82% /
/==
/ 83% /
/===
/ 83% /
/===
/ 84% /
/==
/ 84% /
/==
/ 85% /
/===
/ 85% /
/===
/ 86% /

/==
/   86%   /
/==
/   87%   /
/===
/   87%   /
/===
/   88%   /
/==
/   88%   /
/==
/   89%   /
/===
/   89%   /
/===
/   90%   /
/==
/   90%   /
/==
/   91%   /
/===
/   91%   /
/===
/   92%   /
/==
/   92%   /
/==
/   93%   /
/===
/   93%   /
/===
/   94%   /
/==
/   94%   /
/==
/   95%   /
/===
/   95%   /
/===
/   96%   /
/==
/   96%   /
/==
/   97%   /
/==
=   /   97%   /
/==
=   /   98%   /
/==
==   /   98%   /
/==
==   /   99%   /

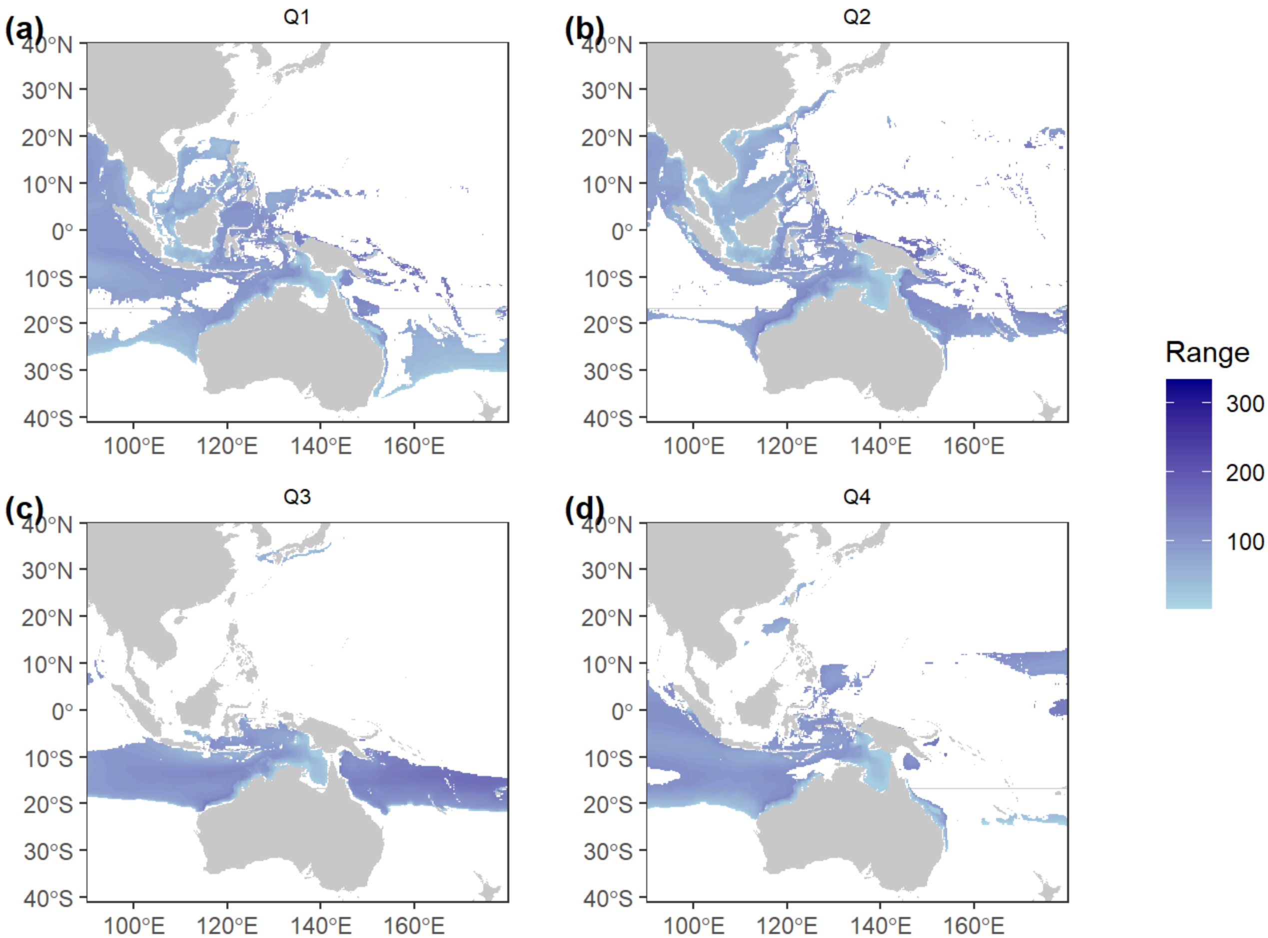
```
|==========================================================================================
=== |  99%  |
|==========================================================================================
=== | 100%  |
|==========================================================================================
====| 100%
#>  Group Q4: 17602 cells have valid depth ranges
#>  Group Q4 processing complete, 3 files saved
#>  Saved temperature profile plot: Temperature_profile_Q4.tif
#>
#> TB_depth processing complete! Generated 12 files
## For Preferred Depth (PD), run the following code:
# TPD_quarter<-TB_depth(temp_bin_path, PTR, time_type = "quarter",
#                        layer_option="top",
#                        save_path_TPD, na_value = -9999,
#                        use_makima = TRUE,
#                        N_cells = 50)
```

3.4 Plot habitat volume and area with plot_stat_4D()

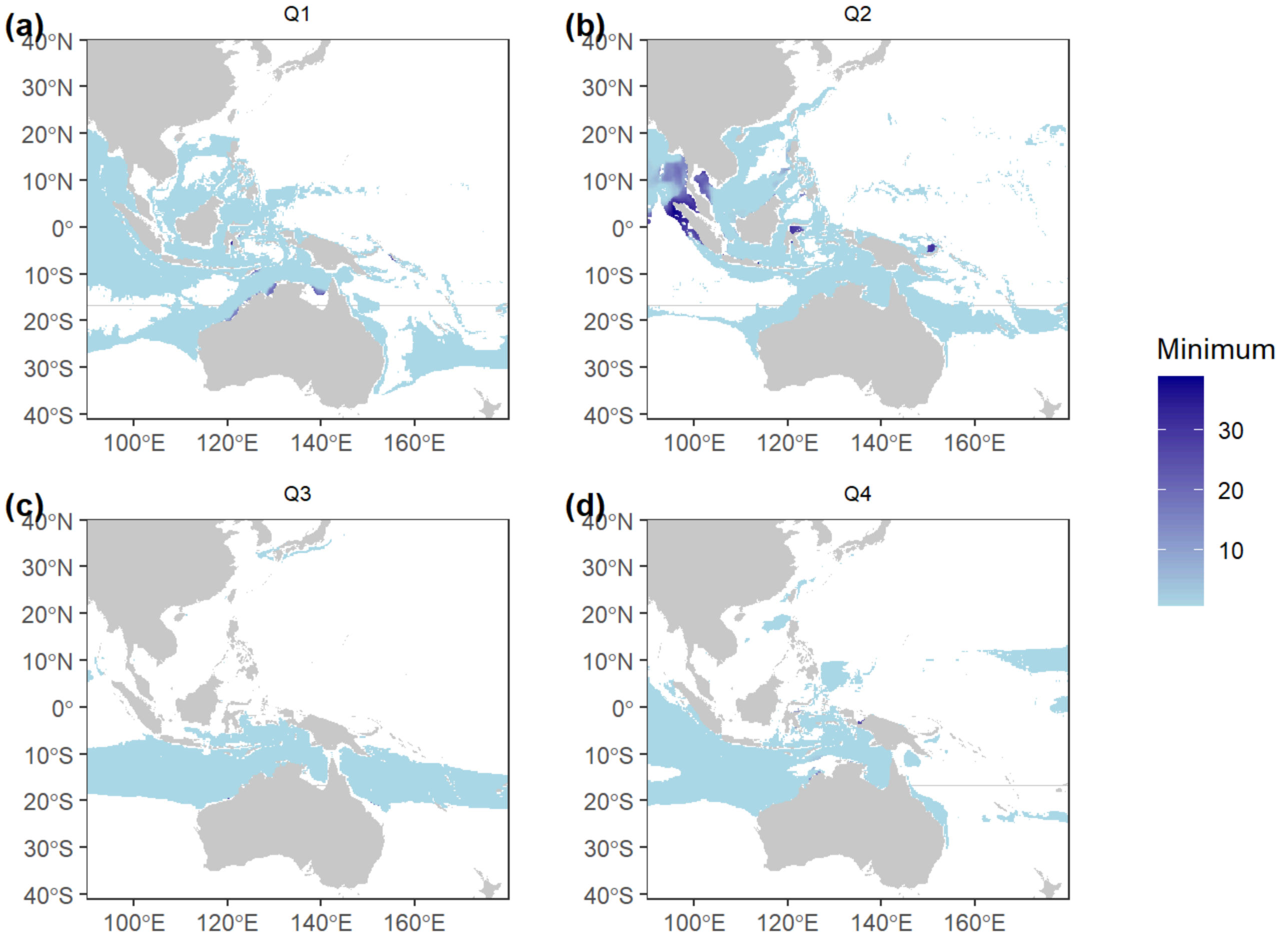
plot_stat_4D() maps the statistics (Max, Min, Range) of TD and PD across predicted 2D presence maps, generates boxplots, and tests for significant differences among temporal bins using Kruskal Wallis followed by Dunn’s post-hoc tests. It also create a figure to showcase the total suitable habitat volume based on the TD or PD, and estimates total suitable habitat area based on the 2D presence map of each temporal bin.

```
## Run the whole code in subsection 3.3 to estimate three parameters (minimum, maximum, range) of thermally preferred depth or thermally tolerable depth. Here I load the input data I already prepared.

TPD_quarter<-list(
system.file("extdata","4D/TPD/Q1/Q1_min_depth.asc",package="OceanSDM"),
  system.file("extdata","4D/TPD/Q1/Q1_max_depth.asc",package="OceanSDM"),
  system.file("extdata","4D/TPD/Q1/Q1_depth_range.asc",package="OceanSDM"),
  system.file("extdata","4D/TPD/Q2/Q2_min_depth.asc",package="OceanSDM"),
  system.file("extdata","4D/TPD/Q2/Q2_max_depth.asc",package="OceanSDM"),
  system.file("extdata","4D/TPD/Q2/Q2_depth_range.asc",package="OceanSDM"),
  system.file("extdata","4D/TPD/Q3/Q3_min_depth.asc",package="OceanSDM"),
  system.file("extdata","4D/TPD/Q3/Q3_max_depth.asc",package="OceanSDM"),
  system.file("extdata","4D/TPD/Q3/Q3_depth_range.asc",package="OceanSDM"),
  system.file("extdata","4D/TPD/Q4/Q4_min_depth.asc",package="OceanSDM"),
  system.file("extdata","4D/TPD/Q4/Q4_max_depth.asc",package="OceanSDM"),
  system.file("extdata","4D/TPD/Q4/Q4_depth_range.asc",package="OceanSDM")
)
TTD_quarter<-list(
  system.file("extdata","4D/TTD/Q1/Q1_min_depth.asc",package="OceanSDM"),
  system.file("extdata","4D/TTD/Q1/Q1_max_depth.asc",package="OceanSDM"),
  system.file("extdata","4D/TTD/Q1/Q1_depth_range.asc",package="OceanSDM"),
  system.file("extdata","4D/TTD/Q2/Q2_min_depth.asc",package="OceanSDM"),
  system.file("extdata","4D/TTD/Q2/Q2_max_depth.asc",package="OceanSDM"),
  system.file("extdata","4D/TTD/Q2/Q2_depth_range.asc",package="OceanSDM"),
  system.file("extdata","4D/TTD/Q3/Q3_min_depth.asc",package="OceanSDM"),
  system.file("extdata","4D/TTD/Q3/Q3_max_depth.asc",package="OceanSDM"),
  system.file("extdata","4D/TTD/Q3/Q3_depth_range.asc",package="OceanSDM"),
  system.file("extdata","4D/TTD/Q4/Q4_min_depth.asc",package="OceanSDM"),
  system.file("extdata","4D/TTD/Q4/Q4_max_depth.asc",package="OceanSDM"),
  system.file("extdata","4D/TTD/Q4/Q4_depth_range.asc",package="OceanSDM")
)
save_path_TPD<-"F:/whaleshark_sdm/4D/TPD" ## for Preferred Depth (PD)
save_path_TTD<-"F:/whaleshark_sdm/4D/TTD" ## for Tolerable Depth (TD)
quarter_files<-c(
  Q1=system.file("extdata","result_sdm/Q1/predicted_presence.asc",package="OceanSDM"),
  Q2=system.file("extdata","result_sdm/Q2/predicted_presence.asc",package="OceanSDM"),
  Q3=system.file("extdata","result_sdm/Q3/predicted_presence.asc",package="OceanSDM"),
  Q4=system.file("extdata","result_sdm/Q4/predicted_presence.asc",package="OceanSDM"))

result_TPD <- plot_stat_4D(
  file_list = TPD_quarter,
  save_path = save_path_TPD,
  time_type = "quarter",
  presence_files = quarter_files
)
#> Reading 12 raster files...
#>  Reading Q1_min_depth.asc
#>  Reading Q1_max_depth.asc
#>  Reading Q1_depth_range.asc
#>  Reading Q2_min_depth.asc
#>  Reading Q2_max_depth.asc
```

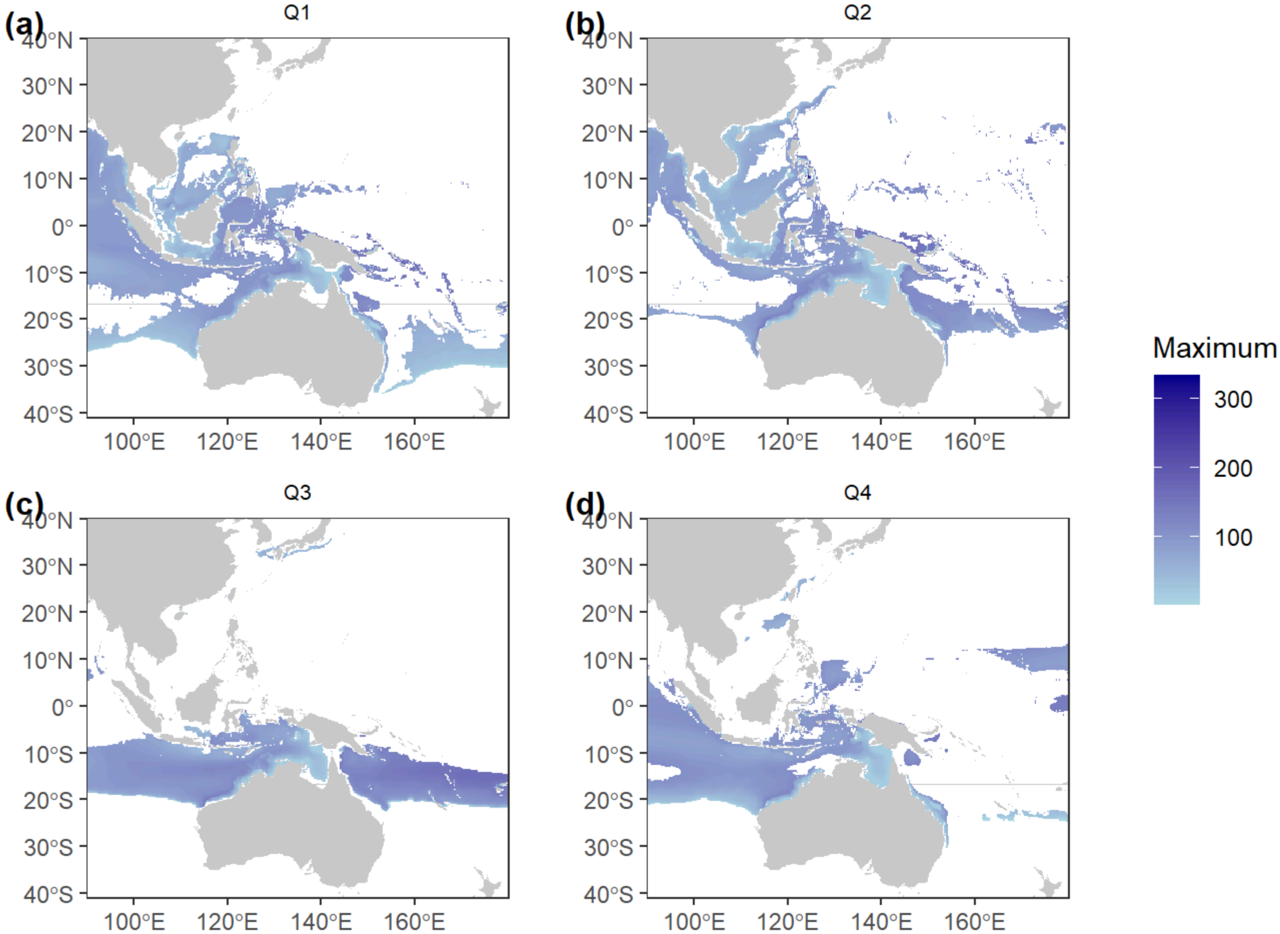
```
#>  Reading Q2_depth_range.asc
#>  Reading Q3_min_depth.asc
#>  Reading Q3_max_depth.asc
#>  Reading Q3_depth_range.asc
#>  Reading Q4_min_depth.asc
#>  Reading Q4_max_depth.asc
#>  Reading Q4_depth_range.asc
#> All rasters loaded.
#>
#> Computing habitat volume for each quarter ...
#>  Processing Q1 depth_range raster...
#>  Volume: 1140652827611852 km³
#>  Processing Q2 depth_range raster...
#>  Volume: 1039132546769828 km³
#>  Processing Q3 depth_range raster...
#>  Volume: 1042251992171751 km³
#>  Processing Q4 depth_range raster...
#>  Volume: 889068095860444 km³
#>
#> === Habitat Volume by quarter ===
#> Q1 : 1140653 cubic kilometers
#> Q2 : 1039133 cubic kilometers
#> Q3 : 1042252 cubic kilometers
#> Q4 : 889068.1 cubic kilometers
#> ================================
#>
#> Geographic projection detected. Preparing land background...
#>  Land data prepared.
#> Extracting raster values...
#>  Creating map panels for Minimum ...
#>  Creating map panels for Maximum ...
#>  Creating map panels for Range ...
#> Generating boxplots with significance tests...
#>  Processing Minimum ...
#>  Kruskal-Wallis p-value: 0
#>  Significant differences detected, running Dunn's post-hoc test.
#>  Processing Maximum ...
#>  Kruskal-Wallis p-value: 0
#>  Significant differences detected, running Dunn's post-hoc test.
#>  Processing Range ...
#>  Kruskal-Wallis p-value: 0
#>  Significant differences detected, running Dunn's post-hoc test.
#>
#> Calculating area from presence rasters...
#>   Q1 area: 20999863 km²
#>   Q2 area: 14088001 km²
#>   Q3 area: 11722848 km²
#>   Q4 area: 13031528 km²
#> Overlay plot saved.
#> All plots and report saved to: F:/whaleshark_sdm/4D/TPD
#> Done.
result_TPD$depth_range
```



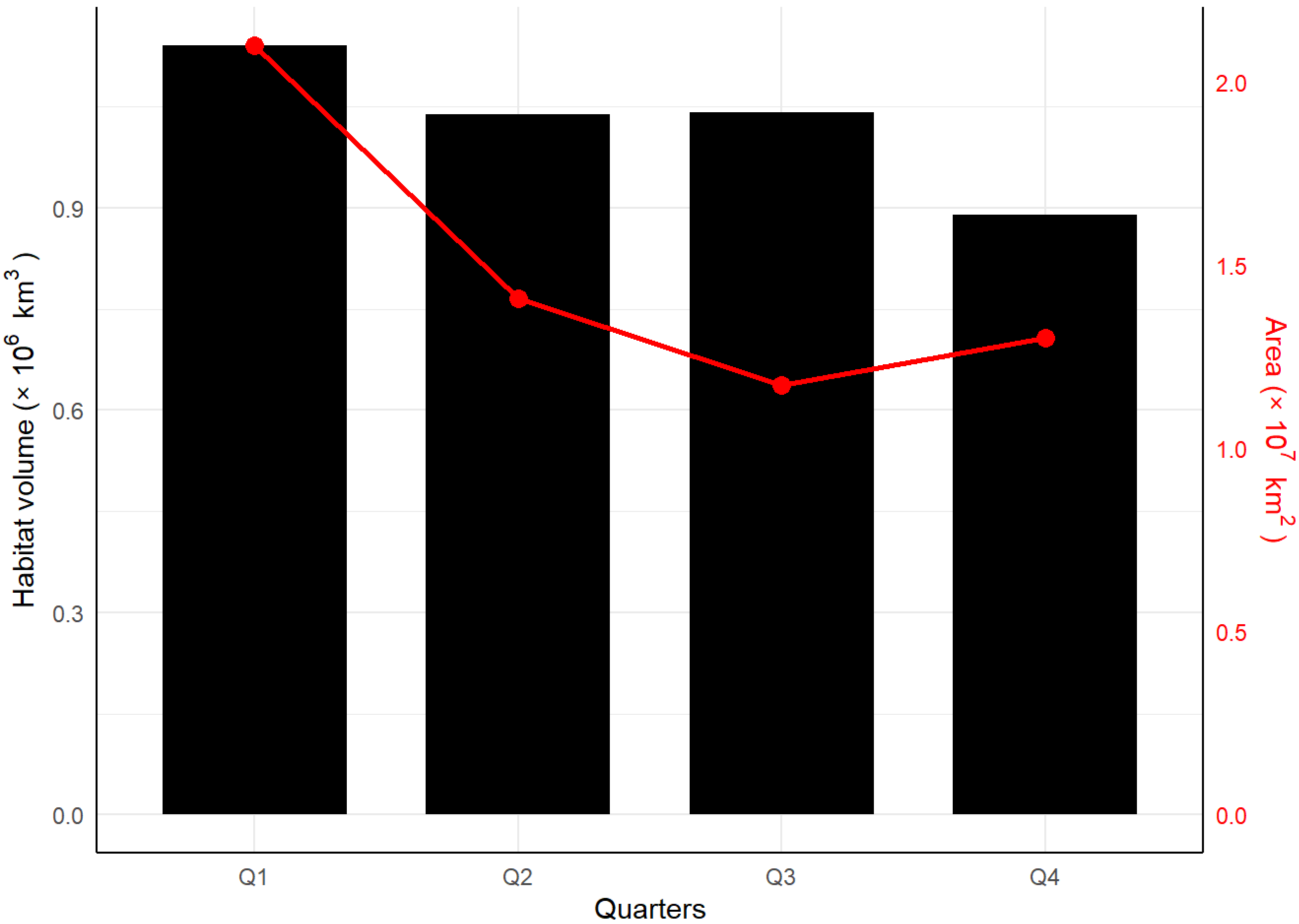

```
result_TPD$depth_min
```




```
result_TPD$depth_max
```



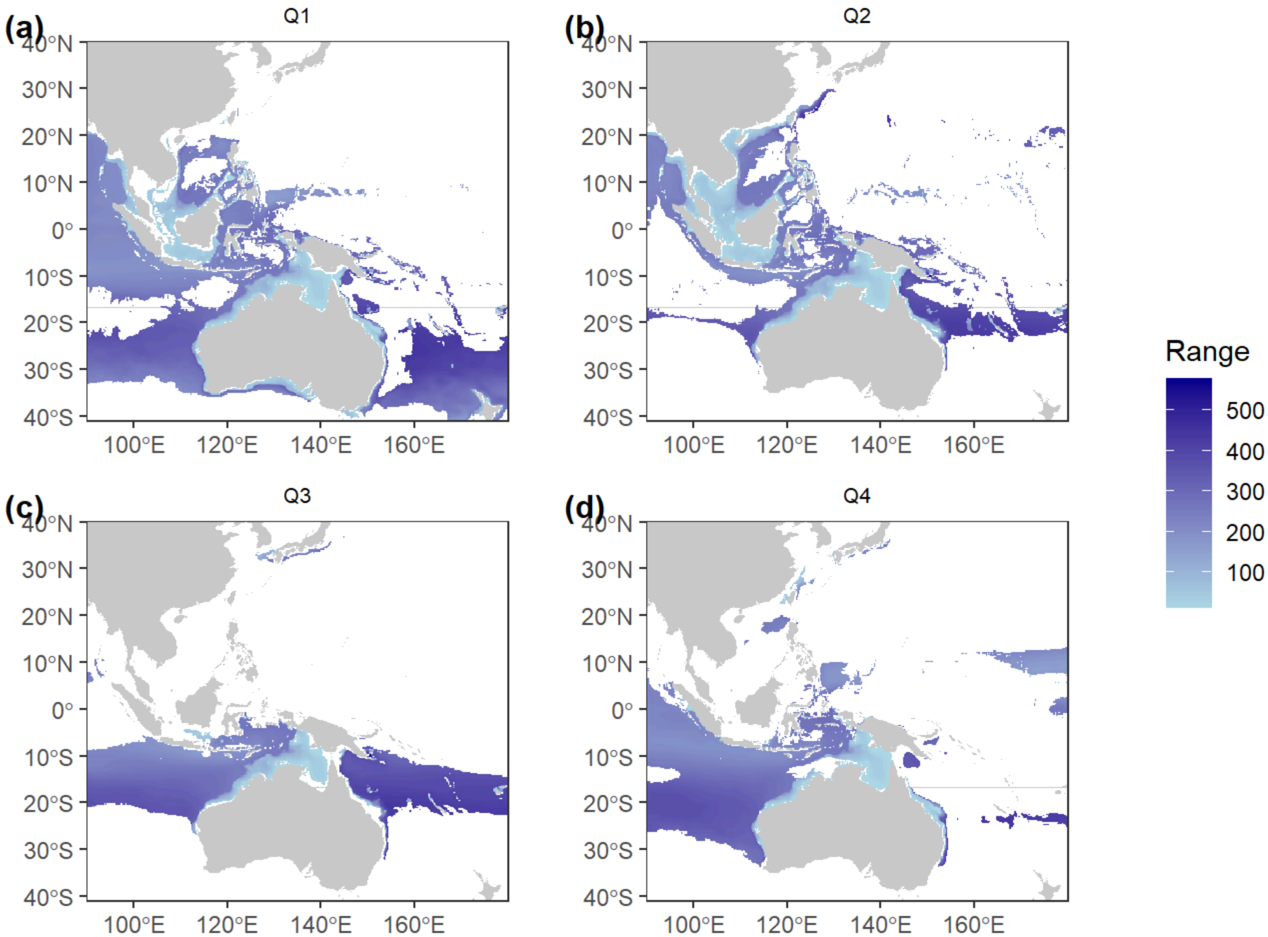

```
result_TPD$overlay_plot
#> Warning: No shared levels found between `names(values)` of the manual scale and the data's fill
values.
```



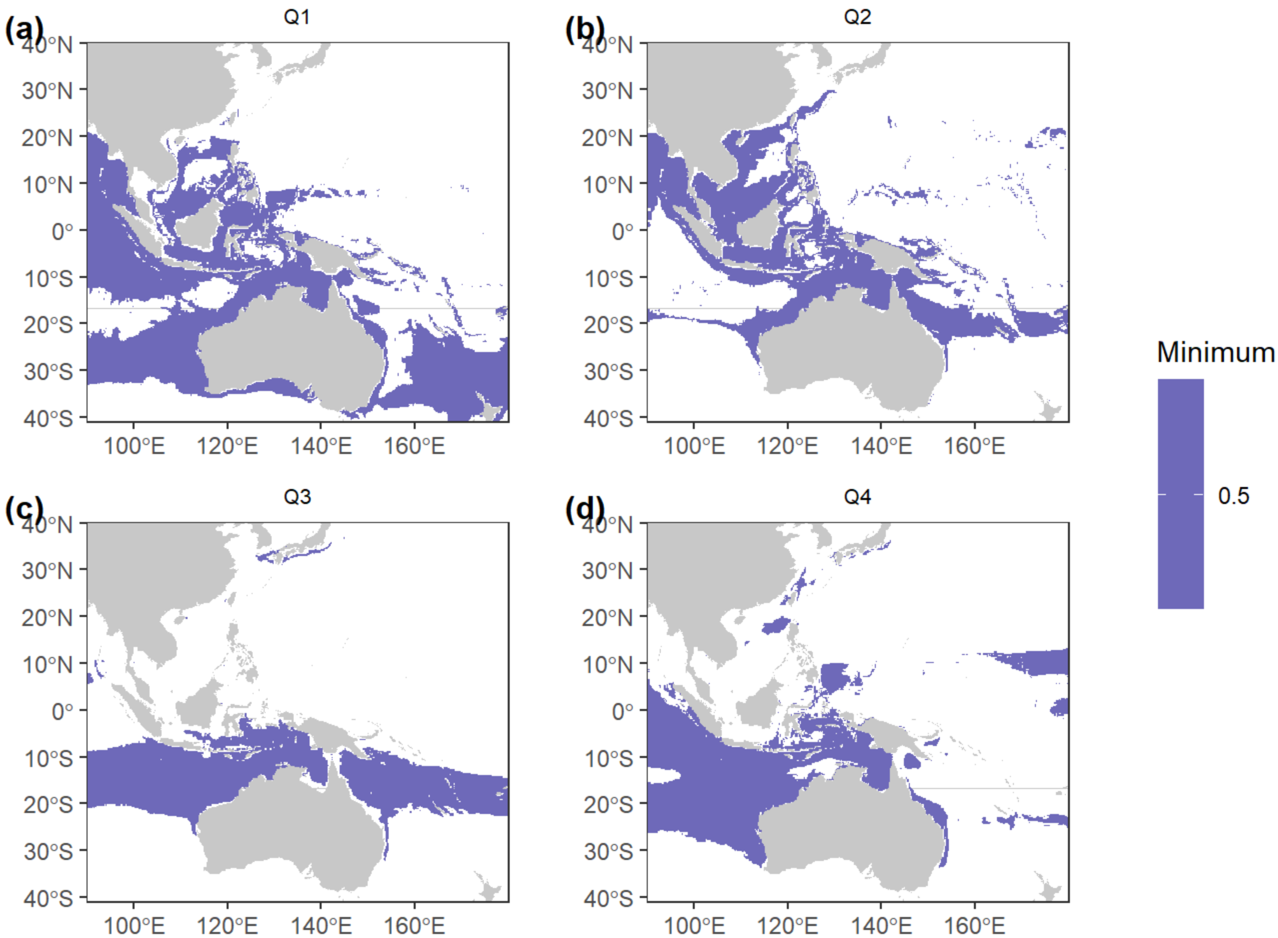
Habitat volume (× 10^6 km^3)
0.0
0.3
0.6
0.9
Area (× 10^7 km^2)
0.0
0.5
1.0
1.5
2.0
Q1
Q2
Q3
Q4
Quarters

```r
result_TTD <- plot_stat_4D(
  file_list = TTD_quarter,
  save_path = save_path_TTD,
  time_type = "quarter",
  presence_files = quarter_files
)
#> Reading 12 raster files...
#>  Reading Q1_min_depth.asc
#>  Reading Q1_max_depth.asc
#>  Reading Q1_depth_range.asc
#>  Reading Q2_min_depth.asc
#>  Reading Q2_max_depth.asc
#>  Reading Q2_depth_range.asc
#>  Reading Q3_min_depth.asc
#>  Reading Q3_max_depth.asc
#>  Reading Q3_depth_range.asc
#>  Reading Q4_min_depth.asc
#>  Reading Q4_max_depth.asc
#>  Reading Q4_depth_range.asc
#> All rasters loaded.
#>
#> Computing habitat volume for each quarter ...
#>  Processing Q1 depth_range raster...
#>  Volume: 4878510509266754 km³
#>  Processing Q2 depth_range raster...
#>  Volume: 2884376718297078 km³
#>  Processing Q3 depth_range raster...
#>  Volume: 3262322321583319 km³
#>  Processing Q4 depth_range raster...
#>  Volume: 3110711625978076 km³
#>
#> === Habitat Volume by quarter ===
#> Q1 : 4878511 cubic kilometers
#> Q2 : 2884377 cubic kilometers
#> Q3 : 3262322 cubic kilometers
#> Q4 : 3110712 cubic kilometers
#> ===============================
#>
#> Geographic projection detected. Preparing land background...
#>  Land data prepared.
#> Extracting raster values...
#>  Creating map panels for Minimum ...
#>  Creating map panels for Maximum ...
#>  Creating map panels for Range ...
#> Generating boxplots with significance tests...
#>  Processing Minimum ...
#>  Kruskal-Wallis p-value is NaN/NA. Skipping post-hoc tests.
#>  Processing Maximum ...
#>  Kruskal-Wallis p-value: 0
#>  Significant differences detected, running Dunn's post-hoc test.
#>  Processing Range ...
#>  Kruskal-Wallis p-value: 0
```

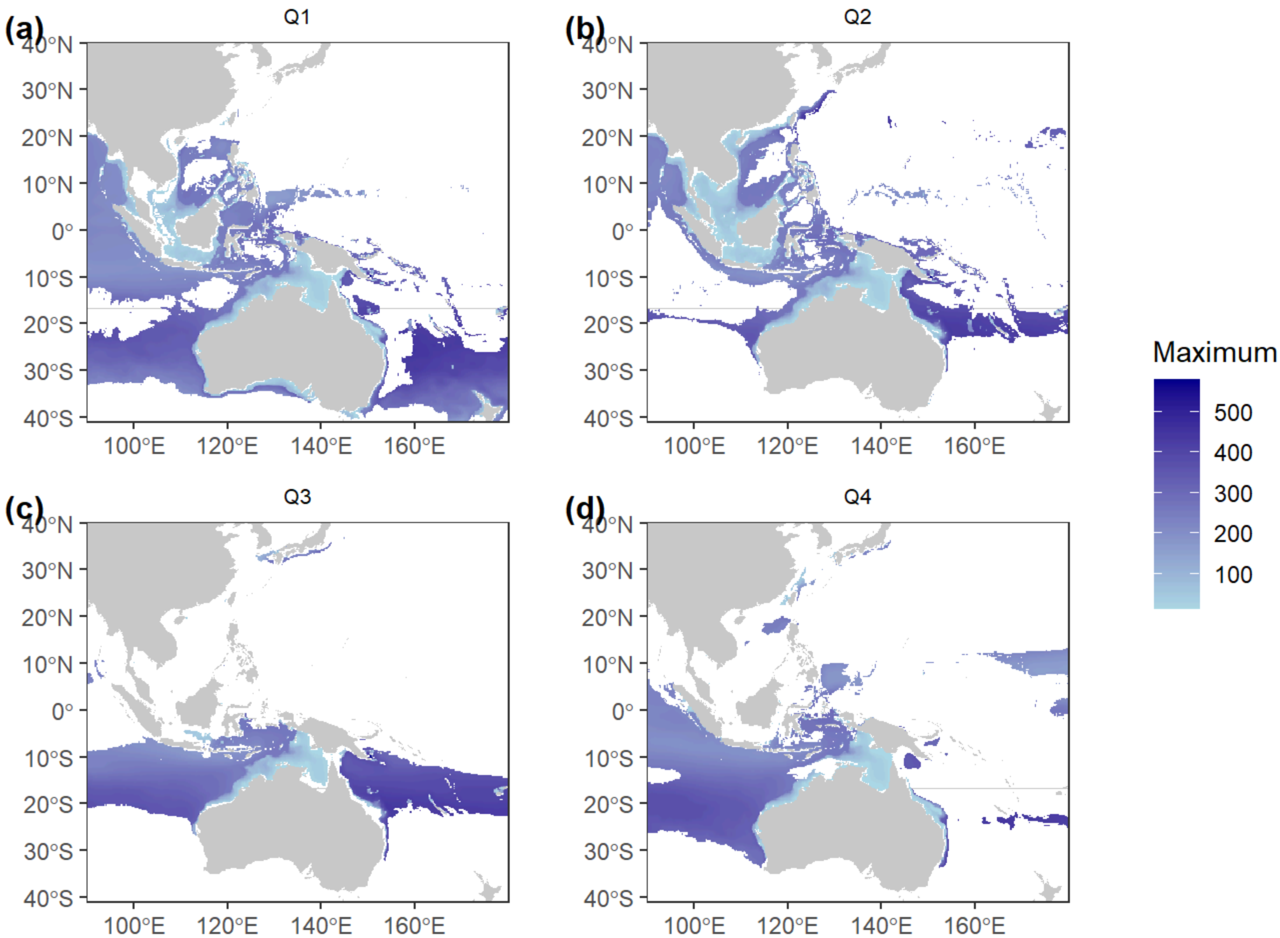
```
#>  Significant differences detected, running Dunn's post-hoc test.
#>
#> Calculating area from presence rasters...
#>   Q1 area: 20999863 km²
#>   Q2 area: 14088001 km²
#>   Q3 area: 11722848 km²
#>   Q4 area: 13031528 km²
#> Overlay plot saved.
#> All plots and report saved to: F:/whaleshark_sdm/4D/TTD
#> Done.
result_TTD$depth_range
```



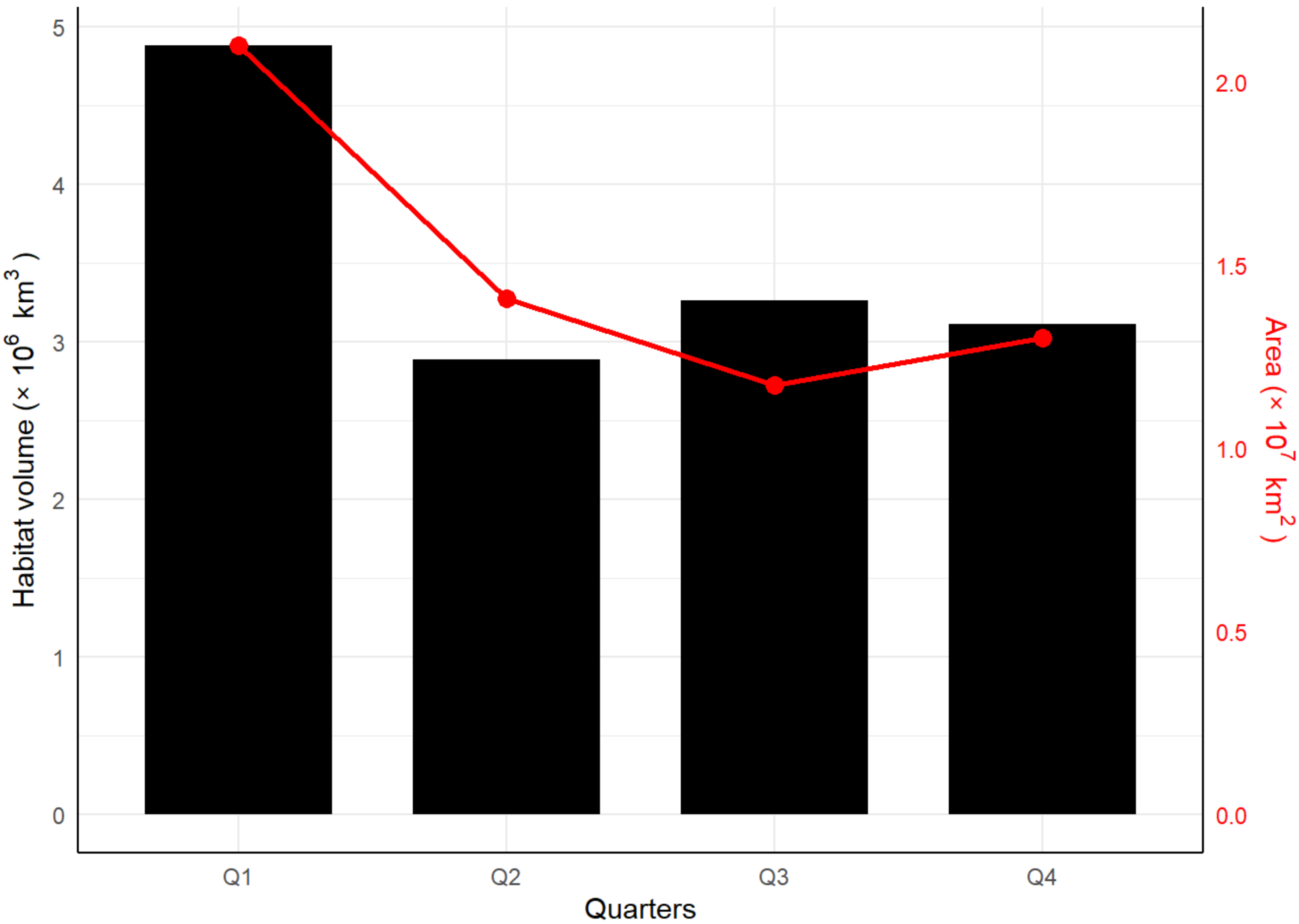

```
result_TTD$depth_min
```




```
result_TTD$depth_max
```




```
result_TTD$overlay_plot
#> Warning: No shared levels found between `names(values)` of the manual scale and the data's fill
values.
```



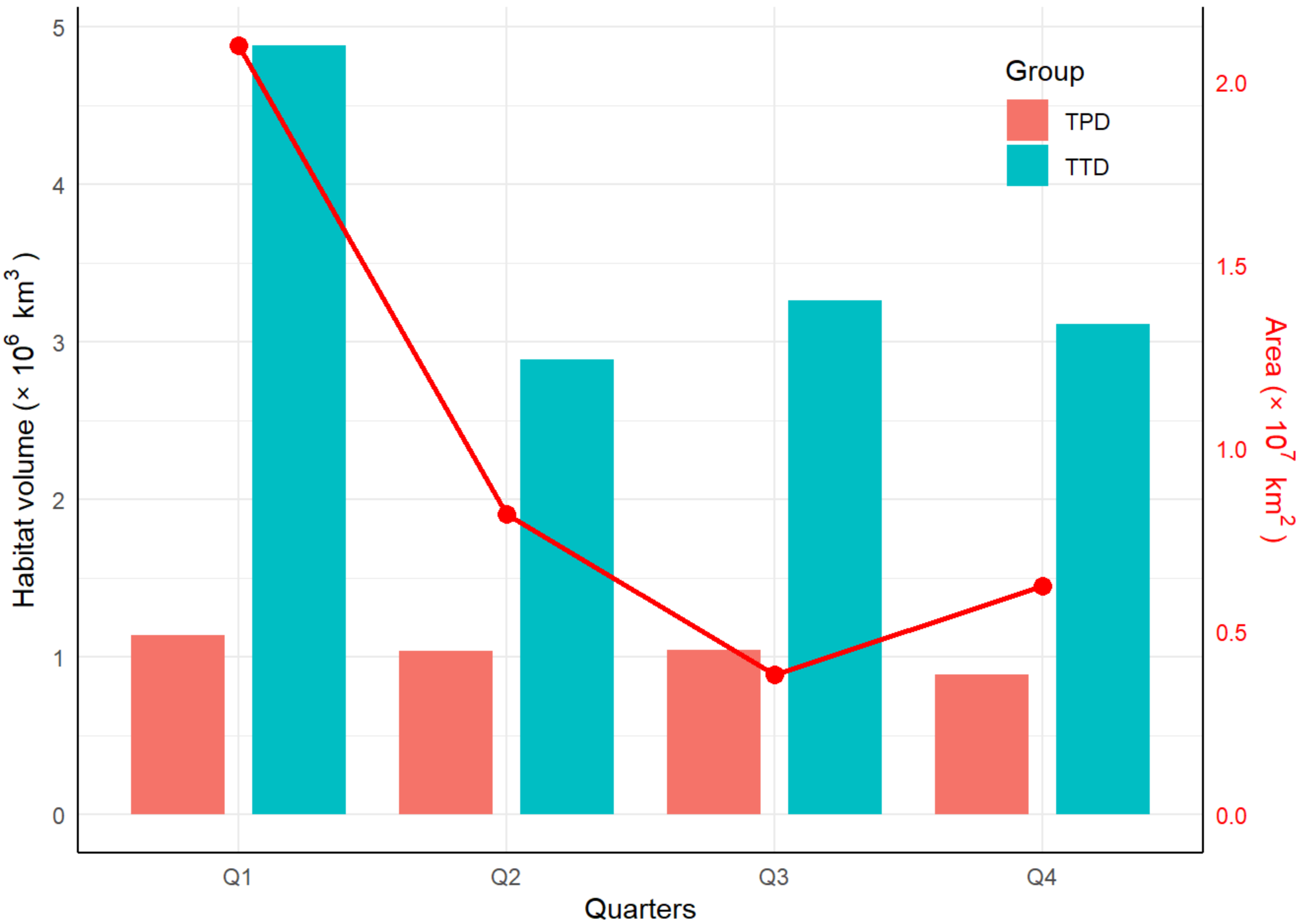

```
volum<-rbind(result_TPD$volum,result_TTD$volum)
volum$Group<-c(rep("TPD",4),rep("TTD",4))
area<-result_TPD$area
# write.csv(volum,"F:/whaleshark_sdm/4D/volume_data.csv",row.names = F)
# write.csv(area,"F:/whaleshark_sdm/4D/area_data.csv",row.names = F)
```

3.5 Plot habitat volumes (estimated based on TD and PD) and habitat area in one graph (Optional)

You can use the following codes to plot the two habitat volume estimates (based on TD and PD) and habitat area on one graph.

```
library(ggplot2)
volum<-read.csv("F:/whaleshark_sdm/4D/volume_data.csv")
area<-read.csv("F:/whaleshark_sdm/4D/area_data.csv")
# convert raw values to [0,1]
choose_factor <- function(x) {
if (all(x == 0)) return(1)
max_val <- max(x, na.rm = TRUE)
e <- floor(log10(max_val))
factor <- 10^e
while (max_val / factor > 10) factor <- factor * 10
while (max_val / factor < 1 && factor > 1) factor <- factor / 10
return(factor)
}
vol_factor <- choose_factor(volum$Volume_km3)
area_factor <- choose_factor(area$Area_km2)
volum$Volume <- volum$Volume_km3 / vol_factor
area$Area <- area$Area_km2 / area_factor
scale_ratio <- max(volum$Volume ) / max(area$Area)
area$Area <- area$Area * scale_ratio
# Prepare plot labels
vol_exp <- log10(vol_factor)
area_exp <- log10(area_factor)
vol_label <- bquote("Habitat volume (×" ~ 10^.(vol_exp) ~ " km"^3 ~ ")")
area_label <- bquote("Area (×" ~ 10^.(area_exp) ~ " km"^2 ~ ")")
# Ensure Temporal_bin is an ordered factor (in the order Q1, Q2, Q3, Q4)
volum$Temporal_bin <- factor(volum$Temporal_bin, levels = c("Q1", "Q2", "Q3", "Q4"))
area$Temporal_bin <- factor(area$Temporal_bin, levels = c("Q1", "Q2", "Q3", "Q4"))
# Merge area data into the main dataset (for plotting)
library(dplyr)
area_matched <- area %>%
right_join(volum %>% select(Temporal_bin) %>% distinct(), by = "Temporal_bin")
# Calculate the range of Volume (for automatic scaling of Area)
vol_range <- range(volum$Volume, na.rm = TRUE)
area_range <- range(area_matched$Area, na.rm = TRUE)
# Scaling Factor: The numerical range mapping Area to Volume
# Mapping formula: area_scaled = (Area - area_min) / (area_max - area_min) * (vol_max - vol_min) +
vol_min
area_min <- area_range[1]
area_max <- area_range[2]
vol_min <- vol_range[1]
vol_max <- vol_range[2]
# Avoid division by zero (if the area range is too small)
if (area_max == area_min) {
scaling_factor <- 1
offset <- 0
} else {
scaling_factor <- (vol_max - vol_min) / (area_max - area_min)
offset <- vol_min - scaling_factor * area_min
}
# Add a column for the scaled area
area_matched$Area_scaled <- area_matched$Area * scaling_factor + offset
p <- ggplot(volum, aes(x = Temporal_bin, y = Volume, fill = Group)) +
```

```
  geom_col(position = position_dodge(0.9), width = 0.7) +
  geom_line(
  data = area_matched,
  aes(x = Temporal_bin, y = Area_scaled, group = 1),
  color = "red", size = 1,
  inherit.aes = FALSE
  ) +
  geom_point(
  data = area_matched,
  aes(x = Temporal_bin, y = Area_scaled),
  color = "red", size = 3,
  inherit.aes = FALSE
  ) +
  scale_y_continuous(
  name = vol_label,
  sec.axis = ggplot2::sec_axis(
  ~ . / scale_ratio,
  name = area_label,
  labels = function(x) format(x, digits = 2)
  )
  )+
  labs(x = "Quarters", fill = "Group") +
  theme_minimal() +
  theme(
  axis.line = element_line(),
  axis.title.y.right = element_text(color = "red"),
  axis.text.y.right = element_text(color = "red"),
  legend.position = c(0.93, 0.96),
  legend.justification = c(1, 1)
  )
p
```




```
## save the figure
ggsave(
filename = "F:/whaleshark_sdm/figures/Fig. 7 TB_volume_area.tif",
plot = p,
device = "tiff",
dpi = 300,
compression = "lzw",
width = 6,
height = 6
)
```

The end

References

Andrzejaczek, S., Gleiss, A. C., Pattiaratchi, C. B., & Meekan, M. G. (2019). Patterns and drivers of vertical movements of the large fishes of the epipelagic. Reviews in Fish Biology and Fisheries, 29(2), 335–354. https://doi.org/10.1007/s11160-019-09555-1 (https://doi.org/10.1007/s11160-019-09555-1)

Arrowsmith, L., Sequeira, A., Pattiaratchi, C., & Meekan, M. (2021). Water temperature is a key driver of horizontal and vertical movements of an ocean giant, the whale shark Rhincodon typus. Marine Ecology Progress Series, 679, 101–114. https://doi.org/10.3354/meps13899 (https://doi.org/10.3354/meps13899)

Braun, C. D., Della Penna, A., Arostegui, M. C., Afonso, P., Berumen, M. L., Block, B. A., Brown, C. A., Fontes, J., Furtado, M., Gallagher, A. J., Gaube, P., Golet, W. J., Kneebone, J., Macena, B. C. L., Mucientes, G., Orbesen, E. S., Queiroz, N., Shea, B. D., Schratwieser, J., … Thorrold, S. R. (2023). Linking vertical movements of large pelagic predators with distribution patterns of biomass in the open ocean. Proceedings of the National Academy of Sciences, 120(47), e2306357120. https://doi.org/10.1073/pnas.2306357120 (https://doi.org/10.1073/pnas.2306357120)

Coffey, D. M., Carlisle, A. B., Hazen, E. L., & Block, B. A. (2017). Oceanographic drivers of the vertical distribution of a highly migratory, endothermic shark. Scientific Reports, 7(1), 10434. https://doi.org/10.1038/s41598-017-11059-6 (https://doi.org/10.1038/s41598-017-11059-6)

Dahms, C., & Killen, S. S. (2023). Temperature change effects on marine fish range shifts: A meta-analysis of ecological and methodological predictors. Global Change Biology, 29(16), 4459–4479. https://doi.org/10.1111/gcb.16770 (https://doi.org/10.1111/gcb.16770)

Dobson, R., Challinor, A. J., Cheke, R. A., Jennings, S., Willis, S. G., & Dallimer, M. (2023). dynamicSDM: An R package for species geographical distribution and abundance modelling at high spatiotemporal resolution. Methods in Ecology and Evolution, 14(5), 1190–1199. https://doi.org/10.1111/2041-210X.14101 (https://doi.org/10.1111/2041-210X.14101)

Hertz, P. E., Huey, R. B., & Stevenson. (1993). Evaluating Temperature Regulation by Field-Active Ectotherms: The Fallacy of the Inappropriate Question | The American Naturalist: Vol 142, No 5. The American Naturalist, 142(5). https://www.journals.uchicago.edu/doi/10.1086/285573 (https://www.journals.uchicago.edu/doi/10.1086/285573)

Naimi, B., & Araújo, M. B. (2016). sdm: A reproducible and extensible R platform for species distribution modelling. Ecography, 39(4), 368–375. https://doi.org/10.1111/ecog.01881 (https://doi.org/10.1111/ecog.01881)

Owens, H. L., & Rahbek, C. (2023). voluModel: Modelling species distributions in three-dimensional space. Methods in Ecology and Evolution, 14(3), 841–847. https://doi.org/10.1111/2041-210X.14064 (https://doi.org/10.1111/2041-210X.14064)

Valle, M., Ramírez-Romero, E., Ibaibarriaga, L., Citores, L., Fernandes-Salvador, J. A., & Chust, G. (2024). Pan-Atlantic 3D distribution model incorporating water column for commercial fish. Ecological Modelling, 490, 110632. https://doi.org/10.1016/j.ecolmodel.2024.110632 (https://doi.org/10.1016/j.ecolmodel.2024.110632)